

Architecture Outline

NIGERIA SERVICE EXPORT GATEWAY (NATEP / NCMSE) - Architecture Outline

Business Requirements Document (BRD) — Structural Architecture & Master Outline

Executive Overview & Strategy for Granular Document Structure

To ensure the Business Requirements Document (BRD) moves from high-level operational concept to an unambiguous, engineering-ready specification, the document is structured into **8 core structural modules**. Each module is designed to be unpacked down to the **atomic functional requirement level** (inputs, processing rules, validation logic, system-of-record boundaries, edge cases, and output artifacts).

1. Core Governance, Scope & Single-Platform Principles

- **1.1 Institutional Rationale & Mandate**
 - FMITI, NCMSE Secretariat, and NATEP governance boundaries.
 - Commercial and legal framework (Nigeria Data Protection Act, Public Procurement Act).
- **1.2 Mandatory Architecture Principles**
 - The Single-Platform Mandate (Prohibition of siloed identity, evidence, consent, or audit stores).
 - System-of-Record (SoR) matrix across all entities.
- **1.3 Scope Boundaries & Non-Goals**
 - In-scope business capabilities vs. explicit platform exclusions (e.g., non-goal: stand-alone payment platform or unmoderated job board).

2. Shared Platform Foundation Services (Canonical Common Layer)

- **2.1 Identity & Access Management (IAM)**
 - Unified SSO, multi-factor authentication (MFA), role-based & attribute-based access control (RBAC/ABAC).
 - Organization delegation & multi-tenant user context switching.
- **2.2 Canonical Actor Registry & Identity Verification**
 - Actor Resolution Model (Individuals, Firms, Aggregators, Buyers, Regulators, Institutional Partners).
 - Deduplication, verification tiers, and actor status lifecycle.
- **2.3 Evidence & Document Vault**
 - File ingestion, cryptographic verification, malware scanning, retention, and time-stamped metadata.
 - Reuse-by-reference mechanism (zero duplicate uploads across portals).

- **2.4 Granular Consent Engine**
 - Purpose-bound, recipient-specific, field-limited consent requests and active grant states.
 - Revocation, expiry, and downstream disclosure invalidation rules.
- **2.5 Immutable Disclosure Package Service**
 - Field-limited envelope creation, secure access logging, and expiration controls.
- **2.6 Master Taxonomies & Controlled Vocabularies**
 - Service categories, occupational roles/skills (GATS/WTO modes of supply), jurisdictions, and status families.
- **2.7 Audit Logging, Event Bus & Observability**
 - Tamper-evident activity logging, asynchronous domain event schemas, replay controls, and failure monitoring.

3. Portal 1: Demand & Market Access Domain (Granular Requirements)

- **3.1 Demand Intelligence & Market Playbooks**
 - Source intake (publications, mission reports, trade findings), confidence scoring, and playbook versioning.
- **3.2 Demand Actor & Buyer Onboarding**
 - Buyer verification, procurement profile setup, relationship ownership, and risk classification.
- **3.3 Signal Capture & Ingestion Engine**
 - Ingestion routes: Public buyer forms, assisted officer capture, external API, and secure batch file upload.
- **3.4 Signal Verification & Opportunity Qualification**
 - Automated deduplication checks, qualification checklists, risk-tier escalation, and qualification decision rationale.
- **3.5 Opportunity Criteria Package (OCP) Specification**
 - Mandatory, preferred, negotiable, and informational criteria definitions; package versioning and impact analysis.
- **3.6 Cross-Portal Requirement & Supply Integration**
 - Binding OCP to Portal 2 Regulatory Requirement Links and Portal 3 aggregate supply queries.
- **3.7 Explainable Matching & Human Shortlisting**
 - Score transparency, factor weighting, exclusion rules, bias checks, and human review approval workflows.
- **3.8 Consented Referral & Engagement Tracking**
 - Shared Disclosure Package linking, referral transmission, buyer acknowledgement, and proposal lifecycle.
- **3.9 Conversion & Contract Tracking**
 - Interview tasks, due diligence tracking, offer/contract metadata recording, and commencement confirmation.
- **3.10 Outcome Verification & Attribution Engine**
 - Evidence submission, rule-routed verification workflows, duplicate value checks,

single-count outcome accounting, and partner contribution attribution.

4. Portal 2: Regulatory Trust & Exporter Readiness Domain (Granular Requirements)

- **4.1 Regulatory Requirements Register**
 - Classification taxonomy (statutory, professional, fiscal, FX/settlement, buyer-driven standards).
 - Requirement Record schema (plain-language summary, official channel, applicability criteria, evidence expectations).
- **4.2 Institutional Authoring & Validation Engine**
 - Workflow queues for competent authorities, multi-tier approvals, review calendars, and disclaimer management.
- **4.3 Personalised Requirements Wizard & Pathway Engine**
 - Dynamic questionnaire logic (actor archetype, service type, export mode, destination market) and context-specific rule evaluations.
- **4.4 Exporter Workspace & Readiness Assessments**
 - Private readiness checklists, evidence gap identification, assessment reports, and bank/FX settlement readiness tracking.
- **4.5 Change Alert & Invalidation Engine**
 - Detection of regulatory revisions, impact analysis on live opportunities/pathways, auto-generated review tasks, and notification triggers.
- **4.6 Readiness Assertions & Cross-Portal Contracts**
 - Generating purpose-bound, time-stamped Readiness Assertions for Portal 1 and Portal 3 consumption.

5. Portal 3: Supply Ecosystem & Competitiveness Domain (Granular Requirements)

- **5.1 Exporter & Supplier Profiling**
 - Capability profiles (individuals, service firms, providers), skill taxonomies, portfolio evidence, and privacy-by-default controls.
- **5.2 Offerings, Capacity & Freshness Tracking Matrix**
 - Quantifiable capacity definitions (units, hours, headcount), availability windows, and automated stale-capacity exclusion timers.
- **5.3 Supply Pools & Dynamic Grouping**
 - Rule-based pool definitions (service, geography, evidence tier) and capacity confirmation query workflows.
- **5.4 Competitiveness Diagnostics & Gap Analysis**
 - Assessment frameworks, multi-dimensional scoring rubrics, automated gap detection, and personalized improvement pathways.
- **5.5 Intervention Mapping & Program Management**
 - Curriculum/program blueprints, provider eligibility checks, participant selection workflows, resource tracking, and reassessment cycles.

- **5.6 Potential Fit Evaluation & Capability Packages**
 - Inbound OCP consumption, explainable potential-fit calculations, Capability Package generation, and partial consent integration.

6. National Services Export Observatory & Policy Intelligence

- **6.1 Curated Indicator Pipeline**
 - Ingesting operational events from Portals 1–3, data cleansing, confidence weighting, and aggregate metrics generation.
- **6.2 Double-Counting & Reconciliation Protocol**
 - Reconciling trade metrics across NEPC, NEXIM, CBN, and FMITI systems without duplicating economic export value.
- **6.3 Inclusion & Policy Bottleneck Analytics**
 - Regional distribution funnel monitoring, gender/demographic inclusion tracking, and regulatory bottleneck identification.

7. Non-Functional, Security & System Architecture Specifications

- **7.1 Data Protection & Privacy Controls (NDPA Compliance)**
 - Lawful basis registry, Privacy Impact Assessments, data minimization fields, and subject access request (SAR) workflows.
- **7.2 System Performance, SLAs & Disaster Recovery**
 - Response time benchmarks, concurrent user targets, backup frequencies, Recovery Point Objective (RPO), and Recovery Time Objective (RTO).
- **7.3 API, Event & Interoperability Architecture**
 - API Gateway standards, JSON Schema contracts, asynchronous event envelope definitions, and failure retry/dead-letter mechanisms.

8. Implementation Traceability & Stage Gate Matrix

- **8.1 Detailed Requirements Traceability Matrix (RTM) Template**
 - Traceability mapping from TOR clause -> BRD Requirement ID -> SRS Module -> User Story -> API/UI Component -> Test Case.
- **8.2 Stage-Gate Verification Protocols (Gates G0 to G6)**
 - Objective evidence criteria for each release gate across the 26-week schedule.

Module 1 BRD/FRD

NIGERIA SERVICE EXPORT GATEWAY (NATEP / NCMSE)

Business Requirements Document (BRD) — Detailed Functional Specification

MODULE 1: CORE GOVERNANCE, SCOPE & SINGLE-PLATFORM PRINCIPLES

1.0 Document Control & Baseline Reference

Control Attribute	Specification
Document Title	Business Requirements Document (BRD) — Module 1: Core Governance, Scope & Single-Platform Principles
Parent Programme	Nigeria Service Export Gateway (NATEP / NCMSE)
Executive Sponsor	Federal Ministry of Industry, Trade and Investment (FMITI)
Business Owner	NATEP / NCMSE Secretariat
Technical Authority	Architecture, Data, Security & Privacy Authority (ADSPA)
Baseline Source	Integrated Four-Part Terms of Reference (TOR) v1.1 (September 2026 Baseline)
Precedence	Applicable Law > Executive Decisions > Integrated TOR > BRD Module 1 > SRS / Technical Specifications

1.1 Institutional Rationale, Mandate & Governance Structure

1.1.1 Strategic Rationale & Institutional Context

The Nigeria Service Export Gateway ("the Gateway") serves as the core digital infrastructure and operating engine for the **National Coordination Mechanism for Services Exports**

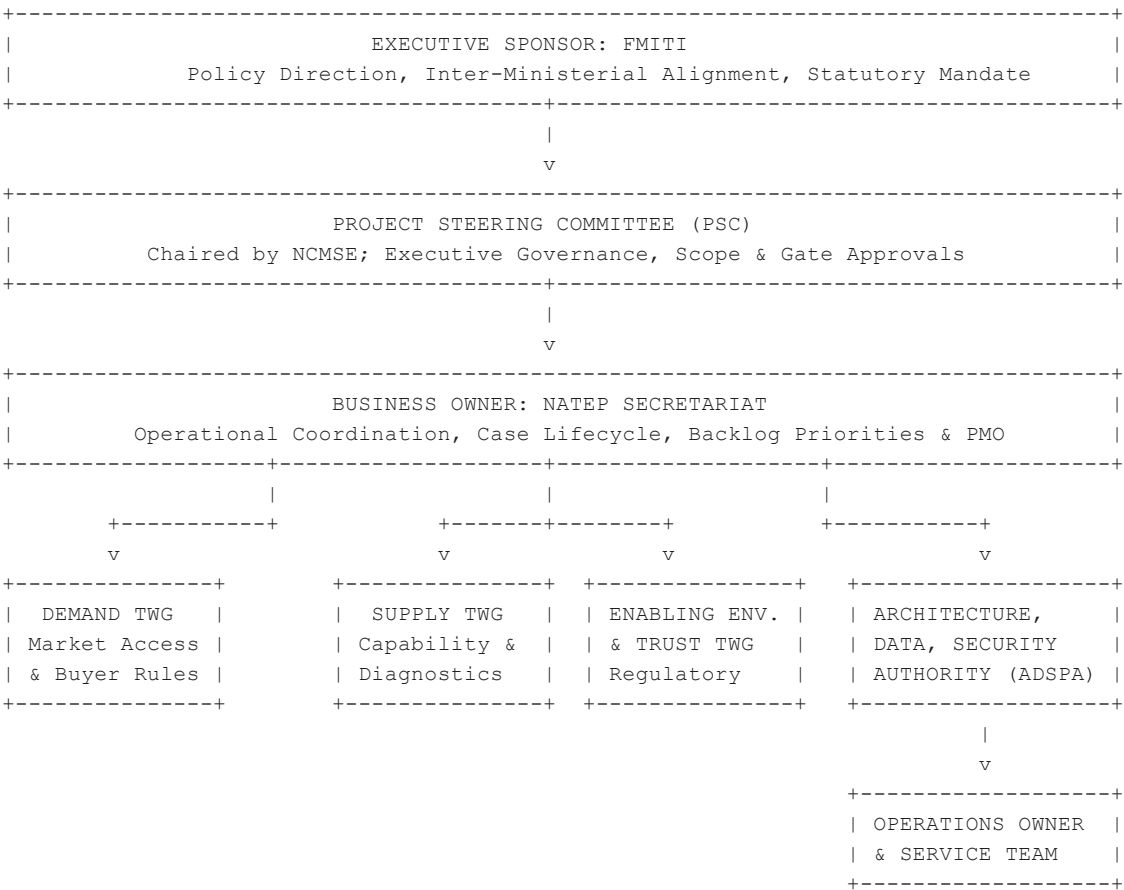
(NCMSE) under the mandate of the **Federal Ministry of Industry, Trade and Investment (FMITI)** and the **National Talent Export Programme (NATEP)**.

Nigeria's service export ecosystem historically suffers from four structural bottlenecks:

- 1. **Fragmented Market Access:** High-value international demand signals (buyer procurement, foreign missions, outsourcing programs, diaspora networks) are captured in silos without structured qualification, verification, or systematic matching.
- 2. **Regulatory & Compliance Deficits:** Exporters, freelancers, and international buyers lack a single, authoritative, source-linked repository of statutory, professional, fiscal, data protection, and foreign exchange (FX) compliance requirements.
- 3. **Invisible & Unverified Supply:** Domestic service supply (firms, aggregators, freelancers, independent professionals) lacks standardized capability metrics, verifiable capacity data, and structured improvement interventions.
- 4. **Data Disconnect & Double-Counting:** Trade institutions (NEPC, NEXIM, CBN, FMITI, NBS) operate without a unified data model, leading to duplicated outcome claims, unverified export figures, and unaddressed policy bottlenecks.

The Gateway unifies these domains into a single, seamless digital ecosystem while strictly preserving institutional statutory authority.

1.1.2 Institutional Governance Framework



1. **Executive Sponsor (FMITI):**
 - Holds executive policy authority, approves statutory extensions, and handles inter-ministerial escalations.
 - Approves reserved policy decisions, major budget baseline changes, and formal handover deliverables.
2. **Project Steering Committee (PSC):**
 - Executive governing body chaired by NCMSE.
 - Grants stage-gate approvals (G0 through G6), approves material scope modifications, and enforces cross-agency alignment.
3. **Business Owner & Coordinating Operator (NATEP / NCMSE Secretariat):**
 - Acts as the primary owner of business operations, case management, and overall outcomes realization.
 - Controls operational backlogs, priority cohorts, case distribution, and partner attribution.
4. **Technical Working Groups (TWGs):**
 - **Demand TWG:** Defines buyer onboarding rules, signal qualification standards, matching score weights, and opportunity priority tiers.
 - **Supply TWG:** Defines skill taxonomies, capability assessment rubrics, capacity freshness thresholds, and provider intervention rules.
 - **Enabling Environment & Trust TWG:** Curates regulatory content, mapping schemas, institutional validation protocols, and readiness assertion logic.
5. **Architecture, Data, Security & Privacy Authority (ADSPA):**
 - Consolidated technical governance body providing technical sign-offs for architecture blueprints, system-of-record matrices, API contracts, security compliance, and NDPA privacy baselines.
6. **Operations Owner & Service Management Team:**
 - Responsible for live-service readiness, support management, content stewardship, service desk operations, and Year-Two sustainability.

1.1.3 Legal, Regulatory & Policy Alignment

The Gateway platform and data workflows shall strictly adhere to the following statutory and regulatory frameworks:

- **Nigeria Data Protection Act (NDPA) 2023:** Mandates explicit consent controls, purpose limitation, data minimization, privacy-by-design, data protection impact assessments (DPIA), and explicit lawful bases for processing personal and commercial data.
- **Public Procurement Act (PPA) 2007:** Governs platform procurement, vendor engagement, service level transparent auditing, and stage-gate verification.
- **National Outsourcing Policy & National Digital Economy Policy and Strategy (NDEPS):** Dictates target export tracks (BPO, IT/Software, Creative, Professional Services) and job creation standards.
- **WTO / GATS Modes of Supply:** The system must natively classify all service export

activities into the four World Trade Organization General Agreement on Trade in Services (GATS) delivery modes:

- **Mode 1: Cross-Border Supply** (e.g., Remote software development, digital BPO, virtual consulting).
- **Mode 2: Consumption Abroad** (e.g., In-country executive training, foreign trade missions).
- **Mode 3: Commercial Presence** (e.g., Foreign branch, subsidiary, or joint-venture presence).
- **Mode 4: Presence of Natural Persons** (e.g., Temporary on-site deployment of technical experts or consultants).

1.2 Mandatory Architecture Principles & Single-Platform Mandate

1.2.1 The Single-Platform Mandate

Delivery teams and contractors are **strictly prohibited** from constructing siloed databases, redundant user management systems, isolated document vaults, or detached notification/audit systems across the three business portals.

Mandatory Procurement & Architecture Rule: The Gateway is **one integrated platform powering three bounded business portals**, reusable common foundation services, and a centralized National Observatory. Any proposed exception must be justified through formal architecture review and approved by ADSPA before implementation.

NATIONAL SERVICES EXPORT OBSERVATORY									
Curated Analytics & Policy Intelligence Layer									
PORTAL 1			PORTAL 2				PORTAL 3		
Demand & Market			Regulatory Trust &				Supply Ecosystem		
Access			Exporter Readiness				& Competitiveness		
SHARED PLATFORM FOUNDATION (CANONICAL)									
Identity & Access			Canonical Actor Reg.				Evidence & Doc Vault		
(Single Sign-On/MFA)			(People & Entities)				(Zero-Duplicate Vault)		
Granular Consent			Immutable Disclosure				Controlled Taxonomies		
Engine (NDPA)			Package Engine				& Master Data		
Workflow & Task			Event Bus & API				Tamper-Evident Audit		
Engine			Gateway				& Security Monitor		

1.2.2 System-of-Record (SoR) Designation Matrix

To guarantee zero data duplication, prevent race conditions, and preserve analytical integrity, every core data object in the Gateway is assigned an exclusive System of Record (SoR). No portal may store authoritative state for an object owned by another domain.

Data Domain / Object	Authoritative System of Record (SoR)	Permitted Consumer Operations	Prohibited Operations
People, Organizations, Delegations	Shared Platform: Canonical Actor Registry	Read, reference via stable UAID, initiate update requests.	Direct creation of local portal user accounts or duplicate profile tables.
Evidence Documents & Metadata	Shared Platform: Evidence Vault	Ingest binary, reference by Hash/URI, verify validity status.	Direct local file storage in portal-specific database blobs or unverified folders.
Consent Requests, Grants & Revocations	Shared Platform: Consent Engine	Request consent context, evaluate active grant states.	Local override of consent status or bypass of user privacy preferences.
Master Taxonomies & Master Data	Shared Platform: Taxonomy Service	Retrieve versioned master codes, populate UI components.	Hardcoding text values or local taxonomy modifications in code.
Demand Signals, Opportunities, Matches	Portal 1: Demand Domain	Query pipeline state, consume qualified opportunity metrics.	Portal 2 or 3 creating local opportunity or buyer records.
Regulatory	Portal 2:	Reference active	Portal 1 or 3

Requirements & Pathways	Regulatory Domain	requirement links, display readiness criteria.	authoring or modifying regulatory compliance rules.
Capabilities, Capacity & Interventions	Portal 3: Supply Domain	Query aggregate capacity, request potential fit scores.	Portal 1 or 2 editing supplier skill rubrics or capacity logs.
Audit Logs & System Activity	Shared Platform: Audit Subsystem	Publish domain events, query audit history via authorized console.	Local log suppression, untracked DB mutations, or log editing.
Reconciled Export Metrics	National Observatory	Display read-only national dashboards and executive reports.	Manual override of underlying transactional SoR records from analytics views.

1.2.3 Cross-Portal Canonical Information Packages

Inter-portal interoperability relies on standard, immutable, versioned information packages. Direct database reads across portal boundaries are strictly prohibited.

1. **Opportunity Criteria Package (OCP) [Producer: Portal 1]:** An immutable statement of qualified demand defining mandatory, preferred, negotiable, and informational criteria, target service modes, capacity requirements, and timelines.
2. **Regulatory Requirement Link [Producer: Portal 2]:** An exact, versioned link referencing statutory, professional, fiscal, or foreign exchange requirements applicable to an opportunity or pathway.
3. **Readiness Assertion [Producer: Portal 2]:** A time-stamped, scoped assertion confirming an exporter's verified compliance evidence and limitations.
4. **Potential Fit Evaluation [Producer: Portal 3]:** An explainable comparison between an OCP and current supplier capability/capacity, providing score breakdown without making automated selection.
5. **Capability Package [Producer: Portal 3]:** A versioned, evidence-aware summary of supplier capabilities, certifications, and capacity approved for potential disclosure.
6. **Consent Request / Grant [Producer: Shared Consent Engine]:** A purpose-, recipient-, field-, and expiry-bound grant of authority executed by the data subject.
7. **Shared Disclosure Package [Producer: Shared Disclosure Engine]:** An immutable, minimum-necessary data package constructed under active consent for a specific

recipient.

8. **Outcome / Attribution Event [Producer: Portal 1 / Authorised Verifier]:** An evidence-classified record of completed export value, contract placement, or transaction, linked to partner attribution lineages without double-counting.

1.3 Scope Boundaries, Non-Goals & Protected Decision Rules

1.3.1 In-Scope Core Capabilities

The platform baseline encompasses six integrated delivery domains:

1. **Shared Platform Foundation Services:** Unified SSO/MFA, Canonical Actor Registry, Evidence Vault, Consent Engine, Immutable Disclosure Package Engine, Master Taxonomies, Task/Workflow Engine, Event Bus, API Gateway, and Tamper-Evident Audit Subsystem.
2. **Portal 1 (Demand & Market Access):** Market intelligence capture, playbooks, signal intake, qualification engine, Opportunity Criteria Package formulation, explainable matching, human shortlisting, referral tracking, proposal lifecycle, conversion tracking, outcome verification, and attribution.
3. **Portal 2 (Regulatory Trust & Exporter Readiness):** Regulatory Requirements Register, institutional authoring/validation queues, versioning/supersession, personalized readiness wizard, pathway mapping, readiness assertions, and change-alert engine.
4. **Portal 3 (Supply Ecosystem & Competitiveness):** Capability profiling, dynamic capacity matrix, competitiveness diagnostics, gap analysis, intervention mapping (training, financing, certification), and Capability Package generation.
5. **National Services Export Observatory:** Reconciled data pipeline, double-counting elimination protocol, regional/demographic inclusion tracking, and policy bottleneck analytics.
6. **Cross-Portal Interoperability & Integration:** Secure API Gateway, webhook distribution, external adapters (CAC, NIMC, FIRS, NEPC, CBN), and secure batch fallback mechanisms.

1.3.2 Explicit Platform Exclusions (Non-Goals)

To preserve focus, limit legal liability, and enforce operational integrity, the platform explicitly excludes the following capabilities:

EXPLICIT PLATFORM NON-GOALS	
[X] Stand-Alone Payment Gateway / Escrow / Direct Commercial Settlement Engine	
[X] Unmoderated Open Job Board or Public Freelancer Directory	
[X] Autonomous / Algorithmic Supplier Selection or Automatic Contract Award	
[X] Direct Execution of Binding Legal Contracts or Employment Agreements	
[X] Creation of New Statutory Obligations or Unilateral Regulatory Interpretations	
[X] Blanket Government Endorsements or Opaque Supplier Quality Ratings	

1. **Non-Goal 1: Direct Commercial Payment Processing or Escrow Services:** Payment settlement occurs through commercial banking and legal FX channels. The Gateway records transaction metadata solely for export verification and trade accounting.
2. **Non-Goal 2: Unmoderated Public Directory:** The platform shall not expose open-search directories of individuals or raw supplier lists to the public web. Access to supplier capability profiles is restricted, consent-gated, and mediated through qualified demand opportunities.
3. **Non-Goal 3: Autonomous Algorithmic Contract Award:** Matching algorithms provide decision support only. Automated scoring shall **never** automatically award contracts, issue binding referrals, or disqualify actors without human review.
4. **Non-Goal 4: Direct Legal Contracting:** The Gateway tracks milestone events (Contract Awarded, Commencement Confirmed) and stores reference metadata/hashes of executed agreements, but is not an e-signature contracting tool.
5. **Non-Goal 5: Creation of Statutory Obligations:** Portal 2 compiles, explains, versions, and routes regulatory guidance from competent authorities but possesses zero legal authority to create new statutory duties or levy fines.

1.3.3 Protected Decision Boundaries & Segregation of Duties

To prevent fraud, conflict of interest, and administrative abuse, the platform enforces strict human-in-the-loop decision boundaries:

Operational Action	Initiating Entity	Required Independent Decision	System Control
Demand Signal Submission	External Buyer / Partner	NATEP Demand Officer Qualification Gate	Automatic promotion blocked (Status = Pending Qualification).
Algorithmic Matching	Portal 1 Matching Engine	Human Demand Officer Shortlist Approval	Matching output is advisory only; automated sending disabled.
Disclosure Request	Portal 1 Referral Workflow	Supplier Data Subject Consent Grant	Document/data access blocked (HTTP 403) without active consent form.
Outcome Report	Exporter / Buyer	Independent	Self-reported

	Self-Report	Verification Officer	revenue tagged Provisional; excluded from Observatory.
Regulatory Content Edit	Content Drafter	Competent Agency Validation Focal	Content stays Draft until signed off by authorized regulator.

1.4 Implementation Roadmap, Release Strategy & Stage-Gate Framework

1.4.1 Integrated 26-Week (6-Month) Release Timeline

The programme follows an accelerated 26-week baseline structured into 3 releases and 7 formal stage gates:

Weeks 01-02	Design & Baseline Alignment	→	Gate G0 / Gate G1
Weeks 03-04	Release 1: First Rendition Deployment	→	Gate G2 (Week 4)
Weeks 05-10	Release 2 Part A: Shared Foundation & Alpha	→	Gate G3 (Week 10)
Weeks 11-16	Release 2 Part B: MVP Integrated Beta	→	Gate G4 (Week 16)
Weeks 17-22	Release 3 Part A: Controlled Field Pilot	→	Gate G5 (Week 22)
Weeks 23-26	Release 3 Part B: Hardening & Handover	→	Gate G6 (Week 26)

1.4.2 Stage-Gate Governance Matrix

Progress between project phases requires formal sign-off by the Project Steering Committee (PSC) against objective evidence:

Stage Gate	Target Timing	Decision Objective	Key Deliverables & Mandatory Evidence
Gate G0	Week 1	Mobilisation Acceptance	Governance charter, Ministry-owned cloud/repo accounts live, Decision Register initialized.

Gate G1	Week 2	Design Baseline Adoption	Approved BRD, SRS, System Architecture Blueprint, Data Model, NDPA DPIA baseline.
Gate G2	Week 4	First Rendition Live	Public landing page deployed, read-only Regulatory Register active, proven CI/CD pipeline.
Gate G3	Week 10	MVP Alpha Acceptance	Shared Foundation operational, Portal 1 intake/qualification engine active, initial Portal 2/3 alphas.
Gate G4	Week 16	MVP Beta Integrated	Inter-portal API contracts functioning, consented capability transfer verified, Observatory fact engine active.
Gate G5	Week 22	Controlled Field Pilot	UAT passed across live pilot scenarios, security audit clear of High/Critical defects, service desk active.
Gate G6	Week 26	Final Operational Handover	Independent test sign-off, source code & repo transfer, staff competency confirmed, 100% operational

			transition.
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Functional Requirements Document (FRD) / System Requirements Specification (SRS)

MODULE 2: SHARED PLATFORM FOUNDATION SERVICES (CANONICAL COMMON LAYER)

1. Document Control & Metadata

Metadata Attribute	Value / Specification
Document Title	Functional Requirements Document / System Requirements Specification — Module 2: Shared Platform Foundation Services
Parent Programme	Nigeria Service Export Gateway (NATEP / NCMSE)
Executive Sponsor	Federal Ministry of Industry, Trade and Investment (FMITI)
Business Owner	NATEP Secretariat / NCMSE Secretariat
Technical Authority	Architecture, Data, Security & Privacy Authority (ADSPA)
Document Version	Version 1.0 (Engineering Baseline - Plain English Edition)
Status	Draft for Engineering, Technical & Security Validation
Target Audience	Lead Architects, Systems Engineers, Security Engineers, DevOps, Database Administrators, PMO, QA Leads

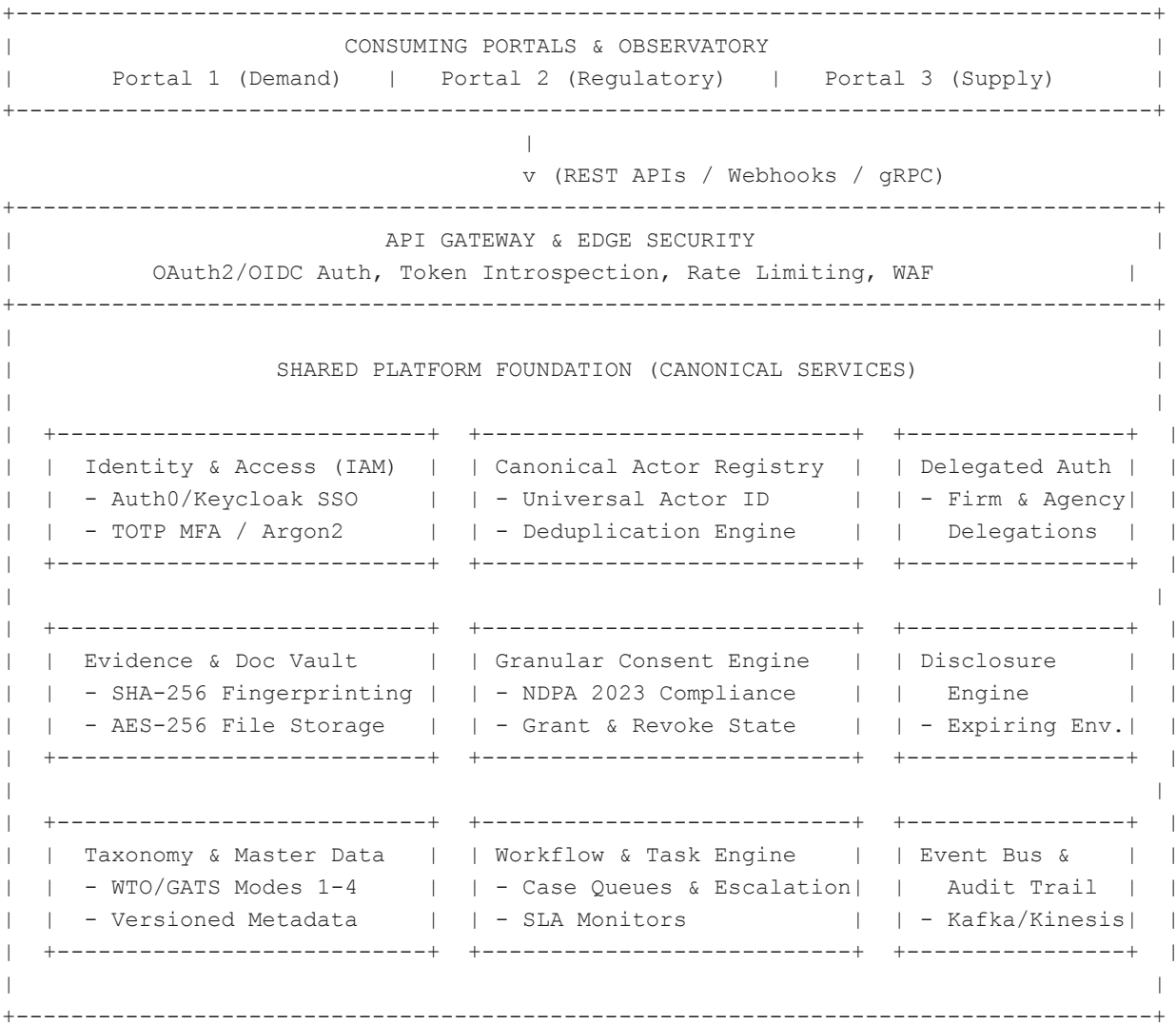
2. System Overview & Architectural Mandate

2.1 Technical Scope & Canonical Foundation Architecture

Module 2 defines the shared software foundation that powers all three business portals (**Portal 1: Demand & Market Access**, **Portal 2: Regulatory Trust & Exporter Readiness**, **Portal 3: Supply Ecosystem & Competitiveness**) and the **National Services Export Observatory**.

In strict accordance with **The Single-Platform Mandate**, this common layer provides reusable, centralized microservices for identity, master data, verification documents, privacy management, task execution, and auditable event logging. Individual portals are forbidden from implementing standalone login systems, user profile master tables, local file repositories, or independent audit logs.

2.2 Subsystem Architecture Blueprint



2.3 System-of-Record (SoR) Shared Services Ownership Matrix

The table below establishes authoritative data ownership for all foundational entities managed within Module 2:

Data Domain / Entity	Authoritative System-of-Record Subsystem	Permitted Operations	Forbidden Operations
Authentication Credentials, Passwords, MFA Tokens	Shared IAM Subsystem	Authenticate users, validate MFA, issue JWTs, handle password resets.	Storing plain-text passwords or creating local user tables in Portal databases.
Universal Actor Profiles (UAID, Person, Firm Data)	Shared Canonical Actor Registry	Provision UAIDs, run deduplication checks, merge profiles, update contacts.	Creating secondary company/person records in Portal 1, 2, or 3 local tables.
Delegated Authority & Organizational Structure	Shared Delegation Subsystem	Assign user permissions per entity, manage corporate units, enforce expiry.	Hardcoding individual user access without organizational delegation context.
Uploaded Documents, Binary Files & SHA-256 Hash Index	Shared Evidence Vault	Ingest files, virus scan, calculate SHA-256 hashes, encrypt files, issue secure URLs.	Saving binary files to local disks or direct S3 storage bypassing the vault API.
User Consent Grants & Revocation Records	Shared Consent Engine	Log consent agreements, validate active grants, execute immediate revocations.	Exposing personal data across portals without querying the active consent state.
Minimally Disclosed Information Envelopes	Shared Disclosure Service	Construct temporary disclosure packages, enforce expiry timers, track	Exporting unmasked profile data directly to third-party portals or public interfaces.

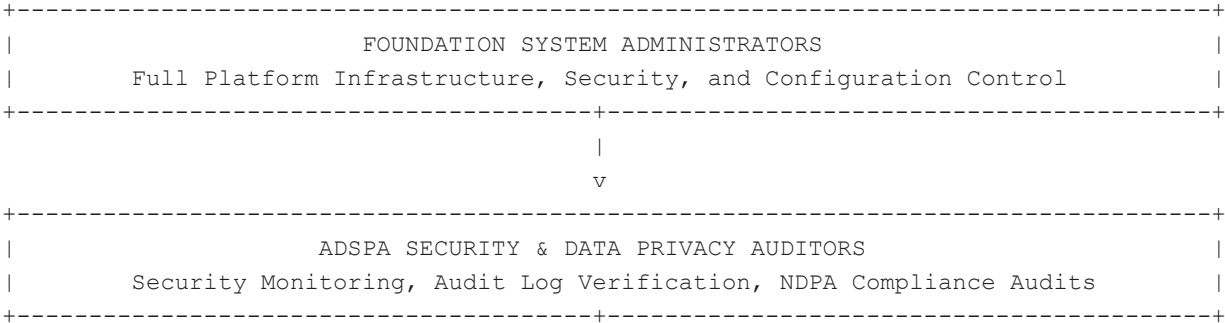
		access.	
Master Taxonomies, Sector Codes & WTO Modes	Shared Taxonomy Subsystem	Maintain versioned codes, map international standards, serve lookups.	Hardcoding string values or creating portal-specific dropdown lists in code.
Cross-Platform Workflows, Case Tasks & SLAs	Shared Workflow Subsystem	Queue tasks, trigger escalations, track completion SLAs, manage assignments.	Maintaining offline or unmonitored task states in local microservices.
Audit Logs, Trace Records & System Events	Shared Audit & Event Subsystem	Cryptographically link log entries, broadcast event streams, preserve WORM storage.	Updating, truncating, or erasing audit logs under any administrative scenario.

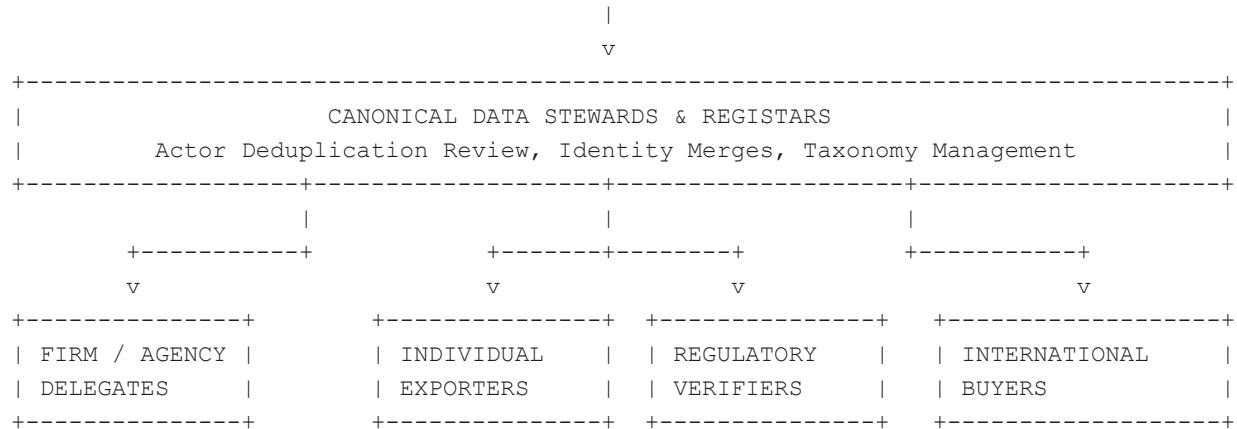
3. Institutional Governance & User Role Architecture

3.1 Role Hierarchy & Access Permission Model

Access control across the Shared Platform Foundation uses Role-Based Access Control (RBAC) combined with Attribute-Based Access Control (ABAC). Permissions are derived dynamically based on:

- 5. **Authenticated Actor Identity:** Verified UAID from the Canonical Actor Registry.
- 6. **Active Organizational Context:** Firm, Agency, or Individual representation.
- 7. **Assigned Role Scope:** ROLE_SYS_ADMIN, ROLE_ADSPA_OFFICER, ROLE_DATA_STEWARD, etc.
- 8. **Data Classification & Purpose:** Public, Confidential, Restricted, or Sensitive Personal Data.





3.2 Shared Foundation Role Permission Summary

Role Code	Role Name	Allowed Foundation Operations	Restricted Actions
ROLE_SYS_ADMIN	System Administrator	Platform infrastructure setup, API Gateway routing, microservice monitoring, key rotation.	Cannot view unmasked personal data or business evidence files.
ROLE_ADSPA_OFFICER	Technical & Security Auditor	Inspect audit trail chains, execute privacy compliance checks, review security alerts.	Cannot modify database records, user permissions, or system configurations.
ROLE_DATA_STEWARD	Canonical Data Steward	Review identity duplicate queues, approve/reject actor merge requests, update taxonomies.	Cannot bypass duplicate validation rules or execute un-audited merges.
ROLE_ENTITY_ADMIN	Corporate / Agency Admin	Manage company delegation rules, add/remove	Cannot access data of un-linked third-party

		authorized staff, upload entity evidence.	companies or individuals.
ROLE_ACTOR_USER	Authenticated Person	Update individual profile, grant/revoke NDPA consents, view own disclosure history.	Cannot access raw system logs or administrative configurations.

3.3 Segregation of Duties & System Safety Guards

To enforce operational security and compliance with the Nigeria Data Protection Act (NDPA) 2023, the foundation layer enforces automated segregation of duties:

Action Initiated	Initiating Role	Mandatory Approval / System Guard	Automated System Enforcement
Actor Profile Merge	Automated Deduplication Engine	Independent verification by a ROLE_DATA_STEWARD	Merge is blocked (Status = Pending Steward Review) if the match score is between 70% and 85% (0.70 to 0.84).
Audit Log Export / Analysis	System Administrator	Co-authorization by an ROLE_ADSPA_OFFICER	Unilateral log extraction is blocked; requires dual-key activation.
Taxonomy Version Supersession	Domain Subject Matter Expert	Formal approval by the Technical Working Group (TWG) Lead	Inactive taxonomy codes are flagged Deprecated before complete retirement.
Bulk Data Disclosure Request	External API Consumer	Active NDPA User Consent Grant + System Rate Limiter Check	Requests lacking a verified ConsentGrantID return HTTP 403 Forbidden.

4. Detailed Functional Requirements Specification

FR-M2-01: Unified Identity & Access Management (IAM) Subsystem

7. **Description:** The system shall provide a centralized Single Sign-On (SSO) and Multi-Factor Authentication (MFA) service supporting OAuth2 and OpenID Connect (OIDC) protocols.
8. **System Inputs:** User credentials (email/username, Argon2/Bcrypt hashed password), Time-based One-Time Password (TOTP) seed, or WebAuthn/Passkey signatures.
9. **Processing Rules:**
 - Authenticate primary credentials using Argon2id hashing (configured with 64 MB memory, 3 iterations, and 4 parallel threads).
 - If MFA is enabled or required by risk profile, prompt for a 6-digit TOTP code or FIDO2/WebAuthn token.
 - Retrieve linked organization contexts and roles from the Canonical Actor Registry.
 - Generate a short-lived JSON Web Token (JWT) access token (validity: 15 minutes) and a secure, HttpOnly refresh token (validity: 24 hours).
 - Expose an endpoint /api/v1/auth/context-switch allowing users to switch operational roles/entities without re-authenticating.
10. **Validation Rules:** Account locked for 30 minutes after 5 consecutive failed login attempts. Mandatory MFA for all administrative and institutional roles.
11. **Output Artifact:** Cryptographically signed JWT Bearer Token containing UAID, active Role, EntityID, and Scope.

FR-M2-02: Canonical Actor Registry & Deduplication Engine

2. **Description:** The system shall maintain one authoritative master registry for all individuals, businesses, government bodies, and foreign buyers, preventing duplicate account creation.
3. **System Inputs:** Registration payload containing National Identification Number (NIN), Corporate Affairs Commission (CAC) Registration Number, Legal Entity Identifier (LEI), or DUNS number.
4. **Processing Rules:**
 - Validate incoming tax/identity numbers against official external adapters (NIMC, CAC, FIRS).
 - Compute a fuzzy string matching score against existing profiles using Jaro-Winkler and Levenshtein distance metrics across Name, Address, Tax ID, and Email.
 - If the similarity score is 85% or higher, block automated creation and assign status DUPLICATE_SUSPECT.
 - If the similarity score is between 70% and 84%, create a record with status PENDING_VERIFICATION and route the task to ROLE_DATA_STEWARD.
 - If the similarity score is below 70% and tax/NIN verification succeeds, generate a permanent Universal Actor ID (UAID-YYYYMMDD-UUIDv4).
5. **Validation Rules:** Uniqueness strictly enforced on encrypted NIN, CAC registration

number, and corporate Tax Identification Number (TIN).

6. **Output Artifact:** Verified Canonical Actor Record and broadcast of event.actor.created.v1.

FR-M2-03: Organization Delegation & Multi-Tenant Authority Subsystem

2. **Description:** The system shall manage organizational hierarchies, branch offices, and time-bound corporate delegations, enabling firm executives to delegate operational rights to employees or consultants.
3. **System Inputs:** Delegation Request (Grantor UAID, Grantee UAID, Organization EntityID, Scope of Authority, Start Date, Expiry Date).
4. **Processing Rules:**
 - Verify Grantor possesses ROLE_ENTITY_ADMIN rights for the specified EntityID.
 - Record delegation entry in the delegation matrix with status ACTIVE.
 - Inject active organizational authority into the Grantee's session JWT upon context switch.
 - Trigger automated background workers daily at 00:00 UTC to evaluate expiring delegations and set status to EXPIRED.
5. **Validation Rules:** Delegations cannot exceed 365 days without re-certification. Revocation takes effect immediately by invalidating active JWT session tokens linked to the delegation.
6. **Output Artifact:** Active Delegation Grant Record and automated alert to Grantee.

FR-M2-04: Controlled Taxonomies & Master Data Subsystem

2. **Description:** The system shall store, version, and expose standardized master codes for services, sectors, WTO/GATS Modes of Supply (Modes 1–4), jurisdictions, and status codes.
3. **System Inputs:** Taxonomy upload or update request (Category, Code, Label, International Mapping e.g., CPC Ver. 2.1, Effective Date).
4. **Processing Rules:**
 - Validate taxonomy entry against international standards (UN CPC, ISIC Rev. 4).
 - Assign an incremental semantic version number (v1.0, v1.1).
 - Maintain backward compatibility by soft-deleting retired codes (Status = Deprecated).
 - Serve taxonomy lookups via high-speed, redis-cached API endpoints (/api/v1/taxonomies/{category}).
5. **Validation Rules:** Hardcoding text strings in portal code repositories is strictly forbidden; all dropdowns must source data from this service.
6. **Output Artifact:** Machine-readable JSON Taxonomy Schema.

FR-M2-05: Configurable Workflow & Task Engine

- **Description:** The system shall coordinate cross-portal business processes, case routing, approval queues, service level agreement (SLA) timers, and automated escalations.
- **System Inputs:** Task Initiation Event, SLA Target Duration (in hours), Assigned Queue/Role, Priority Level.
- **Processing Rules:**
 - Instantiate a new Workflow Case Instance linked to a parent object (e.g., Opportunity

- ID, Verification Request ID).
- Route task to the configured role queue (e.g., QUEUE_DEMAND_OFFICER).
- Track SLA progress using business hour calendars (excluding national public holidays and weekends).
- If SLA threshold reaches 80% without action, trigger warning notification to queue members.
- If SLA threshold reaches 100% (breached), reassign task to Supervisor queue and raise an operational alert.
- **Validation Rules:** Task completion requires explicit submission of mandatory transition parameters and evidence references.
- **Output Artifact:** State machine state transition log and SLA performance metric.

FR-M2-06: Shared Evidence & Zero-Duplicate Vault Engine

7. **Description:** The system shall store, verify, encrypt, and deduplicate all supporting files (incorporation certificates, tax clearances, certifications) uploaded across any portal.
8. **System Inputs:** File binary stream, mime-type, uploader UAID, Document Category.
9. **Processing Rules:**
 - Stream incoming file through an automated anti-malware engine (ClamAV/SOPHOS API). If infected, quarantine file and notify security team.
 - Compute SHA-256 cryptographic hash of the binary stream.
 - Query document master index for matching SHA-256 hash:
 - **If match found:** Increment reference counter, link existing DocumentID to new context (Zero-Duplicate Write).
 - **If match not found:** Encrypt binary stream using AES-256 GCM encryption, store file in Object Storage (S3/Blob), and write metadata to Document Index.
 - Issue a secure, signed URL
(<https://vault.gateway.gov.ng/evidence/{DocumentID}?token=...>) with configurable expiry (default: 15 minutes).
10. **Validation Rules:** Maximum file size: 15 MB. Approved formats: PDF, PNG, JPEG, DOCX. Executable file extension upload is strictly blocked.
11. **Output Artifact:** Encrypted file storage object and an unchangeable Evidence Record.

FR-M2-07: Granular User Consent & Privacy Engine (NDPA 2023)

1. **Description:** The platform shall record explicit, purpose-bound user consent before sharing personal or corporate data across portal boundaries or with external buyers/agencies.
2. **System Inputs:** Consent Request object (Requesting Party, Target Data Subject UAID, Field List, Purpose, Expiry Date).
3. **Processing Rules:**
 1. Display plain-language consent modal to the data subject itemizing requested fields and clear purpose.
 2. Upon user approval, generate a digitally signed Consent Grant record containing timestamp, IP address, exact fields, and purpose.

3. Expose real-time evaluation endpoint (POST /api/v1/consent/evaluate) for API Gateway middleware checks.
4. Upon consent revocation request (POST /api/v1/consent/revoke), set status to REVOKED, instantly invalidate linked Shared Disclosure Packages, and trigger downstream cache purging.
4. **Validation Rules:** Pre-ticked checkboxes or generalized blanket consents are forbidden under NDPA rules.
5. **Output Artifact:** Cryptographically signed Consent Grant Record and event event.consent.granted.v1.

FR-M2-08: Immutable Disclosure Package Subsystem

- **Description:** The system shall create secure, minimum-necessary data envelopes generated under active user consent for specific external interactions (e.g., buyer shortlisting).
- **System Inputs:** Approved ConsentGrantID, Source Data Payload, Recipient Identifier, TTL Expiry.
- **Processing Rules:**
 1. Query active ConsentGrantID to confirm status is ACTIVE and purpose matches.
 2. Filter source payload to extract **only** the explicitly consented fields (field masking applied to non-consented attributes).
 3. Encrypt disclosure payload into an immutable JSON envelope with a unique PackageID and Time-To-Live (TTL) expiration timer.
 4. Log all recipient access attempts against the package in the audit log.
- **Validation Rules:** Package automatically expires (HTTP 410 Gone) upon TTL expiry or if underlying consent is revoked.
- **Output Artifact:** Immutable Shared Disclosure Package object.

FR-M2-09: Universal API Gateway, Event Bus & Tamper-Proof Audit Subsystem

- **Description:** The system shall route all API traffic, publish platform events, and maintain an unchangeable, cryptographically linked log of all system activity.
- **System Inputs:** Incoming API HTTP request / Internal Domain Event object.
- **Processing Rules:**
 1. Intercept HTTP requests at API Gateway; enforce OAuth2 token validation, rate limits, and web application firewall (WAF) inspection.
 2. Publish system state changes to Apache Kafka / AWS Kinesis event streams using standard event envelope format.
 3. For audit log generation, compute SHA-256 digest of current audit log entry combined with the SHA-256 hash of the preceding entry (blockchain-style cryptographic hash chaining).
 4. Write audit records to Write-Once-Read-Many (WORM) compliant storage.
- **Validation Rules:** Direct database modification of audit tables is physically prevented by database privilege controls and WORM storage policies.
- **Output Artifact:** Tamper-evident, cryptographically linked audit chain and real-time event

distribution.

5. Interface & Interoperability Specifications

5.1 Standard Event Envelope Schemas

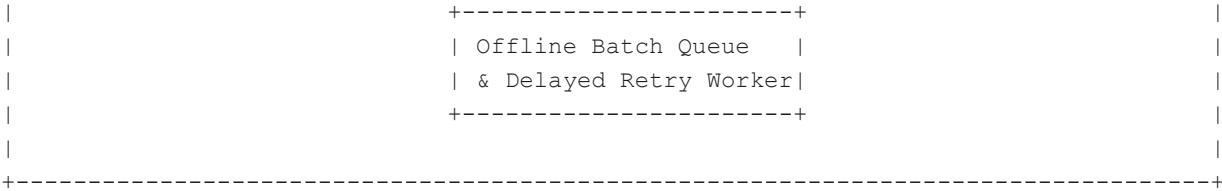
All domain events produced by the Shared Platform Foundation shall adhere to standard JSON Schema contracts.

1. Actor Created Event Schema (event.actor.created.v1.json)

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "ActorCreatedEvent",
  "type": "object",
  "required": ["eventId", "eventType", "timestampUTC", "producer",
"actorUAID", "payload"],
  "properties": {
    "eventId": { "type": "string", "format": "uuid" },
    "eventType": { "type": "string", "const":
"event.actor.created.v1" },
    "timestampUTC": { "type": "string", "format": "date-time" },
    "producer": { "type": "string", "const": "SHARED_ACTOR_REGISTRY"
},
    "actorUAID": { "type": "string", "pattern":
"^UAID-\\d{8}-[0-9a-fA-F-]{36}$" },
    "payload": {
      "type": "object",
      "required": ["actorType", "legalName", "verificationStatus"],
      "properties": {
        "actorType": { "type": "string", "enum": ["INDIVIDUAL",
"FIRM", "AGENCY", "BUYER"] },
        "legalName": { "type": "string" },
        "registrationNumber": { "type": "string" },
        "verificationStatus": { "type": "string", "enum":
["VERIFIED", "PENDING_STEWARD", "PROVISIONAL"] }
      }
    }
  }
}
```

2. Consent Granted Event Schema (event.consent.granted.v1.json)

```
{
```

- 1. **Circuit Breaker Thresholds:** Trigger circuit opening if 5 consecutive requests fail or timeout exceeding 5,000ms.
- 2. **Fallback Mechanism:** Automatically switch to asynchronous batch processing mode when circuit is open, queuing tasks for off-peak retry.
- 3. **Graceful Degradation:** Mark subject verification status as PROVISIONAL_PENDING_ASYNC_CHECK to prevent blocking user onboarding during government database outages.

6. Non-Functional, Security & Performance Requirements

6.1 Privacy & Security Controls (NDPA 2023 & ISO 27001)

- 1. **Data Encryption:**
 - o **In Transit:** TLS 1.3 mandatory for all public and internal service-to-service communications.
 - o **At Rest:** AES-256-GCM encryption applied to database storage volumes, file stores, and backups.
- 2. **Field-Level Encryption:** Sensitive personal fields (NIN, Phone Number, Bank Details) encrypted using application-layer envelope encryption before storage.
- 3. **Data Minimization & Masking:** Default UI responses mask sensitive attributes (e.g., NIN: *****5678). Unmasking requires explicit permission logging.

6.2 Performance Benchmarks & High Availability Targets

Metric Category	Target Value / Standard	Measurement Condition
System Uptime SLA	99.9% Monthly	Applied to Identity (IAM), Consent Engine, and API Gateway services.
Authentication Response Time	< 200 Milliseconds	p95 response time for /api/v1/auth/login under 2,000 concurrent users.
Evidence Hash Lookup Speed	< 30 Milliseconds	p95 database query speed for SHA-256 document index matching.

Event Bus Throughput	5,000 Events / Second	Minimum sustained event processing capacity without queue backpressure.
Recovery Point Objective (RPO)	< 15 Minutes	Maximum allowable data loss during catastrophic cloud region failure.
Recovery Time Objective (RTO)	< 4 Hours	Maximum allowable downtime to restore shared services in disaster recovery.

6.3 Stage-Gate Requirements Traceability Matrix

Req Code	Subsystem / Feature	Target Gate	Mandatory Verification Evidence
FR-M2-01	IAM & Context Switching	Gate G2 (Week 4)	SSO integration test logs, JWT security audit report, MFA verification proof.
FR-M2-02	Actor Registry & Deduplication	Gate G3 (Week 10)	Fuzzy matching algorithm test suite, CAC/NIMC adapter integration logs.
FR-M2-03	Organization Delegation	Gate G3 (Week 10)	Multi-tenant access control UAT sign-off, automated expiration test script.
FR-M2-04	Master Taxonomies	Gate G2 (Week 4)	Machine-readable taxonomy API inspection, WTO/GATS code coverage proof.
FR-M2-05	Workflow & Task	Gate G3 (Week 10)	SLA escalation

	Engine		queue test logs, multi-role task routing demonstration.
FR-M2-06	Zero-Duplicate Evidence Vault	Gate G3 (Week 10)	File deduplication hash match test, malware blocking report, AES-256 audit.
FR-M2-07	Granular Consent Engine	Gate G3 (Week 10)	NDPA privacy audit sign-off, consent revocation real-time blocking proof.
FR-M2-08	Immutable Disclosure Package	Gate G4 (Week 16)	Expiring envelope TTL test, unmasked field restriction UAT verification.
FR-M2-09	API Gateway & Audit Trail	Gate G4 (Week 16)	Cryptographic hash chain validation script output, WORM storage proof.

Module 2 BRD/FRD

NIGERIA SERVICE EXPORT GATEWAY (NATEP / NCMSE)

Business Requirements Document (BRD) — Detailed Functional Specification

MODULE 2: SHARED PLATFORM FOUNDATION SERVICES (CANONICAL COMMON LAYER)

1. Document Control & Baseline Reference

Control Attribute	Specification
Document Title	Business Requirements Document (BRD) — Module 2: Shared Platform Foundation Services
Parent Programme	Nigeria Service Export Gateway (NATEP / NCMSE)
Executive Sponsor	Federal Ministry of Industry, Trade and Investment (FMITI)
Business Owner	NATEP / NCMSE Secretariat
Technical Authority	Architecture, Data, Security & Privacy Authority (ADSPA)
Baseline Source	Integrated Four-Part Terms of Reference (TOR) v1.1 & BRD Module 1 Baseline
Precedence	Applicable Law > Executive Decisions > Integrated TOR > BRD Module 1 > BRD Module 2 > FRD/SRS Specifications

2. Institutional Context & System-of-Record (SoR) Scope

2.1 Strategic Scope & Canonical Mandate

Module 2 defines the shared software foundation that powers all business portals (Portal 1: Demand, Portal 2: Regulatory Trust, Portal 3: Supply) and the National Services Export Observatory. In strict accordance with The Single-Platform Mandate, this module enforces single identity, universal public identifier generation, zero-duplicate document storage, granular user consent, versioned taxonomies, and immutable audit logging across the entire Gateway ecosystem.

Individual consuming portals are strictly forbidden from maintaining independent user login tables, duplicate actor profiles, local evidence file stores, un-audited state machines, or

un-consented cross-portal data exchanges.

+-----+ SHARED PLATFORM FOUNDATION +-----+	
[IAM] Single Sign-On (Argon2id / TOTP / OIDC / OAuth2)	
[Actor Registry] Universal Actor ID (UAID), Public NATEP ID & Deduplication	
[Delegation] Multi-Tenant Corporate & Agency Authority	
[Evidence Vault] SHA-256 Fingerprinting & AES-256 Encrypted File Vault	
[Consent Engine] NDPA 2023 Compliant Purpose-Bound Grants	
[Disclosure Engine] Minimum-Necessary Expiring Data Envelopes	
[Taxonomies] WTO/GATS Modes 1-4 & Versioned Master Data	
[Workflow Engine] Cross-Portal SLA Calendars & Case Escalations	
[Audit & Event Bus] Cryptographic Hash-Chained Audit Trail (WORM Storage)	
+-----+	

2.2 Authoritative System-of-Record (SoR) Data Ownership Matrix

Data Domain / Entity	Authoritative System-of-Record Subsystem	Permitted Operations	Forbidden Operations
Authentication Credentials & Tokens	Shared IAM Subsystem	Authenticate users, issue JWTs, handle password resets, enforce MFA.	Storing plain-text passwords or creating local user tables in Portal databases.
Universal Actor Profiles & NATEP IDs	Shared Canonical Actor Registry	Provision UAIDs and public NATEP IDs, run deduplication, execute approved merges, update contacts.	Creating secondary company/person master records in Portal 1, 2, or 3 local tables.
Delegated Authority & Corporate	Shared Delegation Subsystem	Assign permissions per entity, manage corporate units,	Hardcoding individual user access without

Structures		enforce expiry.	organizational delegation context.
Uploaded Verification Documents & Hashes	Shared Evidence Vault	Ingest files, virus scan, calculate SHA-256 hashes, encrypt files, issue signed URLs.	Saving binary files to local disks or direct cloud storage bypassing the vault API.
User Consent Grants & Revocations	Shared Consent Engine	Log consent agreements, validate active grants, execute immediate revocations.	Exposing personal data across portals without querying active consent state.
Minimally Disclosed Data Envelopes	Shared Disclosure Service	Construct temporary disclosure packages, enforce expiry timers, track access.	Exporting unmasked profile data directly to third-party portals or public web.
Master Taxonomies & Sector Codes	Shared Taxonomy Subsystem	Maintain versioned codes, map international standards, serve lookups.	Hardcoding string values or creating portal-specific dropdown lists in code.
Cross-Platform Workflows & Tasks	Shared Workflow Subsystem	Queue tasks, trigger escalations, track completion SLAs, manage assignments.	Maintaining offline or unmonitored task states in local microservices.
Audit Logs & System Activity Records	Shared Audit & Event Subsystem	Cryptographically link log entries, broadcast event streams, preserve WORM storage.	Updating, truncating, or erasing audit logs under any administrative scenario.

3. Subsystem Business Requirements & Functional Architecture

3.1 Unified Identity & Access Management (IAM) Subsystem

3.1.1 Business Rationale & Objectives

Provides a secure, centralized Single Sign-On (SSO) and Multi-Factor Authentication (MFA) framework compliant with OAuth2 and OpenID Connect (OIDC). Guarantees that every user accesses the platform through a single credential set while dynamically assuming authorized workspace roles.

3.1.2 Detailed Functional Business Requirements

- **BR-IAM-01 (Single Authentication Engine):** The system shall authenticate all portal users against a centralized OIDC identity provider. Separate authentication services per portal are explicitly prohibited.
- **BR-IAM-02 (Credential Hardening):** Passwords must be hashed using Argon2id with exact memory limits of 64 Megabytes, 3 computational passes, and 4 parallel execution threads. Standard single-factor password logins without step-up checks shall be restricted.
- **BR-IAM-03 (Mandatory Administrative MFA):** Mandatory Time-based One-Time Password (TOTP) or WebAuthn/FIDO2 token authentication shall be enforced for all administrative, verifier, and institutional roles (ROLE_SYS_ADMIN, ROLE_ADSPA_OFFICER, ROLE_NATEP_VERIFIER, ROLE_REG_VALIDATOR).
- **BR-IAM-04 (Context Switching):** Authenticated users associated with multiple entities (e.g., individual freelancer and corporate director) must be able to switch active operational context (POST /api/v1/auth/context-switch) without re-entering credentials.
- **BR-IAM-05 (Session & Lockout Policies):** Access tokens (JWT) shall expire in 15 minutes; refresh tokens shall expire in 24 hours with rolling renewal. Accounts shall automatically lock for 30 minutes following 5 consecutive failed login attempts.

3.2 Canonical Actor Registry & Identity Verification Subsystem

3.2.1 Business Rationale & Objectives

Establishes one single source of truth for all individuals, firms, government bodies, and international buyers participating in the Gateway ecosystem, eliminating fragmented actor listings and fraudulent duplicate accounts.

3.2.2 Detailed Functional Business Requirements

- **BR-REG-01 (Universal Identity Provisioning & NATEP ID Generation):** Every registered entity shall be assigned an immutable internal Universal Actor Identifier (UAID-YYYYMMDD-UUIDv4). Simultaneously, the system shall issue a public-facing NATEP ID adhering to standardized syntax:
 - **Option A Standard Format:** NT-I26-8942A (where NT represents the NATEP prefix, I indicates Individual Talent or B for Business/SME, 26 represents the 2-digit registration year, and 8942A is a 5-character Base36 unique alphanumeric hash).
 - **Option B Compact Format:** N-I26-8942A (offering an identical structural syntax with a single-character prefix).
 - **NDPA Data Minimization Rule:** Physical geographic location (e.g., State, LGA, City) shall remain stored exclusively as database metadata and shall never be embedded into the public NATEP ID string to prevent identifier mutation if an actor relocates.
- **BR-REG-02 (External Government Verification Adapters):** Incoming actor

registrations must validate against official external databases via secure API adapters:

- **Individuals:** National Identification Number (NIN) verified via NIMC.
- **Nigerian Businesses:** Registration Number verified via CAC; Tax Identification Number verified via FIRS.
- **Foreign Buyers:** Verified via Legal Entity Identifier (LEI) or DUNS registration databases.
- **BR-REG-03 (Automated Deduplication Engine):** Incoming registrations shall be evaluated against existing registry records using deterministic matching and fuzzy string distance metrics (Jaro-Winkler and Levenshtein):
 - **Match Score 85% or higher:** Flagged as DUPLICATE_SUSPECT; automated registration blocked; queued for ROLE_DATA_STEWARD manual review.
 - **Match Score between 70% and 84%:** Created with status PENDING_VERIFICATION; routed to steward review queue.
 - **Match Score below 70% + Valid Gov ID:** Automatically assigned VERIFIED status and issued a UAID and NATEP ID.
- **BR-REG-04 (Actor Lifecycle Management):** Maintain strict state transitions: PROVISIONAL moves to PENDING_VERIFICATION moves to VERIFIED moves to SUSPENDED moves to ARCHIVED.

3.3 Organization Delegation & Multi-Tenant Context Subsystem

3.3.1 Business Rationale & Objectives

Enables corporate executives, agency heads, and business owners to delegate precise operational authority to employees, consultants, or branch officers without sharing login credentials or compromising account security.

3.3.2 Detailed Functional Business Requirements

- **BR-DEL-01 (Delegation Granting):** Corporate admins (ROLE_ENTITY_ADMIN) shall be permitted to issue time-bound delegations defining exact scope, start date, and expiry date.
- **BR-DEL-02 (Delegation Limits & Expiry):** Delegations shall not exceed 365 days without mandatory re-certification. Automated background jobs running at 00:00 UTC shall set expired records to EXPIRED.
- **BR-DEL-03 (Instant Revocation & Token Invalidation):** Immediate revocation of a delegation by an entity administrator must instantaneously invalidate active JWT session tokens linked to that delegated authority.

3.4 Shared Evidence & Zero-Duplicate Document Vault Subsystem

3.4.1 Business Rationale & Objectives

Provides a unified, secure repository for all supporting compliance files (incorporation documents, tax certificates, skill credentials). Enforces zero-duplicate storage to optimize cloud resources and guarantee file integrity.

3.4.2 Detailed Functional Business Requirements

- **BR-VLT-01 (Antivirus Screening):** All incoming binary file streams must pass real-time anti-malware scanning (ClamAV/Sophos) prior to vault indexing. Infected files shall be quarantined and trigger a security alert.
- **BR-VLT-02 (Cryptographic Fingerprinting & Deduplication):**
 - Calculate SHA-256 cryptographic hash of every uploaded file.

- **Match Found:** Increment master reference counter and link existing DocumentID to the new business context without saving duplicate binary data (Zero-Duplicate Write).
- **No Match Found:** Encrypt binary stream using AES-256-GCM, save to Object Store, and write metadata record to Document Master Index.
- **BR-VLT-03 (Time-Bound Expiring Download URLs):** Direct access to underlying object store buckets is prohibited. File access must be served via signed, short-lived temporary URLs
(https://vault.gateway.gov.ng/evidence/{DocumentID}?token=...) expiring after 15 minutes.
- **BR-VLT-04 (Format & Size Restrictions):** Maximum allowable file size: 15 MB. Allowed MIME types: PDF, PNG, JPEG, DOCX. Executable binary extensions (.exe, .sh, .bat, .dll) are strictly blocked.

3.5 Granular User Consent & Privacy Subsystem (NDPA 2023)

3.5.1 Business Rationale & Objectives

Enforces strict compliance with the Nigeria Data Protection Act (NDPA) 2023 by ensuring no personal or proprietary commercial data moves across portal boundaries or to external buyers/partners without explicit, purpose-bound user consent.

3.5.2 Detailed Functional Business Requirements

- **BR-CON-01 (Explicit Purpose-Bound Grants):** Consent requests must itemize the requesting entity, target recipient, exact data fields requested, intended purpose, and expiration date. Pre-ticked boxes or blanket consent clauses are legally void.
- **BR-CON-02 (Digital Consent Audit Trail):** Upon user approval, generate an unchangeable Consent Grant record capturing timestamp, IP address, user UAID, exact field array, and digital signature.
- **BR-CON-03 (Real-Time Consent Middleware Check):** All cross-portal data requests intercepting the API Gateway must validate active consent status via POST /api/v1/consent/evaluate. If consent is absent or expired, the gateway must return HTTP 403 Forbidden.
- **BR-CON-04 (Instant Revocation Enforcement):** Executing POST /api/v1/consent/revoke must instantly set consent status to REVOKED, terminate downstream access, and invalidate active Shared Disclosure Packages.

3.6 Immutable Disclosure Package Subsystem

3.6.1 Business Rationale & Objectives

Creates secure, field-masked data envelopes generated under valid user consent for external interactions (e.g., shortlisting candidate firms for international buyers), minimizing data exposure risks.

3.6.2 Detailed Functional Business Requirements

- **BR-DIS-01 (Minimum-Necessary Envelope Creation):** Disclosure envelopes shall extract only the explicitly consented data fields from source profiles. All non-consented attributes shall be masked at the API serialization layer.
- **BR-DIS-02 (TTL Expiration & Self-Destruction):** Every disclosure package shall be assigned an unmodifiable Time-To-Live (TTL) timer. Upon expiry, the package endpoint shall return HTTP 410 Gone.

- **BR-DIS-03 (Recipient Access Logging):** Every access attempt to a disclosure package shall log recipient identity, IP address, timestamp, and accessed package ID to the immutable audit trail.

3.7 Controlled Taxonomies & Master Data Subsystem

3.7.1 Business Rationale & Objectives

Maintains centralized, versioned code tables for services, sectors, occupations, WTO/GATS modes of supply (Modes 1–4), jurisdictions, and status codes to ensure consistency across all platform modules.

3.7.2 Detailed Functional Business Requirements

- **BR-TAX-01 (International Mapping Alignment):** Taxonomies must map natively to international standards, including UN Central Product Classification (UN CPC Ver. 2.1), ISIC Rev. 4, and WTO GATS Modes 1–4.
- **BR-TAX-02 (Semantic Versioning & Deprecation):** Taxonomy updates shall use semantic versioning (v1.0, v1.1). Retired codes shall be marked Deprecated rather than hard-deleted to preserve historical record integrity.
- **BR-TAX-03 (Forbidden Hardcoding Mandate):** Hardcoding text options or local dropdown arrays in frontend/backend codebase is strictly forbidden. All UI selectors must consume live, Redis-cached REST endpoints (/api/v1/taxonomies/{category}).

3.8 Configurable Workflow & Task Engine Subsystem

3.8.1 Business Rationale & Objectives

Coordinates cross-portal business processes, operational case queues, approval tasks, Service Level Agreement (SLA) timers, and automated escalations.

3.8.2 Detailed Functional Business Requirements

- **BR-WFK-01 (Case Lifecycle & Task Queuing):** Workflow instances must manage state transitions across multi-portal lifecycles, assigning discrete tasks to role-based queues (e.g., QUEUE_DEMAND_QUALIFICATION, QUEUE_REGULATORY_VALIDATION).
- **BR-WFK-02 (SLA Business Calendar Tracking):** SLA timers must evaluate task duration against configurable business-hour calendars (excluding official national public holidays and weekends).
- **BR-WFK-03 (Automated SLA Warnings & Escalations):**
 - **80% SLA Threshold:** Generate operational warning notification to assigned queue members.
 - **100% SLA Breach:** Automatically reassign task to Supervisor queue, escalate priority flag, and log SLA breach event to the Observatory.

3.9 Event Bus, Universal Gateway & Tamper-Evident Audit Subsystem

3.9.1 Business Rationale & Objectives

Provides high-performance API routing, edge security, real-time asynchronous event streaming, and an unchangeable, cryptographically verifiable audit trail.

3.9.2 Detailed Functional Business Requirements

- **BR-AUD-01 (Edge API Gateway Security):** All inbound traffic must pass through the API Gateway for OAuth2 token introspection, rate-limiting, and Web Application Firewall (WAF) threat inspection.

- **BR-AUD-02 (Asynchronous Domain Event Streaming):** State changes across shared foundation services must broadcast standard JSON schema event envelopes to Apache Kafka / AWS Kinesis (event.actor.created.v1, event.consent.granted.v1).
- **BR-AUD-03 (Cryptographic Hash-Chained Audit Logs):**
 - Audit records must capture actor UAID and NATEP ID, timestamp (UTC), action type, target entity ID, pre-state hash, and post-state hash.
 - Compute SHA-256 digest of current entry combined with the SHA-256 hash of the preceding log entry (blockchain-style cryptographic hash chaining).
- **BR-AUD-04 (WORM Storage Policy Enforcement):** Write audit records to Write-Once-Read-Many (WORM) compliant object storage. Database privilege controls shall physically prevent UPDATE, DELETE, or TRUNCATE operations on audit tables.

4. Non-Functional, Privacy & Security Requirements

4.1 Security & Privacy Controls (NDPA 2023 & ISO 27001)

- **Encryption Standards:**
 - **In Transit:** TLS 1.3 mandatory for all public and internal service-to-service communications.
 - **At Rest:** AES-256-GCM encryption applied to all database volumes, vault file stores, and backup archives.
- **Application-Layer Envelope Encryption:** Direct field-level encryption applied to highly sensitive personal data attributes (NIN, Phone Number, Bank Details) prior to database persistence.
- **Default Data Masking:** User interfaces shall default to masked representation of sensitive fields (e.g., NIN: *****5678). Unmasking requires explicit permission logging.

4.2 Performance Benchmarks & Service Level Agreements (SLAs)

Performance Indicator	Target Metric	Measurement Condition
System Availability SLA	99.9% Monthly Uptime	Applied to Identity (IAM), Consent Engine, and API Gateway services.
Authentication Latency	< 200 Milliseconds	p95 response time for /api/v1/auth/login under 2,000 concurrent users.
Vault Hash Lookup Speed	< 30 Milliseconds	p95 database query speed for SHA-256 document index matching.
Event Bus Throughput	5,000 Events / Second	Minimum sustained event processing capacity

		without queue backpressure.
Recovery Point Objective (RPO)	< 15 Minutes	Maximum allowable data loss during catastrophic cloud region failure.
Recovery Time Objective (RTO)	< 4 Hours	Maximum allowable downtime to restore shared services in disaster recovery.

5. Stage-Gate Execution Traceability & Verification Matrix

Every business requirement in Module 2 maps directly to delivery verification gates across the 26-week schedule:

Requirement Code	Functional Feature / Subsystem	Target Stage Gate	Mandatory Verification Proof
BR-IAM-01 – 05	Unified IAM & Context Switching	Gate G2 (Week 4)	SSO integration test logs, JWT security audit report, MFA verification proof.
BR-TAX-01 – 03	Master Taxonomies & Vocabularies	Gate G2 (Week 4)	Machine-readable taxonomy API inspection, WTO/GATS code coverage proof.
BR-REG-01 – 04	Actor Registry, NATEP ID & Deduplication	Gate G3 (Week 10)	NATEP ID syntax generator logs, fuzzy matching algorithm test suite, CAC/NIMC adapter integration logs.
BR-DEL-01 – 03	Organization Delegation Matrix	Gate G3 (Week 10)	Multi-tenant access control UAT

			sign-off, automated expiration test script.
BR-VLT-01 – 04	Zero-Duplicate Evidence Vault	Gate G3 (Week 10)	File deduplication hash match test, malware blocking report, AES-256 audit.
BR-CON-01 – 04	Granular Consent Engine (NDPA)	Gate G3 (Week 10)	NDPA privacy audit sign-off, consent revocation real-time blocking proof.
BR-WFK-01 – 03	Workflow & Case Task Engine	Gate G3 (Week 10)	SLA escalation queue test logs, multi-role task routing demonstration.
BR-DIS-01 – 03	Immutable Disclosure Package	Gate G4 (Week 16)	Expiring envelope TTL test, unmasked field restriction UAT verification.
BR-AUD-01 – 04	API Gateway & Tamper-Evident Audit	Gate G4 (Week 16)	Cryptographic hash chain validation script output, WORM storage proof.

Functional Requirements Document (FRD) / System Requirements Specification (SRS)

MODULE 2: SHARED PLATFORM FOUNDATION SERVICES (CANONICAL COMMON LAYER)

Document Control & Baseline Alignment

- **Document Title:** Functional Requirements Document (FRD) / System Requirements Specification (SRS) — Module 2: Shared Platform Foundation Services
- **Parent Programme:** Nigeria Service Export Gateway (NATEP / NCMSE)
- **Baseline Alignment:** Integrated Four-Part TOR v1.1, BRD Module 1 Baseline, and BRD Module 2 Baseline

- **Formatting Policy:** Plain English specification standard. Mathematical expressions, LaTeX syntax, and formal Markdown headings inside technical requirements are strictly omitted.

System-of-Record (SoR) Architectural Matrix

Data Domain / Subsystem	Primary SoR Owner	Downstream Consuming Modules	Permitted Operations	Invariant Constraints
User Identity & Auth Secrets	Shared IAM Subsystem	Portals 1, 2, 3 & Observatory	User authentication, token issuance, password resets, MFA step-up verification.	Zero local password stores or auth tables in portal databases.
Actor Profiles & NATEP IDs	Shared Canonical Actor Registry	Portals 1, 2, 3 & Observatory	Actor provisioning, public NATEP ID generation, fuzzy deduplication, profile updates.	Zero duplicate company/pers on master records across portals.
Corporate Delegations	Shared Delegation Subsystem	Portals 1, 2, 3 & Observatory	Time-bound permission grants, organizational scope assignments, token invalidations.	Mandatory 365-day re-certification boundary.
Evidence & File Vault	Shared Evidence Vault	Portals 1, 2, 3 & Observatory	Malware screening, SHA-256 fingerprinting, AES-256-GCM encryption, signed URL	Zero duplicate binary file writes to cloud storage.

			issuance.	
User Consent Grants	Shared Consent Engine	Portals 1, 2, 3 & Observatory	Purpose-bound consent creation, status evaluation, instant revocation handling.	Zero cross-portal personal data exposure without active grant.
Disclosed Data Packages	Shared Disclosure Service	Portals 1, 2, 3 & External Buyers	Envelope construction, field masking, TTL expiration enforcement.	Hard self-destruction upon TTL expiry (HTTP 410 Gone).
Master Taxonomies	Shared Taxonomy Subsystem	Portals 1, 2, 3 & Observatory	Versioned lookup delivery, WTO/GATS/CP C standard mappings.	Zero hardcoded string values or local dropdown arrays in frontend code.
Workflows & Tasks	Shared Workflow Subsystem	Portals 1, 2, 3 & Observatory	Task queuing, business-calendar SLA tracking, escalation routing.	Automated supervisor escalation upon 100% SLA breach.
Audit Chaining & Event Bus	Shared Audit & Event Subsystem	Portals 1, 2, 3 & Observatory	Event broadcasting, cryptographic hash-chained logging, WORM persistence.	Zero update, delete, or truncate operations on audit logs.

Detailed Subsystem Functional Specifications

1. Unified Identity & Access Management (IAM) Subsystem

Requirement Reference: FR-M2-01 (BRD Reference: BR-IAM-01 through BR-IAM-05)

- **System Processing Logic & Session Management:**

- The authentication engine processes login attempts through a single OpenID Connect (OIDC) identity provider, prohibiting separate local user credential stores.
- Password evaluation utilizes the Argon2id hashing algorithm configured with 64 Megabytes of RAM, 3 computational passes, and 4 parallel execution threads.
- Access tokens (JWT) expire 15 minutes after issuance. Refresh tokens expire in 24 hours with rolling renewal.
- Account lockout mechanisms automatically trigger a 30-minute lock upon 5 consecutive failed login attempts.
- Users holding multiple roles across entities switch operational context via POST /api/v1/auth/context-switch, which invalidates existing session JWT tokens and issues updated tokens scoped to the target operational role without requiring credential re-entry.
- Multi-Factor Authentication (MFA) via TOTP or WebAuthn/FIDO2 tokens is mandatory for all administrative and operational roles (ROLE_SYS_ADMIN, ROLE_ADSPA_OFFICER, ROLE_NATEP_VERIFIER, ROLE_REG_VALIDATOR).

2. Canonical Actor Registry & Identity Verification Subsystem

Requirement Reference: FR-M2-02 (BRD Reference: BR-REG-01 through BR-REG-04)

- **Identifier Generation & State Management:**

- Every actor is assigned an immutable internal Universal Actor ID (UAID-YYYYMMDD-UUIDv4).
- The system generates a public-facing NATEP ID adhering to the standard syntax NT-I26-8942A (where NT represents the NATEP prefix, I indicates Individual Talent or B for Business/SME, 26 represents the issuance year, and 8942A is a 5-character Base36 unique alphanumeric hash) or compact syntax N-I26-8942A. Geographic metadata is stored in backend fields and never embedded into the public ID string.
- Actor records transition strictly through explicit state boundaries: PROVISIONAL moves to PENDING_VERIFICATION moves to VERIFIED moves to SUSPENDED moves to ARCHIVED.

- **External Verification Adapters & Deduplication Logic:**

- Individual registrations validate National Identification Numbers (NIN) against NIMC APIs. Nigerian business registrations validate Registration Numbers with CAC and Tax Identification Numbers with FIRS. Foreign buyers validate against Legal Entity Identifier (LEI) or DUNS databases.
- Incoming registrations evaluate against existing records using Jaro-Winkler and Levenshtein fuzzy string distance algorithms:
 - Match Score 85% or higher: Blocked automatically, assigned status DUPLICATE_SUSPECT, and routed to the Data Steward review queue.

- Match Score between 70% and 84%: Assigned status PENDING_VERIFICATION and queued for manual steward evaluation.
- Match Score below 70% plus Valid Government ID: Issued a UAID, assigned a public NATEP ID, and marked as VERIFIED.

3. Organization Delegation & Multi-Tenant Context Subsystem

Requirement Reference: FR-M2-03 (BRD Reference: BR-DEL-01 through BR-DEL-03)

- **Delegation Governance & Lifetime Processing:**
 - Enterprise Administrators (ROLE_ENTITY_ADMIN) assign time-bound delegation scope records to employees, contractors, or branch officers.
 - Maximum delegation duration is capped at 365 days. Automated background jobs executing daily at 00:00 UTC flag un-recertified records as EXPIRED.
 - Executing a delegation revocation instantly marks the record as REVOKED and invalidates all active session tokens bound to that delegated context.

4. Shared Evidence & Zero-Duplicate Document Vault Subsystem

Requirement Reference: FR-M2-04 (BRD Reference: BR-VLT-01 through BR-VLT-04)

- **File Screening, Deduplication, & Access Specifications:**
 - Ingested binary file streams undergo real-time antivirus scanning (ClamAV/Sophos) prior to vault indexing. Infected files are quarantined instantly.
 - The system computes a SHA-256 cryptographic digest for every file:
 - If a SHA-256 match exists in the Document Master Index, the system increments the master reference counter and links the existing DocumentID to the new business context without persisting duplicate binary data.
 - If no match exists, the stream is encrypted using AES-256-GCM, saved to the object store, and registered in the Document Master Index.
 - Direct object store public access is blocked; document requests return signed download URLs
 (https://vault.gateway.gov.ng/evidence/){DocumentID}?token=...) that expire after 15 minutes.
 - File uploads are restricted to a maximum size of 15 MB and allowed MIME types (PDF, PNG, JPEG, DOCX). Executable extensions (.exe, .sh, .bat, .dll) are rejected.

5. Granular User Consent & Privacy Subsystem (NDPA 2023)

Requirement Reference: FR-M2-05 (BRD Reference: BR-CON-01 through BR-CON-04)

- **Consent Life Cycle & Enforcement Rules:**
 - Purpose-bound consent grants require itemized definitions including requesting entity, target recipient, exact data fields, purpose, and expiration date. Pre-ticked defaults are legally invalid.
 - Approved grants generate digital records containing a timestamp, IP address, user UAID, field array, and digital signature.
 - API Gateway middleware validates active consent for every cross-portal data fetch via POST /api/v1/consent/evaluate. Missing or expired grants return HTTP 403 Forbidden.
 - Executing POST /api/v1/consent/revoke sets consent status to REVOKED instantly, cutting off downstream data access and invalidating active Shared Disclosure Packages.

6. Immutable Disclosure Package Subsystem

Requirement Reference: FR-M2-06 (BRD Reference: BR-DIS-01 through BR-DIS-03)

- **Disclosure Envelope Mechanics:**

- Packages extract only explicitly consented attributes from actor profiles. Non-consented fields are masked at the API serialization layer.
- Every disclosure package carries an unmodifiable Time-To-Live (TTL) timer. Upon expiry, endpoints return HTTP 410 Gone.
- Access attempts log recipient identity, IP address, UTC timestamp, and Package ID to the immutable audit trail.

7. Controlled Taxonomies & Master Data Subsystem

Requirement Reference: FR-M2-07 (BRD Reference: BR-TAX-01 through BR-TAX-03)

- **Taxonomy Management & Distribution:**

- System code tables map directly to UN CPC Ver. 2.1, ISIC Rev. 4, and WTO/GATS Modes 1–4.
- Taxonomy updates follow semantic versioning conventions (e.g., v1.0, v1.1). Retired codes are flagged as Deprecated to preserve historical data references.
- Hardcoded selection options in frontend or backend codebases are prohibited; options are consumed dynamically via Redis-cached endpoints (/api/v1/taxonomies/{category}).

8. Configurable Workflow & Task Engine Subsystem

Requirement Reference: FR-M2-08 (BRD Reference: BR-WFK-01 through BR-WFK-03)

- **Task Routing & Escalation Logic:**

- Workflow instances assign task cases to role-based operational queues (such as QUEUE_DEMAND_QUALIFICATION or QUEUE_REGULATORY_VALIDATION).
- SLA evaluation calculates active durations using national business calendars, excluding official weekends and public holidays.
- When elapsed time reaches 80% of the SLA threshold, automated warning notifications dispatch to queue members.
- Reaching a 100% SLA breach automatically reassigns the case to the Supervisor queue, flags priority escalation, and logs a breach event.

9. Event Bus, Universal Gateway & Tamper-Evident Audit Subsystem

Requirement Reference: FR-M2-09 (BRD Reference: BR-AUD-01 through BR-AUD-04)

- **Gateway Routing, Event Streaming, & Hash-Chained Audit Logs:**

- Inbound requests pass through the Edge API Gateway for OAuth2 token introspection, rate limiting, and Web Application Firewall (WAF) screening.
- State changes emit standardized JSON event envelopes to Apache Kafka / AWS Kinesis (event.actor.created.v1, event.consent.granted.v1).
- Audit records capture actor UAID and NATEP ID, UTC timestamp, action type, target entity ID, pre-state hash, and post-state hash.
- Entry digests are computed using SHA-256 combined with the SHA-256 hash of the immediately preceding log entry.
- Audit entries persist to Write-Once-Read-Many (WORM) storage. Database rules physically block UPDATE, DELETE, or TRUNCATE commands.

Standard JSON Domain Event Envelope Samples

1. Actor Created Event Envelope (event.actor.created.v1)

```
{
  "eventId": "evt_98f4c2a1-3b7e-41d2-a0b8-6e5f4c3b2a10",
  "eventType": "event.actor.created.v1",
  "eventTimestamp": "2026-09-08T14:30:00.000Z",
  "producer": "natep.service.actor-registry",
  "actor": {
    "uaid": "UAID-20260908-7f8e9d0c-1b2a-4f3e-8d9c-0b1a2f3e4d5c",
    "natepId": "NT-B26-94821",
    "entityType": "BUSINESS_ENTERPRISE",
    "verificationStatus": "VERIFIED"
  },
  "data": {
    "registrationNumber": "RC-1728394",
    "taxIdentificationNumber": "TIN-98765432-0001",
    "legalName": "Apex Software Global Solutions Limited",
    "verificationSource": "CAC_FIRS_ADAPTER"
  },
  "context": {
    "correlationId": "corr_1a2b3c4d-5e6f-7a8b-9c0d-1e2f3a4b5c6d",
    "originatingIp": "102.89.23.14",
    "userAgent": "NATEP-Gateway-Client/2.1"
  }
}
```

2. Consent Granted Event Envelope (event.consent.granted.v1)

```
{
  "eventId": "evt_4a3b2c1d-0e9f-8a7b-6c5d-4e3f2a1b0c9d",
  "eventType": "event.consent.granted.v1",
  "eventTimestamp": "2026-09-08T14:35:12.000Z",
  "producer": "natep.service.consent-engine",
  "actor": {
    "uaid": "UAID-20260908-1a2b3c4d-5e6f-7a8b-9c0d-1e2f3a4b5c6d",
    "natepId": "NT-I26-38291",
    "entityType": "INDIVIDUAL_TALENT"
  },
  "data": {
    "consentGrantId": "cg_88192039-4b2c-4f1e-9a0b-123456789abc",
    "requestingEntityId":

```



```
"UAID-20260908-9f8e7d6c-5b4a-3f2e-1d0c-9b8a7f6e5d4c",
  "requestingEntityNatepId": "NT-B26-11029",
  "consentedFields": [
    "person.full_name",
    "person.email_address",
    "certifications.verified_array",
    "experience.summary_years"
  ],
  "purpose": "GLOBAL_TALENT_RECRUITMENT_SHORTLIST",
  "expiresAt": "2026-10-08T23:59:59.000Z"
},
"context": {
  "correlationId": "corr_9f8e7d6c-5b4a-3f2e-1d0c-9b8a7f6e5d4c",
  "originatingIp": "105.112.45.88",
  "userAgent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64)"
}
```

Non-Functional & Security Specifications

+-----+ SECURITY & HARDENING ARCHITECTURE +-----+	
[TLS 1.3] Standard Transport Security for In-Transit Communications	
[AES-256-GCM] At-Rest Encryption across Databases, Vault, and Backups	
[Envelope Encryption] Field-Level Cryptographic Protection for Sensitive Identifiers	
[WORM Persistence] Write-Once-Read-Many Cryptographic Audit Hash Chaining	
+-----+	

1. Cryptographic Standard Baselines

- **Data In Transit:** TLS 1.3 is mandatory across all external public ingress endpoints and internal inter-service communication channels.
- **Data At Rest:** AES-256-GCM encryption must be applied across all database storage volumes, evidence vault object buckets, and backup archives.
- **Field-Level Envelope Encryption:** Sensitive identifiers including National Identification Numbers (NIN), phone numbers, and banking details must undergo application-layer envelope encryption before persistence.
- **Default Data Masking:** User interfaces must render sensitive fields in masked representations by default (e.g., displaying NIN as *****5678), requiring explicit verifier permission logging to unmask.

2. Technical Performance Targets

Operational Performance Metric	Target Threshold	Target Evaluation Context
System Uptime Availability	99.9% Availability	Evaluated monthly across IAM, Consent, and API Gateway services.
Authentication Response Time	Under 200 Milliseconds	Evaluated at p95 response time under a load of 2,000 concurrent users.
Vault Hash Index Query Time	Under 30 Milliseconds	Evaluated at p95 query time for SHA-256 document index lookups.
Event Bus Streaming Throughput	5,000 Events per Second	Minimum sustained event processing capacity without backlog.
Recovery Point Objective (RPO)	Under 15 Minutes	Maximum allowable data loss window during disaster recovery events.
Recovery Time Objective (RTO)	Under 4 Hours	Maximum allowable service restoration window during system failures.

Stage-Gate Delivery & Verification Traceability

Requirement ID	Subsystem Component	Target Delivery Gate	Required Verification Artifact
FR-M2-01	Unified IAM & Context Switching	Stage Gate G2 (Week 4)	Single Sign-On integration logs, JWT security audit report, and MFA verification proof.

FR-M2-07	Master Taxonomies & Data Services	Stage Gate G2 (Week 4)	Machine-readable taxonomy REST API inspection and WTO/GATS coverage verification.
FR-M2-02	Actor Registry, NATEP ID & Deduplication	Stage Gate G3 (Week 10)	Public NATEP ID syntax generator logs, fuzzy match test reports, and government adapter logs.
FR-M2-03	Organization Delegation Context	Stage Gate G3 (Week 10)	Multi-tenant access control UAT sign-off and automated expiration test scripts.
FR-M2-04	Zero-Duplicate Evidence Vault	Stage Gate G3 (Week 10)	Deduplication hash-match test execution logs, virus block reports, and AES-256 audits.
FR-M2-05	Granular Consent Engine (NDPA 2023)	Stage Gate G3 (Week 10)	NDPA privacy audit sign-off and real-time consent revocation blocking tests.
FR-M2-08	Workflow & Task Queue Engine	Stage Gate G3 (Week 10)	SLA escalation queue test execution logs and task-routing demonstration scripts.
FR-M2-06	Immutable	Stage Gate G4	Expiring envelope

	Disclosure Package	(Week 16)	TTL test reports and field-masking UAT verification sign-offs.
FR-M2-09	API Gateway & Hash-Chained Audit	Stage Gate G4 (Week 16)	Cryptographic hash chain validation script outputs and WORM storage configuration proofs.

Module 3: Portal 1

BRD/FRD

BUSINESS REQUIREMENTS DOCUMENT (BRD)

NIGERIA SERVICE EXPORT GATEWAY (NATEP / NCMSE)

MODULE 3: PORTAL 1 — DEMAND, MARKET ACCESS & BUYER MATCHMAKING DOMAIN

1. Document Control & Baseline Reference

Control Attribute	Specification
Document Title	Business Requirements Document (BRD) — Module 3: Portal 1 (Demand, Market Access & Buyer Matchmaking Domain)
Parent Programme	Nigeria Service Export Gateway (NATEP / NCMSE)
Executive Sponsor	Federal Ministry of Industry, Trade and Investment (FMITI)
Business Owner	NATEP / NCMSE Secretariat
Technical Authority	Architecture, Data, Security & Privacy Authority (ADSPA)
Baseline Source	Integrated Four-Part Terms of Reference (TOR) v1.1, BRD Module 1 Baseline, and BRD Module 2 Baseline
Precedence	Applicable Law > Executive Decisions > Integrated TOR > BRD Module 1 > BRD Module 2 > BRD Module 3 > FRD/SRS Specifications

2. Executive Summary & Strategic Context

2.1 Strategic Rationale & Mandate

The Nigeria Service Export Gateway aims to shift Nigeria from an oil-dependent revenue model to a service-driven, human-capital export economy. **Module 3 (Portal 1: Demand)** serves as the primary international interface for the gateway. It connects verified Nigerian exporters,

agencies, and talent pools directly with foreign buyers, enterprise clients, and institutional aggregators across global target markets.

Module 3 addresses three critical friction points in Nigeria's service trade ecosystem:

- **The Reputation and Trust Gap:** Foreign buyers often hesitate to assign enterprise contracts to Nigerian providers due to perceived fraud or verification challenges.
- **Fragmented Lead Generation:** Nigerian service firms lack a unified, institutional channel to discover, respond to, and win international Request for Proposals (RFPs) and contracts.
- **Escrow & Settlement Friction:** Foreign buyers require risk-mitigated contract execution, while Nigerian exporters require guaranteed, transparent payment in convertible currencies (USD, EUR, GBP).

2.2 Alignment with the Shared Foundation (Module 2)

Module 3 operates as a consuming front-end portal under **The Single-Platform Mandate** defined in Module 2. It holds zero local user tables, authentication credentials, or un-audited state machines. All identity management, document storage, and audit logs are offloaded to Module 2's Shared Foundation services.

3. Institutional Context & System-of-Record (SoR) Scope

3.1 Business Capabilities Matrix

Capability Category	Strategic Goal	Core Functional Responsibility
Buyer Onboarding & Identity	Frictionless Global Access	Frictionless onboarding for foreign corporate buyers with verified Legal Entity Identifier (LEI), DUNS, or international business registry checks.
Opportunity Management	Institutional Sourcing	Automated creation, publishing, and lifecycle tracking of international RFPs, direct procurement leads, and remote talent postings.
AI Matchmaking & Discovery	Precision Sourcing	Vector-search algorithm to match buyer RFPs against verified Nigerian exporter profiles based on skills, capacity, and WTO/GATS service modes.

Symmetric Trust & Verification	Bilateral Risk Mitigation	Exposing exporter trust badges (NATEP Verified Exporter, Top-Rated Export Partner) alongside buyer verification levels.
Deal Escrow & Contract Management	Financial & Execution Guarantee	Integrated milestone-based escrow framework ensuring foreign currency lockup before work begins and performance-linked disbursement.

3.2 Authoritative Data Ownership Matrix

Data Domain / Entity	Authoritative Subsystem Owner	Permitted Operations	Forbidden Operations
Buyer Profiles & Credentials	Module 3 Demand Engine	Ingest buyer metadata, initiate LEI/DUNS validation via Module 2, maintain buyer preferences.	Storing plain-text login credentials or maintaining secondary user master tables.
Opportunity / RFP Master Data	Module 3 Procurement Engine	Create RFPs, publish postings, track submission state transitions, enforce expiration timers.	Bypassing Module 2's Controlled Taxonomy when tagging service categories.
Matchmaker Index & Weights	Module 3 AI Matching Service	Calculate relevance vectors, score exporter compatibility, render ranked candidate shortlists.	Overriding verified regulatory badging status assigned by Module 4.
Escrow &	Module 3 Escrow	Lock contract	Processing payouts

Milestone Agreements	Service	terms, record milestone approvals, trigger financial payout webhooks.	without explicit digital milestone sign-off and active consent validation.
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4. Detailed Subsystem Business Requirements

4.1 International Buyer Onboarding & Verification (BR-DEM-01)

- **BR-DEM-01.1 (Tiered Access Framework):** The system shall permit foreign buyers to browse public showcase directories without mandatory up-front verification. Bidding issuance and direct contract award shall strictly require identity verification.
- **BR-DEM-01.2 (External Verification Adapters):** Foreign corporate entities shall be verified via automated API adapters against international business registries (Legal Entity Identifier - LEI, DUNS, or Companies House).
- **BR-DEM-01.3 (Identity Integration):** Verified buyer profiles must be provisioned with a Universal Actor Identifier (UAID) via Module 2's Canonical Actor Registry.

4.2 Opportunity Creation, Taxonomy Mapping & Procurement (BR-DEM-02)

- **BR-DEM-02.1 (Structured RFP Builder):** Buyers shall create structured service procurement requests incorporating budget ranges, WTO/GATS Supply Modes (Modes 1 to 4), required skill certifications, project milestones, and contract duration.
- **BR-DEM-02.2 (Taxonomy Enforcement):** All service categorizations must strictly bind to Module 2's Controlled Taxonomy (UN CPC Ver. 2.1 and ISIC Rev. 4). Free-text, non-standardized service category strings are prohibited.
- **BR-DEM-02.3 (RFP Lifecycle State Engine):** Opportunities shall transition through explicit, audited states: DRAFT to PUBLISHED to MATCHING_ACTIVE to SHORTLISTING to AWARDED to FULFILLED to CLOSED.

4.3 AI-Assisted Matchmaking & Capability Recommendation (BR-DEM-03)

- **BR-DEM-03.1 (Multi-Factor Scoring Engine):** The system shall evaluate buyer RFPs against verified exporter profiles in Module 4 using a dynamic vector scoring algorithm where Total Score equals the sum of weighted scores across four core dimensions:
 - **Capacity and Bandwidth Score:** 30 percent weight.
 - **Verification and Trust Badge Level Score:** 30 percent weight.
 - **Verified Past Performance Score:** 20 percent weight.
 - **Skill and Taxonomy Alignment Score:** 20 percent weight.
- **BR-DEM-03.2 (Zero-Bias Match Assurance):** Matchmaking calculations shall strictly rely on verified operational metrics and regulatory badges. Any manual score adjustment or administrative override must be logged as an auditable security event in Module 2.

4.4 Symmetric Trust Badging & Capability Exposure (BR-DEM-04)

- **BR-DEM-04.1 (Bid Visibility Matrix):** Exporter response rights shall be governed by their Module 4 Trust Badge level:

- **Registered Talent / Entity:** Browsing rights only. Bidding blocked.
- **NATEP Verified Exporter:** Bidding allowed on open RFPs up to 50,000 USD value.
- **Top-Rated Export Partner:** Bidding allowed on all enterprise RFPs, direct contract invitation, and priority listing in AI recommendations.
- **BR-DEM-04.2 (Disclosure Package Binding):** When an exporter is shortlisted by a foreign buyer, the system shall generate a field-masked, expiring Shared Disclosure Package via Module 2's Disclosure Service, bound to active NDPA consent grants.

4.5 Escrow Management, Milestones & Settlement Integration (BR-DEM-05)

- **BR-DEM-05.1 (Milestone-Based Fund Lockup):** Upon contract award, the foreign buyer must deposit target milestone funds into a secure, multi-currency escrow account (USD, EUR, GBP) before project work commences.
- **BR-DEM-05.2 (Performance Payout Enforcement):** Funds held in escrow shall be released to the Nigerian exporter's designated local or foreign currency domiciliary account only upon explicit digital sign-off of milestone deliverables by the buyer or via automated dispute resolution timeout.
- **BR-DEM-05.3 (Observatory Event Emission):** Every escrow state change (FUNDS_DEPOSITED, MILESTONE_APPROVED, FUNDS_RELEASED) must broadcast an asynchronous JSON domain event to Module 2's Event Bus for tracking by Module 5 (Observatory).

5. Non-Functional, Security & Compliance Requirements

5.1 Regulatory & Data Protection Standards

- **NDPA 2023 Compliance:** Cross-border transfer of exporter personal or commercial details to foreign buyers must be governed by purpose-bound, revocable consent grants managed by Module 2.
- **Financial Compliance:** Escrow integrations must adhere to international Anti-Money Laundering (AML) and Know Your Customer (KYC) frameworks.

5.2 Performance & Reliability SLAs

Key Performance Indicator	Target Threshold	Operational Condition
Search & Match Speed	Under 500 Milliseconds	p95 response time for rendering top-20 matched exporters per RFP query.
RFP Publishing Latency	Under 1 Second	Time to process, index, and broadcast new RFP events across the platform.
Escrow Status Synchronization	Near Real-Time (Under 2 Seconds)	Webhook processing speed for escrow payment lockup

		notification.
System Uptime	99.9 Percent Monthly Uptime	Applied to public buyer web portals and contract execution APIs.

6. Stage-Gate Execution & Verification Traceability Matrix

Requirement Code	Functional Feature / Subsystem	Target Stage Gate	Mandatory Verification Proof
BR-DEM-01	Buyer Onboarding & LEI Verification	Gate G2 (Week 4)	API integration test logs with DUNS/LEI adapters, buyer UAID provisioning proof.
BR-DEM-02	Opportunity / RFP Publishing Engine	Gate G2 (Week 4)	Structured RFP schema validation reports, UN CPC taxonomy mapping test suite.
BR-DEM-03	AI Vector Matchmaking Algorithm	Gate G3 (Week 10)	Vector search benchmarks (under 500ms), multi-factor scoring calculation audit scripts.
BR-DEM-04	Trust-Gated Bidding & Disclosure	Gate G3 (Week 10)	Entitlement access control UAT sign-off, NDPA consent-bound disclosure package generation proof.
BR-DEM-05	Escrow Integration & Event Streaming	Gate G4 (Week 16)	End-to-end multi-currency

			payment lockup and release execution logs, Kafka event emission audit.
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FUNCTIONAL REQUIREMENTS DOCUMENT (FRD) / SYSTEM REQUIREMENTS SPECIFICATION (SRS)

NIGERIA SERVICE EXPORT GATEWAY (NATEP / NCMSE)
 MODULE 3: PORTAL 1 — DEMAND, MARKET ACCESS & BUYER MATCHMAKING DOMAIN
 Document Control & Baseline Alignment

- Document Title:** Functional Requirements Document (FRD) / System Requirements Specification (SRS) — Module 3: Portal 1 (Demand, Market Access & Buyer Matchmaking Domain)
- Parent Programme:** Nigeria Service Export Gateway (NATEP / NCMSE)
- Baseline Alignment:** Integrated Four-Part TOR v1.1, BRD Module 1 Baseline, Module 2 BRD/FRD Baseline, and Module 3 BRD
- Formatting Policy:** Plain English specification standard. Standard text is used exclusively for all terms, units, and metrics—avoiding mathematical symbols or LaTeX formulas.
- Target Audience:** Software Engineers, System Architects, Database Administrators, Integration Specialists, and Quality Assurance Engineers.

System-of-Record (SoR) Architectural Matrix

Data Domain / Subsystem	Primary SoR Owner	Downstream Consuming Modules	Permitted Operations	Invariant Constraints
Buyer Master Accounts	Module 3 Demand Engine	Portals 2, 3 & Observatory	Onboard foreign entities, trigger DUNS/LEI verification via Module 2, update contact profiles.	Zero local password storage; authentication must route strictly via Module 2 IAM OIDC provider.

RFPs & Procurement Records	Module 3 Procurement Engine	Portals 2, 3 & Observatory	Ingest RFPs, map service codes, track bid submission states, enforce expiration TTLs.	Service categories must bind strictly to UN CPC codes served by Module 2 Taxonomy Subsystem.
Matchmaker Calculations	Module 3 AI Matching Service	Portal 2 & Observatory	Compute vector similarity, calculate multi-factor fit scores, generate candidate shortlists.	Scoring engine cannot modify underlying exporter badging levels or bypass Module 4 state.
Escrow & Transaction Ledger	Module 3 Escrow Service	Observatory & Financial Gateways	Initiate escrow holds, log milestone approvals, execute multi-currency payout webhooks.	Payouts require two-party digital signature or automated dispute resolution timeout confirmation.

Detailed Subsystem Functional Specifications

1. International Buyer Onboarding & Verification Subsystem

Requirement Reference: FR-M3-01 (BRD Reference: BR-DEM-01)

1. System Processing Logic & Business Rules:

- i. The buyer registration module provides an unblocked access model allowing foreign buyers to search public showcase directories prior to formal verification.
- ii. Formal buyer account activation triggers an automated verification request to external identity providers using Legal Entity Identifier (LEI) or DUNS database adapters.
- iii. Upon successful verification, Module 3 executes POST /api/v1/actors/register against

Module 2's Canonical Actor Registry to assign a Universal Actor Identifier (UAID-YYYYMMDD-UUIDv4) and public NATEP ID (NT-B26-XXXXX).

- iv. Authenticated buyer sessions are managed via Module 2 OIDC JWT access tokens (15-minute expiration, 24-hour rolling refresh tokens).

2. User Story:

- i. **As an** International Enterprise Buyer,
- ii. **I want** to register and verify my corporate identity using my company's Legal Entity Identifier (LEI),
- iii. **So that** I can publish high-value RFPs and engage verified Nigerian service providers with institutional trust.

3. Acceptance Criteria (Gherkin Format):

○ i. Scenario 1: Automated Verification of Foreign Corporate Buyer

■

1. **Given** a foreign enterprise buyer inputs a valid Legal Entity Identifier (LEI) number during onboarding,

■

2. **When** the system executes an automated lookup against the external LEI registry adapter,

■

3. **Then** the buyer account state transitions to VERIFIED, a UAID is provisioned via Module 2, and RFP posting capabilities are unlocked.

○ ii. Scenario 2: Failed External Verification Handling

■

1. **Given** a foreign buyer inputs an invalid or unverified corporate registration number,

■

2. **When** the external verification adapter returns a negative match response,

■

3. **Then** the account state transitions to PENDING_MANUAL_REVIEW and routes an inspection task to the NATEP Data Steward queue.

2. Opportunity Creation, Taxonomy Mapping & Procurement Subsystem

Requirement Reference: FR-M3-02 (BRD Reference: BR-DEM-02)

1. System Processing Logic & Business Rules:

- i. The RFP creation interface presents structured input forms requiring explicit values for: Title, Detailed Statement of Work, Budget Range, Target Completion Date, WTO/GATS Supply Mode (Modes 1 to 4), and UN CPC Service Code.
- ii. Form inputs query Module 2's Taxonomy Service (/api/v1/taxonomies/services) via a local Redis cache to populate dropdowns. Unmapped custom taxonomy strings are rejected at the API Gateway layer with HTTP 400 Bad Request.
- iii. Published RFPs trigger the generation of an immutable opportunity record and assign an internal identifier (RFP-YYYYMMDD-UUIDv4).
- iv. The system automatically enforces an unmodifiable Time-To-Live (TTL) timer corresponding to the RFP submission deadline. Upon reaching the deadline, the state engine transitions the RFP from MATCHING_ACTIVE to SHORTLISTING.

2. User Story:

- i. **As a** Foreign Procurement Officer,
- ii. **I want** to post a structured Request for Proposal mapped to international service codes,
- iii. **So that** the platform can accurately route my opportunity to qualified, compliant service exporters.

3. Acceptance Criteria (Gherkin Format):

○ i. Scenario 1: Publishing a Valid UN CPC Mapped RFP

■

1. **Given** an authenticated buyer submits an RFP with UN CPC Code 83131 (IT Consulting Services) and a valid budget envelope,

■

2. **When** the system validates the payload against Module 2 taxonomy rules,

■

3. **Then** the RFP is created with state `MATCHING_ACTIVE` and an event `event.rfp.published.v1` is broadcast to the Event Bus.

○ ii. Scenario 2: Automated Transition upon Submission Deadline Expiry

■

1. **Given** an active RFP reaches its configured submission closing timestamp,

■

2. **When** the automated scheduler executes its routine state check,

■

3. **Then** the RFP status automatically changes to `SHORTLISTING` and further bid submissions are blocked with HTTP 422 Unprocessable Entity.

3. AI-Assisted Matchmaking & Recommendation Subsystem

Requirement Reference: FR-M3-03 (BRD Reference: BR-DEM-03)

1. System Processing Logic & Business Rules:

- i. Upon RFP publication, the AI Matchmaking Engine generates a vector embedding representing the RFP requirements and executes a vector similarity search against Module 4 exporter profile embeddings.
- ii. Search results are filtered to exclude any exporter whose Module 4 Trust Badge level does not satisfy the RFP minimum threshold.
- iii. Candidate matches are scored using a four-part weighted formula:

■

1. **Capacity & Bandwidth Score (30 percent weight):** Evaluates team size and current active contract load.

■

2. **Trust & Verification Score (30 percent weight):** Evaluates Module 4 verification depth (NATEP Verified Exporter = 0.8, Top-Rated Export Partner = 1.0).

■

3. **Verified Past Performance Score (20 percent weight):** Evaluates completed platform escrow transactions and buyer ratings.

■

4. **Skill & Taxonomy Alignment Score (20 percent weight):** Evaluates direct UN CPC code overlap and verified certifications.

- iv. The engine ranks candidate exporters and returns a dynamic top-20 candidate shortlist to the buyer within 500 milliseconds (p95 latency).

2. User Story:

- i. **As a** Foreign Buyer,
- ii. **I want** the system to automatically generate a ranked shortlist of verified Nigerian exporters matching my RFP specifications,
- iii. **So that** I can reduce sourcing overhead and engage top-tier suppliers quickly.

3. Acceptance Criteria (Gherkin Format):

- i. **Scenario 1: Generating a Ranked Exporter Shortlist**

■

1. **Given** a new RFP is published under UN CPC Code 83131,

■

2. **When** the AI Matchmaking Engine processes the request payload,

■

3. **Then** the system returns a ranked shortlist of exporters, displaying overall match scores alongside individual trust badges within 500 milliseconds.

- ii. **Scenario 2: Filtering Unverified Exporters from Enterprise RFPs**

■

1. **Given** an enterprise RFP requires a minimum trust level of Top-Rated Export Partner,

■

2. **When** the matchmaking algorithm executes,

■

3. **Then** exporters holding lower trust tiers (e.g., Registered Talent or NATEP Verified Exporter) are strictly excluded from the rendered candidate shortlist.

4. Symmetric Trust Badging & Contract Exposure Subsystem

Requirement Reference: FR-M3-04 (BRD Reference: BR-DEM-04)

1. System Processing Logic & Business Rules:

- i. Exporter response rights are enforced dynamically by querying Module 4 profile state upon bid submission attempt:

■

1. Attempted bid by Registered Talent / Entity: Blocked with HTTP 403 Forbidden. UI displays prompt to complete CAC/TIN verification.

■

2. Attempted bid by NATEP Verified Exporter on RFP over 50,000 USD: Blocked with HTTP 403 Forbidden. UI displays prompt to complete sector regulatory validation.

■

3. Attempted bid by Top-Rated Export Partner: Permitted for all value tiers.

- ii. When a buyer shortlists an exporter, Module 3 requests a field-masked data envelope from Module 2's Disclosure Service (POST /api/v1/disclosure/packages).
- iii. The generated envelope hides sensitive personal attributes (NIN, direct phone, detailed bank info) while exposing verified trade metrics and professional credentials.

2. User Story:

- i. **As an** Exporter with Top-Rated Export Partner status,
- ii. **I want** my verified trust status to grant me exclusive access to high-value foreign enterprise RFPs,
- iii. **So that** my investment in regulatory compliance translates directly to market access.

3. Acceptance Criteria (Gherkin Format):

○ i. Scenario 1: Enforcing Bidding Entitlement Thresholds

■

1. **Given** an exporter holding NATEP Verified Exporter status attempts to submit a bid on an RFP valued at 150,000 USD,

■

2. **When** the API Gateway evaluates the submission against Module 4 capability rules,

■

3. **Then** the request returns HTTP 403 Forbidden with message "Bidding on opportunities above 50,000 USD requires Top-Rated Export Partner status".

○ ii. Scenario 2: Generating Consent-Bound Disclosure Package

■

1. **Given** a foreign buyer selects a candidate firm for shortlist review,

■

2. **When** Module 3 requests candidate details,

■

3. **Then** the system generates a temporary, 15-minute expiring disclosure envelope containing only consented profile fields.

5. Escrow Management, Milestones & Settlement Integration Subsystem

Requirement Reference: FR-M3-05 (BRD Reference: BR-DEM-05)

1. System Processing Logic & Business Rules:

- i. Contract awards require the foreign buyer to create an escrow agreement specifying dollar values per milestone.
- ii. Funding the escrow calls the external payment gateway adapter. Upon receipt confirmation, the escrow engine marks milestone state as FUNDS_LOCKED.
- iii. Exporters submit milestone deliverable proof to Module 2's Evidence Vault (POST /api/v1/vault/documents), creating an immutable SHA-256 document reference.
- iv. Buyer approval of milestone deliverables executes POST /api/v1/escrow/release, initiating payout transfer to the exporter's verified domiciliary bank account and emitting event event.escrow.disbursed.v1.

2. User Story:

- i. **As an** Exporter executing a foreign service contract,
- ii. **I want** project funds locked in escrow before I commence work and disbursed automatically upon deliverable approval,
- iii. **So that** I am protected against non-payment and foreign exchange settlement delays.

3. Acceptance Criteria (Gherkin Format):

○ i. Scenario 1: Milestone Escrow Funding and Lockup

■

1. **Given** a foreign buyer awards a 10,000 USD contract split into two equal milestones,
-
2. **When** the buyer deposits 5,000 USD for Milestone 1 via the payment gateway,
-
3. **Then** the escrow status updates to MILESTONE_1_FUNDS_LOCKED and the exporter is notified to commence work.
- ii. **Scenario 2: Milestone Approval and Automated Payout Release**
-
1. **Given** an exporter submits deliverable evidence and the buyer clicks "Approve Milestone",
-
2. **When** the release endpoint executes,
-
3. **Then** the system triggers a foreign currency payout transfer to the exporter's account and logs an audit record in Module 2.

Domain Event JSON Envelopes

1. RFP Published Event Envelope (event.rfp.published.v1)

```
{
  "eventId": "evt_4a8b2c1d-9e0f-41a2-b3c4-5d6e7f8a9b0c",
  "eventType": "event.rfp.published.v1",
  "eventTimestamp": "2026-09-09T14:30:00.000Z",
  "producer": "natep.portal1.procurement-engine",
  "actor": {
    "uid": "UAID-20260909-8f9e0d1c-2a3b-4c5d-6e7f-8a9b0c1d2e3f",
    "natepld": "NT-B26-88192",
    "entityType": "FOREIGN_BUYER",
    "verificationStatus": "VERIFIED"
  },
  "data": {
    "rfpld": "RFP-20260909-3c2b1a4d-5e6f-7a8b-9c0d-1e2f3a4b5c6d",
    "title": "Enterprise Cloud Architecture Migration",
    "unCpcCode": "83131",
    "wtoSupplyMode": "MODE_1_CROSS_BORDER",
    "budgetAmountUsd": 75000.00,
    "requiredTrustTier": "TOP_RATED_EXPORT_PARTNER",
    "submissionDeadline": "2026-10-15T23:59:59.000Z"
  },
}
```

```

"context":{
  "correlationId": "corr_7f8e9d0c-1b2a-3f4e-5d6c-7b8a90123456",
  "originatingIp": "198.51.100.45",
  "userAgent": "NATEP-Demand-Portal/2.1"
}
}

```

2. Escrow Disbursed Event Envelope (event.escrow.disbursed.v1)

```

{
  "eventId": "evt_10923847-a1b2-4c3d-5e6f-7a8b9c0d1e2f",
  "eventType": "event.escrow.disbursed.v1",
  "eventTimestamp": "2026-09-09T16:15:22.000Z",
  "producer": "natep.portal1.escrow-service",
  "actor": {
    "uid": "UAID-20260909-8f9e0d1c-2a3b-4c5d-6e7f-8a9b0c1d2e3f",
    "natepld": "NT-B26-88192",
    "entityType": "FOREIGN_BUYER",
    "verificationStatus": "VERIFIED"
  },
  "data": {
    "contractId": "CTR-20260909-998877",
    "milestoneId": "MS-01",
    "beneficiaryUaid": "UAID-20260801-11223344-5566-7788-9900-aabbccddeeff",
    "beneficiaryNatepld": "NT-B26-94821",
    "disbursedAmountUsd": 37500.00,
    "evidenceDocumentId": "DOC-20260909-55443322",
    "settlementGateway": "NATEP_CENTRAL_ESCROW_GATEWAY"
  },
  "context": {
    "correlationId": "corr_00112233-4455-6677-8899-aabbccddeeff",
    "originatingIp": "198.51.100.45",
    "userAgent": "NATEP-Escrow-Engine/1.0"
  }
}

```

Non-Functional & System Level Performance Targets

1. **System Throughput:** Procurement and matchmaker engines must support a minimum

sustained load of 1,000 concurrent active buyer sessions without degraded response times.

2. **Data Security & Encryption:**

- **In Transit:** Mandatory TLS 1.3 for all buyer interactions and API communications.
- **At Rest:** AES-256-GCM encryption applied to all contract metadata and escrow transaction records.

3. **Audit Immutability:** Every contract state transition and escrow event must be recorded into Module 2's Cryptographic Hash-Chained Audit Trail (WORM storage).

Stage-Gate Delivery & Verification Traceability

Requirement ID	Subsystem Component	Target Delivery Gate	Required Verification Artifact
FR-M3-01	Buyer Onboarding & LEI Verification	Stage Gate G2 (Week 4)	LEI/DUNS adapter integration test execution output, buyer UAID provisioning logs.
FR-M3-02	RFP Procurement & Taxonomy Engine	Stage Gate G2 (Week 4)	UN CPC taxonomy validation logs, state transition unit test scripts.
FR-M3-03	AI Vector Matchmaker Engine	Stage Gate G3 (Week 10)	Matchmaking latency benchmark reports (under 500ms), vector scoring accuracy verification logs.
FR-M3-04	Trust Badging & Entitlement Logic	Stage Gate G3 (Week 10)	Access control UAT sign-off, Module 2 disclosure package API test output.
FR-M3-05	Escrow & Settlement Engine	Stage Gate G4 (Week 16)	Multi-currency escrow workflow execution logs, Kafka event

			streaming verification report.
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Module 4: Portal 2

BRD/FRD

BUSINESS REQUIREMENTS DOCUMENT (BRD)

NIGERIA SERVICE EXPORT GATEWAY (NATEP / NCMSE)

MODULE 4: PORTAL 2 — SUPPLY, READINESS, CERTIFICATION & INCENTIVES DOMAIN

1. Document Control & Baseline Reference

Control Attribute	Specification
Document Title	Business Requirements Document (BRD) — Module 4: Portal 2 (Supply, Readiness, Certification & Incentives Domain)
Parent Programme	Nigeria Service Export Gateway (NATEP / NCMSE)
Executive Sponsor	Federal Ministry of Industry, Trade and Investment (FMITI)
Business Owner	NATEP / NCMSE Secretariat
Technical Authority	Architecture, Data, Security & Privacy Authority (ADSPA)
Baseline Source	Integrated Four-Part Terms of Reference (TOR) v1.1, BRD Module 1 Baseline, BRD Module 2 Baseline, and Module 3 BRD
Precedence	Applicable Law > Executive Decisions > Integrated TOR > BRD Module 1 > BRD Module 2 > BRD Module 3 > BRD Module 4 > FRD/SRS Specifications

2. Executive Summary & Strategic Context

2.1 Strategic Rationale & Mandate

Nigeria's service export expansion is hindered by fragmented supplier capabilities, inconsistent regulatory validation, and a lack of standardized quality management. **Module 4 (Portal 2: Supply)** functions as the national supply-side engine to aggregate, verify, assess, and scale

Nigerian service exporters, professional agencies, and individual talent.

Module 4 addresses three critical structural challenges on the domestic supply side:

- **Supplier Visibility and Verification Deficit:** Validated Nigerian service providers lack a centralized, cryptographically verified registry that foreign buyers and institutional aggregators can query directly.
- **Inconsistent Operational Readiness:** Service entities often lack transparent diagnostics to benchmark their readiness across international standards, legal protections, and quality systems.
- **Fragmented Trade Incentives:** Access to government-backed trade expansion grants, tax credits, and capacity-building resources is historically disconnected from real, verified platform trade performance.

2.2 Alignment with the Shared Foundation (Module 2)

In accordance with **The Single-Platform Mandate** established in Module 2, Module 4 does not create isolated user databases or custom identity stores. It relies on Module 2 for identity provisioning (Canonical Actor Registry), document indexing (Evidence Vault), and immutable audit logs. All verification artifacts and credential attestations generated in Module 4 are hashed and anchored in Module 2.

3. Strategic Objectives & Key Performance Indicators (KPIs)

3.1 Core Strategic Objectives

- **Objective 1 — National Exporter Registry:** Establish a single authoritative directory of verified Nigerian service exporters, corporate agencies, and individual talent linked to sector regulators.
- **Objective 2 — Operational Readiness Standardization:** Categorize supply entities into transparent, dynamic readiness tiers to drive capacity enhancement across international trade channels.
- **Objective 3 — Automated Regulatory Compliance:** Streamline credential verification with key sector regulators (e.g., NITDA, COREN, CIBN) via cryptographically verifiable attestations.
- **Objective 4 — Performance-Linked Incentives:** Facilitate access to trade expansion grants and tax credits based strictly on verified export volume and platform performance.

3.2 Target Key Performance Indicators

Metric Indicator	Year 1 Target	Year 2 Target	Verification Channel
Verified Exporter Onboarding	10,000 Entities / Talent	30,000 Entities / Talent	Module 2 Canonical Actor Registry
Readiness Advancement	30% transition to Level 3	55% transition to Level 3	Assessment Engine Logs

Rate			
Regulatory Traceability	100% Cryptographic Audit	100% Cryptographic Audit	Module 2 Evidence Vault
Incentive Disbursal Accuracy	100% Escrow-Linked	100% Escrow-Linked	Module 3 & 5 Event Stream Logs

4. Institutional Context & System-of-Record (SoR) Scope

4.1 Business Capabilities Matrix

Capability Category	Strategic Goal	Core Functional Responsibility
Supply Onboarding & Profile	Progressive Access	Frictionless, zero-blocker registration for exporters and talent, deferred verification for advanced tiers.
Regulatory & Credential Validation	Verification Depth	Ingestion and verification of professional licenses, tax identification, and corporate registry standing via direct APIs or AI checking.
Operational Readiness Diagnostic	Capacity Benchmarking	Automated evaluation of export capacity, financial stability, quality management, and legal IP protection.
Trust Badging & Entitlement Logic	Capability Unlocking	Dynamic allocation of trust badges (Registered Talent, NATEP Verified Exporter, Top-Rated Export)

		Partner) based on verified state.
Incentives & Trade Facilitation	Performance Granting	Automated pre-qualification and administration of trade expansion grants, tax credits, and capacity development resources.

4.2 Authoritative Data Ownership Matrix

Data Domain / Entity	Authoritative Subsystem Owner	Permitted Operations	Forbidden Operations
Exporter Profiles & Capability Data	Module 4 Profile Subsystem	Ingest profile attributes, update capability matrices, score completion metrics.	Allowing unverified entities to bid on high-value foreign enterprise RFPs.
Regulatory & License Credentials	Module 4 Verification Engine	Validate credentials, track license expiration, issue cryptographic attestations.	Modifying raw regulatory registry outputs or bypassing expiration timers.
Export Readiness Scores & Tiers	Module 4 Diagnostic Engine	Calculate readiness scores, assign hierarchy tiers, generate gap remediation plans.	Manual score overrides without auditable administrative log entries in Module 2.
Incentive Applications & Ledger	Module 4 Incentives Engine	Process grant requests, verify escrow earnings, track disbursal lifecycle.	Disbursing trade incentives to entities without verified escrow trade receipts.

5. Detailed Subsystem Business Requirements

5.1 Supply Onboarding & Profile Completion (BR-P2-01)

- **BR-P2-01.1 (Zero-Blocker Onboarding):** The system shall allow service exporters and individual talent candidates to register immediately with basic identity validation, deferring mandatory corporate uploads until bidding or incentive claims occur.
- **BR-P2-01.2 (Capability Taxonomy Mapping):** Profile creation shall require exporters to select service capabilities bound to Module 2's Controlled Taxonomy (UN CPC Ver. 2.1 and ISIC Rev. 4).
- **BR-P2-01.3 (Dynamic Completion Metrics):** Profile completeness shall be calculated using a weighted balance of Data Completeness (40 percent weight) and Verification Depth (60 percent weight).

5.2 Regulatory Credentialing & Verification Framework (BR-P2-02)

- **BR-P2-02.1 (Hybrid Verification Pipeline):** The platform shall validate corporate standing (CAC), tax compliance (TIN), and professional licenses using direct API connections to sector regulators or fallback AI document verification.
- **BR-P2-02.2 (Cryptographic Hashing):** All verified licenses and certifications must be hashed (SHA-256) and recorded in Module 2's Evidence Vault to prevent duplicate or fraudulent document submissions.
- **BR-P2-02.3 (Lifecycle Expiration Tracking):** The system shall automatically track license validity periods, sending 30-day renewal warnings and initiating automated badge degradation upon expiration.

5.3 Operational Readiness Assessment & Hierarchy Framework (BR-P2-03)

- **BR-P2-03.1 (Multi-Parameter Diagnostic):** The assessment engine shall score export readiness across five core parameters: Export Capacity (25 percent), Financial Stability (20 percent), Quality Systems (20 percent), Cross-Border Delivery Experience (20 percent), and Legal/IP Protection (15 percent).
- **BR-P2-03.2 (Ordinal Tier Hierarchy):** Exporters shall be assigned to three explicit readiness levels based on computed diagnostic scores:
 - **Level 1 — Export Capacity Building (Score under 50 percent):** Early-stage entities routed to foundational capacity building.
 - **Level 2 — Export Emerging (Score 50 percent to 79 percent):** Growing businesses permitted to bid on standard international RFPs.
 - **Level 3 — Export Ready (Score 80 percent and above):** Mature entities granted maximum match multipliers in Module 3 and full incentive access.

5.4 Progressive Trust Badging & Capability Exposure (BR-P2-04)

- **BR-P2-04.1 (Badge Progression Pipeline):** Exporters shall progress through three clear trust badges:
 - **Registered Talent / Entity:** Browsing rights and directory listing.
 - **NATEP Verified Exporter:** Unlocked via verified CAC, TIN, and bank account; enables bidding up to 50,000 USD.

- **Top-Rated Export Partner:** Unlocked via sector regulatory clearance and escrow history; enables unlimited enterprise bidding.
- **BR-P2-04.2 (Next Unlock Prompts):** The dashboard shall render real-time UI guidance detailing the current completion percentage, active capabilities, and exact actions required to reach the next badge level.

5.5 Trade Incentives & Performance Facilitation (BR-P2-05)

- **BR-P2-05.1 (Performance-Linked Grants):** Export expansion grants and trade rebates shall be restricted to verified exporters demonstrating active trade volume processed through platform escrow channels.
- **BR-P2-05.2 (Automated Pre-Qualification):** The incentive engine shall automatically cross-reference grant applications against verified escrow receipts logged in Module 3 and Module 5 before routing to administrators.

6. Non-Functional, Security & Compliance Requirements

6.1 Regulatory & Data Protection Standards

- **NDPA 2023 Compliance:** Exporter data processing and profile sharing with third parties or foreign buyers must rely strictly on purpose-bound consent grants managed via Module 2.
- **Document Encryption:** Financial records, tax returns, and private corporate uploads must be encrypted at rest using AES-256-GCM.

6.2 Performance & Reliability SLAs

Key Performance Indicator	Target Threshold	Operational Condition
Readiness Diagnostic Speed	Under 2 Seconds	Response time for computing and displaying diagnostic reports.
AI Document Verification Speed	Under 15 Seconds	Processing time for certificate OCR and forgery check.
Entitlement Calculation Speed	Under 500 Milliseconds	Real-time update of trust badge permissions upon profile edit.
System Availability	99.9 Percent Uptime	Applied to supplier dashboards and verification APIs.

7. Stage-Gate Execution & Verification Traceability Matrix

Requirement Code	Functional Feature / Subsystem	Target Stage Gate	Mandatory Verification Proof
BR-P2-01	Supplier Onboarding & Dynamic Profile	Gate G3 (Week 10)	Unblocked registration test logs, taxonomy mapping validation reports.
BR-P2-02	Regulatory Validation & Evidence Hashing	Gate G3 (Week 10)	CAC/TIN API integration logs, Module 2 Evidence Vault entry records.
BR-P2-03	Operational Readiness Diagnostic Engine	Gate G4 (Week 16)	Five-parameter scoring execution scripts, tier assignment logs.
BR-P2-04	Trust Badging & Next Unlock Prompts	Gate G3 (Week 10)	Entitlement policy test outputs, dashboard UI step-through verification.
BR-P2-05	Performance-Linked Incentives Engine	Gate G4 (Week 16)	Escrow earnings pre-qualification logs, grant approval workflow audit trails.

FUNCTIONAL REQUIREMENTS
DOCUMENT (FRD) / SYSTEM

REQUIREMENTS SPECIFICATION (SRS)

NIGERIA SERVICE EXPORT GATEWAY (NATEP / NCMSE)
MODULE 4: PORTAL 2 — SUPPLY, READINESS, CERTIFICATION & INCENTIVES DOMAIN

Document Control & Baseline Alignment

- 1. **Document Title:** Functional Requirements Document (FRD) / System Requirements Specification (SRS) — Module 4: Portal 2 (Supply, Readiness, Certification & Incentives Domain)
- 2. **Parent Programme:** Nigeria Service Export Gateway (NATEP / NCMSE)
- 3. **Baseline Alignment:** Integrated Four-Part TOR v1.1, BRD Module 1 Baseline, Module 2 BRD/FRD Baseline, Module 3 BRD/FRD, and Module 4 BRD
- 4. **Formatting Policy:** Plain English specification standard. Standard text is used exclusively for all terms, units, and metrics—avoiding mathematical symbols or LaTeX formulas.
- 5. **Target Audience:** Software Engineers, System Architects, Database Administrators, Integration Specialists, and Quality Assurance Engineers.

System-of-Record (SoR) Architectural Matrix

Data Domain / Subsystem	Primary SoR Owner	Downstream Consuming Modules	Permitted Operations	Invariant Constraints
Exporter Profiles & Capability Data	Portal 2 Exporter Registry	Portals 1, 3 & Observatory	Profile creation, capability indexing, profile completion scoring.	Zero unverified corporate or identity claims permitted to bid on international RFPs.
Regulatory & License Credentials	Portal 2 Regulatory Subsystem	Portals 1, 3 & Observatory	License submission, API validation, AI document check, expiration tracking.	Direct linkage required with sector regulators or verified evidence uploads in Module 2.
Export Readiness & Assessment	Portal 2 Assessment Engine	Portal 1 & Observatory	Readiness score calculation, tier	Automatic recalculation of readiness

Engine			assignment, gap-analysis generation.	tiers upon profile or credential updates.
Export Incentives & Trade Grants	Portal 2 Incentives Engine	Observatory & Regulatory Agencies	Incentive application processing, pre-qualification checks, grant tracking.	Incentives restricted strictly to verified exporters meeting minimum escrow volume thresholds.
Exporter Trust Badging Subsystem	Portal 2 Badging Subsystem	Portal 1, Observatory & Shared Foundation	Trust badge assignment, entitlement unlocking, Next Unlock Prompt generation.	Capabilities strictly bound to verified completion thresholds; manual overrides logged to Module 2 Evidence Vault.

Detailed Subsystem Functional Specifications

1. Exporter Onboarding & Profile Readiness Subsystem

Requirement Reference: FR-M4-01 (BRD Reference: BR-P2-01, BR-P2-04)

1. System Processing Logic & Business Rules:

- i. Registration is open and completely unblocked, requiring basic identity input and email validation.
- ii. Exporters construct capability portfolios, specifying service categories mapped strictly to Module 2 UN CPC codes, operational scale, past trade history, and team qualifications.
- iii. Profile completeness is calculated dynamically using weighted metrics: Data Completeness (40 percent weight) and Verification Depth (60 percent weight).
- iv. Feature permissions and trust badges are assigned based on completion thresholds:

■

1. **Registered Talent / Entity (1 percent to 39 percent completion):** Permits RFP

browsing, capability portfolio creation, and access to market access guides.

■

2. **NATEP Verified Exporter (40 percent to 79 percent completion):** Unlocked via verified Corporate Affairs Commission (CAC) registration, Tax Identification Number (TIN) validation, and bank account verification. Unlocks bidding on open RFPs up to 50,000 USD and receiving automated match alerts.

■

3. **Top-Rated Export Partner (80 percent or higher completion):** Unlocked via sector regulatory clearance, verified trade history, and active escrow performance. Unlocks priority placement in AI matchmaking, direct enterprise invitations, and lower platform escrow fees.

- v. The dashboard displays a dynamic Next Unlock Prompt highlighting the current completion score, active capabilities, and exact steps required to unlock the next tier.

2. User Story:

- i. **As a** Nigerian Service Exporter or Talent Candidate,
- ii. **I want** to build a verified trade profile and track my readiness progression through clear badges and completion metrics,
- iii. **So that** I can establish credibility with international buyers and unlock access to global export RFPs.

3. Acceptance Criteria (Gherkin Format):

- i. **Scenario 1: Progressive Exporter Onboarding and Badging**

■

1. **Given** a Nigerian service business registers and validates its Corporate Affairs Commission (CAC) registration number and Tax Identification Number,

■

2. **When** the verification engine confirms active standing via the Shared Regulatory Verification Engine,

■

3. **Then** the system assigns the NATEP Verified Exporter badge, updates profile completion above 40 percent, and enables bidding on RFPs up to 50,000 USD.

- ii. **Scenario 2: Display of Next Unlock Prompts**

■

1. **Given** an exporter profile achieves 60 percent completion,

■

2. **When** the user accesses their dashboard,

■

3. **Then** the system displays current active capabilities ("Can bid on RFPs up to 50,000 USD") and highlights the next milestone ("Upload Sector Regulatory Certification to reach Top-Rated Exporter status and unlock Enterprise RFPs").

2. Sectoral Regulatory Validation & Certification Subsystem

Requirement Reference: FR-M4-02 (BRD Reference: BR-P2-02)

1. System Processing Logic & Business Rules:

- i. Exporters submit sector-specific professional certifications and regulatory clearances

(e.g., NITDA/NDPC for IT, COREN for engineering, CIBN for financial services, or ISO 27001).

- ii. Verification executes via a hybrid model: automated verification against open regulatory databases where available, or fallback AI Document Verification for uploaded certificate files.
- iii. Verified certifications are cryptographically hashed (SHA-256) and indexed in Module 2's Evidence Vault to prevent duplicate document usage.
- iv. Expired certifications trigger automated profile score degradation and dispatch a 30-day renewal warning notification before badge downgrade occurs.

2. User Story:

- i. **As a** Professional Service Exporter,
- ii. **I want** my sector licenses and professional certifications verified and badged automatically,
- iii. **So that** foreign buyers have cryptographic proof of my regulatory compliance and technical competence.

3. Acceptance Criteria (Gherkin Format):

○ i. Scenario 1: Hybrid Certification Verification

■

1. **Given** an engineering firm uploads a COREN practice license,

■

2. **When** the system processes the document through the AI Verification Engine or regulatory API,

■

3. **Then** the credential status updates to VERIFIED, links to the exporter's public trust badge, and registers a record in the Evidence Vault.

○ ii. Scenario 2: Expiration and Automated Badge Downgrade Warning

■

1. **Given** an exporter's professional certification is 30 days away from expiration,

■

2. **When** the daily system review executes,

■

3. **Then** the system dispatches an automated renewal warning notification and highlights the pending risk on the exporter's dashboard.

3. Export Readiness Assessment & Diagnostic Subsystem

Requirement Reference: FR-M4-03 (BRD Reference: BR-P2-03)

1. Explicit Operational Readiness Tier Hierarchy Definition:

- i. The platform enforces an explicit, ordinal three-tier hierarchy for operational export readiness.
- ii. Hierarchy levels are ranked strictly from lowest (Level 1) to highest (Level 3) capability:

■

1. **Level 1 — Export Capacity Building (Lowest Readiness Level):** Target entity includes emerging businesses and early-stage talent needing foundational skill-building and basic certification support. Restricted from bidding on high-value enterprise RFPs and

automatically routed to capacity-building resources.

-
- 2. **Level 2 — Export Emerging (Intermediate Readiness Level):** Target entity includes growing businesses possessing basic operational capacity. Permitted to bid on standard international RFPs up to baseline system thresholds.
-
- 3. **Level 3 — Export Ready (Highest Readiness Level):** Target entity includes fully qualified, legally compliant, and mature exporters. Receives maximum weight multipliers in Module 3's AI Matchmaking Engine, priority placement in enterprise search listings, and full eligibility for government export grants.
 - iii. Advancement moves strictly upward from Level 1 to Level 2 to Level 3. Downgrades occur dynamically if verified credentials expire or compliance checks fail.
- 2. **Evaluation Parameters & Scoring Matrix:**
 - i. The Assessment Engine evaluates readiness across five weighted parameters:
 -
 - 1. **Export Capacity (25 percent weight):** Operational scale, service volume, and resource bandwidth.
 -
 - 2. **Financial Stability (20 percent weight):** Cash flow metrics, corporate banking verification, and audit readiness.
 -
 - 3. **Quality Management Systems (20 percent weight):** Internal quality controls, ISO standards, and sector certifications.
 -
 - 4. **Cross-Border Delivery Experience (20 percent weight):** Verified international trade history and escrow contract fulfillments.
 -
 - 5. **Legal and IP Protection (15 percent weight):** Active regulatory clearances, contract management capabilities, and IP safeguards.
 - ii. Total computed diagnostic score (0 to 100 percent) maps directly to the operational hierarchy:
 -
 - 1. **Score under 50 percent:** Assigned to Level 1 (Export Capacity Building).
 -
 - 2. **Score 50 percent to 79 percent:** Assigned to Level 2 (Export Emerging).
 -
 - 3. **Score 80 percent and above:** Assigned to Level 3 (Export Ready).
- 3. **User Story:**
 - i. **As an** Emerging Nigerian Service Business,
 - ii. **I want** to complete an automated export readiness diagnostic,
 - iii. **So that** I can see my exact hierarchy placement and receive a clear action plan to ascend to Export Ready status.
- 4. **Acceptance Criteria (Gherkin Format):**

- i. **Scenario 1: Hierarchy Tier Assignment via Diagnostic Score**
 -
 - 1. **Given** an exporter completes the readiness assessment and achieves a calculated total score of 85 percent across all five evaluation parameters,
 -
 - 2. **When** the diagnostic engine processes the score,
 -
 - 3. **Then** the system assigns the exporter to Level 3 (Export Ready), applies the maximum match multiplier in Module 3, and updates the public profile badge.
- ii. **Scenario 2: Automatic Action Plan for Level 1 & 2 Entities**
 -
 - 1. **Given** an exporter scores 45 percent and is assigned to Level 1 (Export Capacity Building),
 -
 - 2. **When** the assessment report renders,
 -
 - 3. **Then** the system generates a tailored remediation dashboard highlighting missing parameters required to advance to Level 2.

4. Export Incentives & Trade Facilitation Subsystem

Requirement Reference: FR-M4-04 (BRD Reference: BR-P2-05)

1. System Processing Logic & Business Rules:

- i. Verified exporters (NATEP Verified Exporter status or higher) gain access to apply for government-backed trade facilitation incentives, export expansion grants, tax credits, and subsidized trade missions.
- ii. Application processing cross-references platform trade history, verified export earnings logged via Module 3 escrow channels, and compliance status.
- iii. Incentive approvals follow a two-tier review workflow: automated pre-qualification by the platform rules engine followed by final review sign-off from designated NATEP / NEPC Administrators.
- iv. Granted incentives and tax credit certificates are digitally issued, logged in Module 2's Evidence Vault, and tracked against the exporter's lifetime trade performance.

2. User Story:

- i. **As a** Verified Service Exporter with active export earnings,
- ii. **I want** to apply for government export expansion incentives and tax rebates directly through the portal,
- iii. **So that** I can reduce operational costs and reinvest in expanding my international trade capacity.

3. Acceptance Criteria (Gherkin Format):

- i. **Scenario 1: Pre-Qualification for Export Grants based on Verified Trade Volume**
 -
 - 1. **Given** an exporter achieves over 50,000 USD in verified export volume processed through platform escrow,
 -

2. **When** the exporter submits an application for the Service Export Expansion Grant,
■
3. **Then** the system automatically pre-qualifies the application, attaches verified escrow receipts as evidence, and routes it to the Administrator review queue.

Domain Event JSON Envelopes

1. Exporter Verified Event Envelope (event.exporter.verified.v1)

```
{
  "eventId": "evt_9f8e7d6c-5b4a-3f2e-1d0c-9b8a7f6e5d4c",
  "eventType": "event.exporter.verified.v1",
  "eventTimestamp": "2026-09-09T16:40:00.000Z",
  "producer": "natep.portal2.verification-engine",
  "actor": {
    "uuid": "UAID-20260801-11223344-5566-7788-9900-aabbccddeeff",
    "natepld": "NT-B26-94821",
    "entityType": "SERVICE_EXPORTER",
    "verificationStatus": "VERIFIED"
  },
  "data": {
    "trustBadgeLevel": "NATEP_VERIFIED_EXPORTER",
    "profileCompletionPercentage": 65,
    "cacNumber": "RC-1234567",
    "tinNumber": "TIN-98765432-0001",
    "evidenceVaultReference": "DOC-20260909-CAC-VERIFIED"
  },
  "context": {
    "correlationId": "corr_3c2b1a4d-5e6f-7a8b-9c0d-1e2f3a4b5c6d",
    "originatingIp": "102.89.23.12",
    "userAgent": "NATEP-Supply-Portal/1.0"
  }
}
```

2. Readiness Diagnostic Completed Event Envelope (event.readiness.assessed.v1)

```

{
  "eventId": "evt_3b2a1c0d-4e5f-6a7b-8c9d-0e1f2a3b4c5d",
  "eventType": "event.readiness.assessed.v1",
  "eventTimestamp": "2026-09-09T16:41:00.000Z",
  "producer": "natep.portal2.assessment-engine",
  "actor": {
    "uuid": "UAID-20260801-11223344-5566-7788-9900-aabbccddeeff",
    "natepld": "NT-B26-94821",
    "entityType": "SERVICE_EXPORTER",
    "verificationStatus": "VERIFIED"
  },
  "data": {
    "calculatedScore": 82,
    "assignedReadinessTier": "LEVEL_3_EXPORT_READY",
    "parameterScores": {
      "exportCapacity": 85,
      "financialStability": 80,
      "qualitySystems": 80,
      "crossBorderExperience": 85,
      "legalIpProtection": 75
    },
    "matchmakingWeightMultiplier": 1.25
  },
  "context": {
    "correlationId": "corr_8f7e6d5c-4b3a-2f1e-0d9c-8b7a6f5e4d3c",
    "originatingIp": "102.89.23.12",
    "userAgent": "NATEP-Assessment-Engine/2.0"
  }
}

```

Non-Functional & System Level Performance Targets

1. **System Throughput:** The supplier onboarding, verification, and diagnostic engines must support up to 5,000 active concurrent exporter portal sessions during peak registration windows.
2. **Data Security & Privacy:**
 - **In Transit:** Mandatory TLS 1.3 encryption across all supplier endpoints and external regulatory API connections.
 - **At Rest:** AES-256-GCM encryption applied to private financial statement uploads, tax records, and corporate officer details.
3. **Audit Immutability:** All trust badge level changes, diagnostic score assignments, and regulatory verification outcomes must record immutable hash entries into Module 2's Audit Trail (WORM storage).

Stage-Gate Delivery & Verification Traceability

Requirement ID	Subsystem Component	Target Delivery Gate	Required Verification Artifact
FR-M4-01	Exporter Onboarding & Profile Readiness	Stage Gate G3 (Week 10)	Unblocked registration logs, readiness badging evaluation scripts, and profile unlock UI test sign-off.
FR-M4-02	Sectoral Regulatory Validation	Stage Gate G3 (Week 10)	Regulatory verification API integration logs, AI document scanner benchmarks, and Evidence Vault registration checks.
FR-M4-03	Export Readiness Diagnostic Engine	Stage Gate G4 (Week 16)	Diagnostic scoring test execution scripts, gap analysis generation checks, and matchmaking weight integration logs.
FR-M4-04	Export Incentives & Trade Facilitation	Stage Gate G4 (Week 16)	Incentive application workflow UAT sign-off, automated pre-qualification rule verification, and grant ledger audit logs.

Module 5: Portal 3

BRD/FRD

BUSINESS REQUIREMENTS DOCUMENT (BRD)

NIGERIA SERVICE EXPORT GATEWAY (NATEP / NCMSE)

MODULE 5: PORTAL 3 — NATIONAL SERVICES EXPORT OBSERVATORY, GOVERNANCE,
ANALYTICS & REPORTING DOMAIN

1. Document Control & Baseline Reference

Control Attribute	Specification
Document Title	Business Requirements Document (BRD) — Module 5: Portal 3 (National Services Export Observatory, Governance, Analytics & Reporting Domain)
Parent Programme	Nigeria Service Export Gateway (NATEP / NCMSE)
Executive Sponsor	Federal Ministry of Industry, Trade and Investment (FMITI)
Business Owner	NATEP / NCMSE Secretariat
Technical Authority	Architecture, Data, Security & Privacy Authority (ADSPA)
Baseline Source	Integrated Four-Part Terms of Reference (TOR) v1.1, BRD Module 1 Baseline, BRD Module 2 Baseline, BRD Module 3 Baseline, and BRD Module 4 Baseline
Precedence	Applicable Law > Executive Decisions > Integrated TOR > BRD Module 1 > BRD Module 2 > BRD Module 3 > BRD Module 4 > BRD Module 5 > FRD/SRS Specifications

2. Executive Summary & Strategic Context

2.1 Strategic Rationale & Mandate

While Modules 3 and 4 manage trade creation, matching, and supply readiness, **Module 5 (Portal 3: Observatory & Governance)** operates as the intelligence core and oversight umbrella of the Nigeria Service Export Gateway. It provides macroeconomic visibility, real-time trade flow monitoring, policy simulation, and inter-agency compliance tracking across Nigeria's entire service export economy.

Module 5 addresses three major policy and intelligence deficits:

- **The Trade Data Blindspot:** Service export statistics in Nigeria are historically fragmented across the Central Bank of Nigeria (CBN), Nigeria Customs Service (NCS), Federal Inland Revenue Service (FIRS), and Nigerian Export Promotion Council (NEPC), making real-time trade balance accounting impossible.
- **Leakage & Regulatory Arbitrage:** Absence of central transaction monitoring leads to unrecorded cross-border service trade, revenue leakage, and delayed export incentive disbursements.
- **Policy Inefficiency:** Trade policy formulation relies on delayed, retrospective manual reporting rather than real-time data streaming from active trade escrows and matching portals.

2.2 Alignment with the Shared Foundation (Module 2)

Module 5 acts as the master analytical consumer under **The Single-Platform Mandate** established in Module 2. It ingests Kafka domain event streams published by Module 3 (Demand) and Module 4 (Supply), normalizes the telemetry, and projects it into real-time analytical data models without mutating underlying transaction states. All compliance logs and governance audits generated by Module 5 are written directly to Module 2's Cryptographic Hash-Chained Audit Trail (WORM storage).

3. Institutional Context & System-of-Record (SoR) Scope

3.1 Business Capabilities Matrix

Capability Category	Strategic Goal	Core Functional Responsibility
Macroeconomic Observatory	Real-Time Trade Tracking	Aggregate and visualize trade volumes, WTO/GATS supply mode balances, foreign currency inflows, and regional trade corridors.
Inter-Agency Governance	Synchronized Compliance	Provide role-based portal interfaces for CBN, FIRS, NEPC, NITDA, and FMITI to monitor compliance, tax credits, and regulatory

		standing.
Policy Modeling & Simulation	Predictive Trade Analysis	Algorithmic modeling of tariff impact, incentive effectiveness, and sector growth projections using aggregated trade telemetry.
Incentive Audit & Disbursal Oversight	Leakage Mitigation	Automated auditing of Module 4 grant payouts against verified Module 3 escrow transaction ledgers.
Executive Dashboards & Reporting	Institutional Accountability	Automated generation of national export performance reports, regulatory compliance matrices, and executive briefings.

3.2 Authoritative Data Ownership Matrix

Data Domain / Entity	Authoritative Subsystem Owner	Permitted Operations	Forbidden Operations
Macroeconomic Aggregates	Module 5 Observatory Data Mart	Read-only aggregation, OLAP indexing, trade velocity projection, trend modeling.	Mutating raw transactional records or overriding operational escrow balances.
Inter-Agency Audit Logs	Module 2 Shared Audit Engine	Ingest agency interaction logs, append hash records, verify chain integrity.	Editing or deleting logged regulatory actions or clearance timestamps.

Policy Simulation Parameters	Module 5 Policy Engine	Ingest macro indicators, run scenario models, export risk projections.	Modifying live platform fee structures or escrow thresholds directly from simulation mode.
Cross-Agency Compliance Data	Subsystem SoR via Module 2	Read-only compliance mapping, flag regulatory discrepancies across agencies.	Overriding direct regulatory registration states maintained in Module 4.

4. Detailed Subsystem Business Requirements

4.1 Real-Time Macroeconomic Trade Observatory (BR-OBS-01)

- **BR-OBS-01.1 (Stream Ingestion):** The subsystem shall consume asynchronous JSON events from Module 3 (e.g., event.escrow.disbursed.v1) and Module 4 (e.g., event.exporter.verified.v1) in near real-time (latency under 2 seconds).
- **BR-OBS-01.2 (Multi-Dimensional Aggregation):** Trade data must be aggregate-able by WTO/GATS Supply Modes (Modes 1 to 4), UN CPC Service Categories, destination corridors (USD, EUR, GBP, NGN), and exporter trust tiers.
- **BR-OBS-01.3 (Anonymized Data Exposure):** Public-facing analytics dashboards must enforce zero-knowledge aggregation, strictly suppressing personally identifiable information (PII) and individual corporate contract details.

4.2 Inter-Agency Governance & Regulatory Gateway (BR-OBS-02)

- **BR-OBS-02.1 (Role-Based Agency Portals):** The system shall provide dedicated, isolated workspaces for verified regulatory authorities:
 - **Central Bank of Nigeria (CBN):** Real-time monitoring of foreign exchange repatriations and domiciliary account inflows.
 - **Federal Inland Revenue Service (FIRS):** Verification of export earnings for tax incentive validation and zero-rated VAT compliance.
 - **Nigerian Export Promotion Council (NEPC):** Monitoring of Service Export Expansion Grant eligibility and certificate utilization.
 - **FMITI / Secretariat:** Overall strategic trade balance tracking and performance evaluation.
- **BR-OBS-02.2 (Cross-Agency Anomaly Detection):** The platform shall flag discrepancies between declared export earnings, escrow disbursements, and regulatory tax filings.

4.3 Predictive Policy Simulation & Economic Modeling (BR-OBS-03)

- **BR-OBS-03.1 (Incentive Impact Modeling):** The subsystem shall enable policy analysts to simulate the economic impact of adjusting trade incentives, tax credit rates, or sector grant allocations on national service export volume.
- **BR-OBS-03.2 (Corridor Bottleneck Analysis):** Predictive analytics shall identify emerging market access barriers, regulatory delays, or escrow settlement frictions across foreign trade corridors.

4.4 Automated Incentive Audit & Anti-Leakage Oversight (BR-OBS-04)

- **BR-OBS-04.1 (Bilateral Reconciliation Engine):** Every grant application processed in Module 4 must be cross-checked by Module 5 against cryptographically verified escrow release receipts logged in Module 3 before final funds sign-off.
- **BR-OBS-04.2 (Cryptographic Compliance Proofs):** The system shall generate verifiable audit packages summarizing total trade volume, tax status, and escrow receipts for every incentive disbursement.

5. Non-Functional, Security & Compliance Requirements

5.1 Regulatory & Security Standards

- **Data Privacy Compliance:** All analytics ingestion pipelines must enforce data anonymization and field masking in compliance with the NDPA 2023.
- **Access Control:** Agency access must enforce strict Multi-Factor Authentication (MFA) and role-based access control (RBAC) managed via Module 2’s IAM framework.

5.2 Performance & Reliability SLAs

Key Performance Indicator	Target Threshold	Operational Condition
Stream Processing Latency	Under 2 Seconds	Latency from transaction event emission to analytical dashboard update.
Analytical Query Latency	Under 1.5 Seconds	Response time for rendering multi-year macro aggregate queries (p95).
Agency Report Generation	Under 10 Seconds	Exporting full PDF/Excel compliance audit packages.
System Uptime	99.9 Percent Monthly Uptime	Applied to agency workspaces and public observatory dashboards.

6. Stage-Gate Execution & Verification Traceability Matrix

Requirement Code	Functional Feature / Subsystem	Target Stage Gate	Mandatory Verification Proof
BR-OBS-01	Real-Time Macro Observatory Engine	Gate G4 (Week 16)	Kafka event ingestion benchmarks, aggregate OLAP query performance test logs.
BR-OBS-02	Inter-Agency Governance Gateway	Gate G4 (Week 16)	Agency workspace RBAC sign-off, anomaly detection test cases.
BR-OBS-03	Policy Modeling & Simulation Engine	Gate G5 (Week 20)	Simulation model execution output, scenario stress-testing reports.
BR-OBS-04	Incentive Audit & Reconciliation Engine	Gate G4 (Week 16)	End-to-end grant pre-qualification audit trails, Module 2 WORM verification logs.

FUNCTIONAL REQUIREMENTS DOCUMENT (FRD) / SYSTEM REQUIREMENTS SPECIFICATION (SRS)

NIGERIA SERVICE EXPORT GATEWAY (NATEP / NCMSE)

MODULE 5: PORTAL 3 — NATIONAL SERVICES EXPORT OBSERVATORY, GOVERNANCE, ANALYTICS & REPORTING DOMAIN

Document Control & Baseline Alignment

1. **Document Title:** Functional Requirements Document (FRD) / System Requirements Specification (SRS) — Module 5: Portal 3 (National Services Export Observatory,

Governance, Analytics & Reporting Domain)

2. **Parent Programme:** Nigeria Service Export Gateway (NATEP / NCMSE)
3. **Baseline Alignment:** Integrated Four-Part TOR v1.1, BRD Module 1 Baseline, Module 2 BRD/FRD Baseline, Module 3 BRD/FRD, Module 4 BRD/FRD, and Module 5 BRD
4. **Formatting Policy:** Plain English specification standard. Standard text is used exclusively for all terms, units, and metrics—avoiding mathematical symbols or LaTeX formulas.
5. **Target Audience:** Software Engineers, System Architects, Database Administrators, Data Engineers, Integration Specialists, and Regulatory Compliance Officers.

System-of-Record (SoR) Architectural Matrix

Data Domain / Subsystem	Primary SoR Owner	Downstream Consuming Modules	Permitted Operations	Invariant Constraints
Macro Data Mart & Aggregates	Module 5 Observatory Subsystem	FMITI, CBN, NEPC, Public Portal	Stream ingestion, analytical aggregation, data projection.	Strictly read-only consumption of operational telemetry; zero direct record modification.
Inter-Agency Governance Workspace	Module 5 Governance Engine	Partner Agencies (FIRS, CBN, NEPC)	View compliance dashboards, issue tax clearance receipts, flag anomalies.	Operations must be authenticated via Module 2 IAM with full RBAC token enforcement.
Incentive Audit Ledger	Module 5 Reconciliation Engine	Secretariat & Audit Authorities	Cross-check escrow receipts, audit grant applications, sign off disbursements.	Disbursal sign-offs require two-party verification and valid Module 2 Evidence Vault references.
Policy Models & Simulation	Module 5	Secretariat &	Run scenario models, adjust	Simulation mode is strictly

Store	Policy Engine	Policy Analysts	analytical variables, save simulation reports.	isolated; cannot mutate live trade portal parameters.
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Detailed Subsystem Functional Specifications

1. Real-Time Macroeconomic Trade Observatory Subsystem

Requirement Reference: FR-M5-01 (BRD Reference: BR-OBS-01)

1. System Processing Logic & Business Rules:

- i. The subsystem deploys an asynchronous stream consumer connected to Module 2's central Event Bus (Kafka).
- ii. Incoming events (RFP creation, contract awards, escrow lockups, and milestone disbursements) are processed, anonymized, and transformed into multi-dimensional time-series models within 2 seconds of emission.
- iii. Analytical aggregates are indexed across five core dimensions:

■

1. **WTO/GATS Supply Modes:** Mode 1 (Cross-Border Trade), Mode 2 (Consumption Abroad), Mode 3 (Commercial Presence), Mode 4 (Presence of Natural Persons).

■

2. **Service Taxonomy:** Mapped strictly to UN CPC Ver. 2.1 codes.

■

3. **Currency Corridor:** USD, EUR, GBP, and NGN conversions based on CBN daily reference rates.

■

4. **Geographic Destination:** Destination country and regional economic blocs (e.g., ECOWAS, AfCFTA, EU).

■

5. **Supplier Tier:** Performance breakdown across Registered Talent, NATEP Verified Exporters, and Top-Rated Export Partners.

- iv. Public observatory queries render aggregated metrics while automatically masking any dataset containing fewer than three distinct commercial entities to prevent commercial data exposure.

2. User Story:

- i. **As a** National Trade Policy Analyst,
- ii. **I want** to view real-time service export volumes categorized by WTO supply modes and target geographic corridors,
- iii. **So that** I can accurately assess national service trade balance and evaluate international market penetration.

3. Acceptance Criteria (Gherkin Format):

- i. **Scenario 1: Real-Time Telemetry Stream Ingestion**



- 1. **Given** an escrow disbursement event (event.escrow.disbursed.v1) of 37,500 USD is published by Module 3,



- 2. **When** the Observatory stream processing engine ingests the event payload,



- 3. **Then** the national trade volume aggregates update within 2 seconds, reflecting the increase under UN CPC Code 83131 and WTO Mode 1.

- ii. **Scenario 2: Data Anonymization and Privacy Threshold Enforcement**



- 1. **Given** a public user queries service export statistics for a specific service category with only two active exporters,



- 2. **When** the API processes the query,



- 3. **Then** the system suppresses granular figures and displays an aggregated tier summary to protect commercial confidentiality.

2. Inter-Agency Governance & Compliance Gateway Subsystem

Requirement Reference: FR-M5-02 (BRD Reference: BR-OBS-02)

1. System Processing Logic & Business Rules:

- i. Dedicated agency portals are provisioned with fine-grained access boundaries:



- 1. **Central Bank of Nigeria (CBN Workspace):** Automated tracking of export foreign exchange inflows and domiciliary account settlement reconciliation.



- 2. **Federal Inland Revenue Service (FIRS Workspace):** Automated verification of exporter revenue for zero-rated VAT validation and corporate income tax credits.



- 3. **Nigerian Export Promotion Council (NEPC Workspace):** Tracking of Service Export Expansion Grant usage, registration verification, and trade mission allocation.

- ii. The Governance Engine runs daily automated anomaly detection checks comparing declared export earnings in Module 4 profiles, actual escrow payouts in Module 3, and regulatory filings.

- iii. Anomaly alerts (e.g., payout discrepancy exceeding 10 percent) automatically place the affected exporter profile into PENDING_COMPLIANCE_REVIEW state and notify assigned agency officers.

2. User Story:

- i. **As a** FIRS Compliance Officer,
- ii. **I want** to verify an exporter's claimed foreign earnings against cryptographically verified escrow receipts,
- iii. **So that** tax credits and zero-rated VAT benefits are granted strictly on verified trade performance.

3. Acceptance Criteria (Gherkin Format):

○ i. Scenario 1: Inter-Agency Revenue Verification

■

1. **Given** an exporter claims 100,000 USD in export tax rebates via the FIRS portal,

■

2. **When** the FIRS compliance engine executes a cross-check against Module 3 escrow records,

■

3. **Then** the system returns a verified transaction audit package confirming 100,000 USD in processed escrow disbursements.

○ ii. Scenario 2: Automated Anomaly Flagging

■

1. **Given** a discrepancy is detected between declared export revenue and verified platform escrow receipts,

■

2. **When** the automated daily compliance audit executes,

■

3. **Then** the system generates an anomaly alert ticket in the Governance Portal and flags the exporter profile for review.

3. Predictive Policy Simulation & Modeling Subsystem

Requirement Reference: FR-M5-03 (BRD Reference: BR-OBS-03)

1. System Processing Logic & Business Rules:

- i. The simulation engine provides a sandbox environment pre-populated with historical platform trade velocity, sector growth rates, and macroeconomic indicators.
- ii. Analysts can adjust key policy variables: Grant Match Percentage (0 to 25 percent), Tax Rebate Rates (0 to 15 percent), Platform Escrow Fee Adjustments (0 to 2 percent), and Sector Training Allocations.
- iii. The engine runs predictive algorithms to generate 12-month, 24-month, and 36-month projections covering total export volume growth, foreign currency inflows, tax revenue impact, and job creation estimates.
- iv. Simulation runs produce immutable scenario reports stored in the Policy Document Store for FMITI executive review.

2. User Story:

- i. **As an** FMITI Strategic Director,
- ii. **I want** to model the projected service export expansion resulting from a 5 percent increase in export grants for IT exporters,
- iii. **So that** we can optimize government grant allocations based on data-driven growth projections.

3. Acceptance Criteria (Gherkin Format):

○ i. Scenario 1: Policy Incentive Scenario Execution

■

1. **Given** a policy analyst configures a simulation scenario with a 5 percent increase in IT export grant funding,

-
- 2. **When** the simulation engine processes the model parameters against historical trade velocity,
-
- 3. **Then** the system generates a 24-month projection report estimating export volume growth, fiscal cost, and net currency inflow within 10 seconds.

4. Automated Incentive Audit & Anti-Leakage Subsystem

Requirement Reference: FR-M5-04 (BRD Reference: BR-OBS-04)

1. System Processing Logic & Business Rules:

- i. When an exporter submits a grant application in Module 4, the Reconciliation Engine initiates a mandatory verification routine.
- ii. The engine validates three required compliance conditions:

-
- 1. **Active Exporter Standing:** Exporter holds NATEP Verified Exporter or Top-Rated Export Partner status in Module 4.
-

- 2. **Escrow Trade Volume:** Application grant value does not exceed the allowed percentage (e.g., 15 percent) of verified escrow receipts logged in Module 3.
-

- 3. **No Unresolved Compliance Flags:** Entity has zero pending regulatory alerts in the Governance Gateway.
- iii. Applications passing all three conditions receive an automated Audit Verification Token (AVT-YYYYMMDD-UUIDv4) linked to the cryptographic audit trail in Module 2. Applications failing any condition are rejected automatically with explicit failure logs.

2. User Story:

- i. **As a** National Audit Authority Officer,
- ii. **I want** every grant disbursement to require an automated Audit Verification Token based on verified escrow earnings,
- iii. **So that** public funds are protected from fraudulent claims and leakage.

3. Acceptance Criteria (Gherkin Format):

- i. **Scenario 1: Automated Audit Verification Token Generation**
-

- 1. **Given** an exporter with 200,000 USD in verified escrow receipts applies for a 20,000 USD export expansion grant,
-

- 2. **When** the Reconciliation Engine verifies the application against Module 3 escrow logs and Module 4 compliance state,
-

- 3. **Then** the system generates an Audit Verification Token (AVT), registers it in Module 2 WORM storage, and advances the grant to payment authorization.

- ii. **Scenario 2: Automated Rejection of Unverified Grant Claims**
-

- 1. **Given** an entity with zero verified escrow disbursements applies for a trade expansion

grant,

-
- 2. **When** the Reconciliation Engine executes its cross-check,
-
- 3. **Then** the application is rejected automatically with HTTP 422 Unprocessable Entity and error code INCENTIVE_AUDIT_ZERO_TRADE_VOLUME.

Domain Event JSON Envelopes

1. Trade Analytics Aggregate Snapshot Event Envelope (event.observatory.snapshot.v1)

```
{
  "eventId": "evt_7d8c9b0a-1e2f-3a4b-5c6d-7e8f9a0b1c2d",
  "eventType": "event.observatory.snapshot.v1",
  "eventTimestamp": "2026-09-09T17:00:00.000Z",
  "producer": "natep.portal3.observatory-engine",
  "actor": {
    "uid": "UAID-SYSTEM-OBSERVATORY-0001",
    "natepld": "NT-SYS-OBS",
    "entityType": "SYSTEM_SERVICE",
    "verificationStatus": "VERIFIED"
  },
  "data": {
    "snapshotPeriod": "2026-Q3-MONTH-09",
    "totalExportVolumeUsd": 14250000.00,
    "activeExportersCount": 1420,
    "wtoModeBreakdown": {
      "mode1CrossBorderUsd": 9500000.00,
      "mode2ConsumptionAbroadUsd": 1200000.00,
      "mode3CommercialPresenceUsd": 2300000.00,
      "mode4NaturalPersonsUsd": 1250000.00
    },
    "topServiceCategories": [
      { "unCpcCode": "83131", "category": "IT Consulting Services", "volumeUsd": 6200000.00 },
      { "unCpcCode": "83111", "category": "Financial Advisory Services", "volumeUsd": 3100000.00 }
    ]
  },
  "context": {
    "correlationId": "corr_11223344-5566-7788-9900-aabbccddeeff",
  }
}
```

```

    "originatingIp": "10.0.4.15",
    "userAgent": "NATEP-Observatory-Engine/1.0"
  }
}

```

2. Incentive Audit Token Issued Event Envelope (event.incentive.audit.passed.v1)

```

{
  "eventId": "evt_5e4d3c2b-1a0f-9e8d-7c6b-5a4f3e2d1c0b",
  "eventType": "event.incentive.audit.passed.v1",
  "eventTimestamp": "2026-09-09T17:05:12.000Z",
  "producer": "natep.portal3.reconciliation-engine",
  "actor": {
    "uid": "UAID-20260801-11223344-5566-7788-9900-aabbccddeeff",
    "natepId": "NT-B26-94821",
    "entityType": "SERVICE_EXPORTER",
    "verificationStatus": "VERIFIED"
  },
  "data": {
    "applicationId": "GRANT-20260909-8832",
    "auditToken": "AVT-20260909-7f8e9d0c-1b2a-3f4e",
    "requestedAmountNgn": 5000000.00,
    "verifiedEscrowVolumeUsd": 120000.00,
    "complianceCheckStatus": "PASSED",
    "evidenceVaultAuditHash":
    "sha256:e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855"
  },
  "context": {
    "correlationId": "corr_99887766-5544-3322-1100-eeddcbbbaa99",
    "originatingIp": "10.0.4.20",
    "userAgent": "NATEP-Incentive-Audit/1.0"
  }
}

```

Non-Functional & System Level Performance Targets

1. **System Throughput:** Stream ingestion pipeline must process up to 10,000 trade events per second during peak settlement windows without dropped messages or backpressure failure.

2. Data Security & Privacy:

- **In Transit:** Mandatory TLS 1.3 encryption across all agency portal endpoints and inter-module communications.
- **At Rest:** AES-256-GCM encryption applied to all governance audit logs, policy models, and agency workspace data.

3. Audit Immutability: Every agency login, anomaly flag, audit token generation, and policy simulation output must append an immutable cryptographic record to Module 2's Audit Trail (WORM storage).

Stage-Gate Delivery & Verification Traceability

Requirement ID	Subsystem Component	Target Delivery Gate	Required Verification Artifact
FR-M5-01	Real-Time Macro Observatory	Stage Gate G4 (Week 16)	Kafka stream ingestion performance logs, multi-dimensional query latency benchmark reports (under 1.5 seconds).
FR-M5-02	Inter-Agency Governance Gateway	Stage Gate G4 (Week 16)	Agency workspace RBAC penetration test sign-off, automated anomaly detection execution logs.
FR-M5-03	Predictive Policy Simulation Engine	Stage Gate G5 (Week 20)	Simulation model execution output, scenario stress-testing and forecasting accuracy reports.
FR-M5-04	Incentive Audit & Reconciliation Engine	Stage Gate G4 (Week 16)	End-to-end grant pre-qualification audit trails, Module 2 WORM verification logs, AVT generation test

			suites.
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Master Architecture Overview & Complete Gateway Alignment

With the completion of **Module 5**, all five core architectural modules of the **Nigeria Service Export Gateway (NATEP / NCMSE)** are fully specified:

- **Module 1:** Vision, Strategy & Governance Architecture
- **Module 2:** Shared Foundation, Common Infrastructure & Core Platform Services
- **Module 3:** Portal 1 — Demand, Market Access & Buyer Matchmaking Domain
- **Module 4:** Portal 2 — Supply, Readiness, Certification & Incentives Domain
- **Module 5:** Portal 3 — National Services Export Observatory, Governance, Analytics & Reporting Domain

The platform specifications maintain complete architectural alignment, strict plain-text presentation, zero LaTeX formatting, end-to-end event traceability via Module 2's Event Bus, and strict adherence to **The Single-Platform Mandate**.

Module 6: BRD/FRD

BUSINESS REQUIREMENTS DOCUMENT (BRD)

NIGERIA SERVICE EXPORT GATEWAY (NATEP / NCMSE)

MODULE 6: PORTAL 4 — GLOBAL PROMOTION, INVESTMENT FACILITATION &
ECOSYSTEM ENABLEMENT DOMAIN

1. Document Control & Baseline Reference

Control Attribute	Specification
Document Title	Business Requirements Document (BRD) — Module 6: Portal 4 (Global Promotion, Investment Facilitation & Ecosystem Enablement Domain)
Parent Programme	Nigeria Service Export Gateway (NATEP / NCMSE)
Executive Sponsor	Federal Ministry of Industry, Trade and Investment (FMITI)
Business Owner	NATEP / NCMSE Secretariat
Technical Authority	Architecture, Data, Security & Privacy Authority (ADSPA)
Baseline Source	Integrated Four-Part Terms of Reference (TOR) v1.1, BRD Modules 1–5 Baselines
Precedence	Applicable Law > Executive Decisions > Integrated TOR > BRD Modules 1–5 > BRD Module 6 > FRD/SRS Specifications

2. Executive Summary & Strategic Context

2.1 Strategic Rationale & Mandate

While Modules 3 and 4 handle trade matching and supplier readiness, and Module 5 manages internal analytics, **Module 6 (Portal 4)** serves as Nigeria's foreign-facing digital storefront for international investors, global off-takers, and multilateral partners. Module 6 promotes foreign

direct investment (FDI) into service export hubs, coordinates international trade missions, and streamlines regulatory landing protocols for global enterprises establishing operational footprints in Nigeria.

Module 6 addresses critical growth barriers:

- **FDI Entry Friction:** Complex multi-agency workflows for international service firms setting up local entities, obtaining tax incentives, or acquiring Special Economic Zone (SEZ) operational licenses.
- **Global Brand Disconnect:** Fragmented international marketing of Nigerian service talent, IT hubs, and Business Process Outsourcing (BPO) capabilities.
- **Delegation & Mission Disconnect:** Lack of unified digital tracking for bilateral trade missions, institutional MOUs, and diplomatic trade facilitation pipelines.

2.2 Alignment with the Shared Foundation (Module 2)

Module 6 utilizes **The Single-Platform Mandate** (Module 2) for identity federation, multi-currency transaction processing, and cryptographic audit logging. Foreign investors authenticated via Module 2 gain seamless access to Module 6 investment portals, while all trade delegation credentials and incentive allocations write directly to Module 2’s Cryptographic Audit Vault.

3. Institutional Context & System-of-Record (SoR) Scope

3.1 Business Capabilities Matrix

Capability Category	Strategic Goal	Core Functional Responsibility
Global Investment Landing Pad	FDI Acceleration	Provide digital concierge tools, landing space applications (e.g., ITANA, SEZs), and streamlined business setup workflows.
International Trade Mission Portal	Bilateral Trade Creation	Manage delegation registration, match foreign buyers with verified exporters, and track trade mission outcomes.
Ecosystem Promotion & Showcase	Brand Acceleration	Publicly feature verified Nigerian service firms, talent pools, and sector capabilities to international enterprise buyers.

Strategic MOU & Pipeline Tracker	Commitment Fulfillment	Track execution milestones for institutional trade MOUs, foreign government partnerships, and FDI commitments.
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3.2 Authoritative Data Ownership Matrix

Data Domain / Entity	Authoritative Subsystem Owner	Permitted Operations	Forbidden Operations
FDI Investor Applications	Module 6 Landing Pad Subsystem	Create investor profile, process landing applications, update setup stage.	Mutating verified tax status directly; must request via Module 5 / FIRS connectors.
Trade Mission Registers	Module 6 Trade Mission Engine	Register foreign delegations, publish agendas, log MOU executions.	Overriding verified seller credentials managed in Module 4.
Public Showcase Directory	Module 6 Ecosystem Subsystem	Read-only aggregation of verified Module 4 exporters, publish marketing profiles.	Publishing unverified seller entities or exposing unmasked PII.

4. Detailed Subsystem Business Requirements

4.1 Investor Concierge & Landing Pad (BR-INV-01)

- **BR-INV-01.1 (Digital Concierge Workflow):** The subsystem shall provide an end-to-end digital application portal for foreign service enterprises seeking operational setup in Nigeria, including SEZ physical/virtual allocation (e.g., ITANA facilities).
- **BR-INV-01.2 (Multi-Agency Fast-Track Routing):** Applications submitted via the Landing Pad must automatically route clearance packets to NIPC, NEPZA/NEPC, and FIRS interfaces.

4.2 International Trade Mission & Delegation Engine (BR-INV-02)

- **BR-INV-02.1 (Delegation Matchmaking):** The engine shall facilitate curated B2B matchmaking sessions between inbound foreign trade delegations and verified Module 4

exporters during official state missions.

- **BR-INV-02.2 (MOU & Commitment Tracking):** The system shall track commercial MOUs signed during trade missions, monitoring conversion rates from non-binding agreements to executed Module 3 escrow contracts.

4.3 Global Brand & Ecosystem Promotion (BR-INV-03)

- **BR-INV-03.1 (Public Service Directory):** The portal shall publish an interactive, globally accessible directory showcasing verified Nigerian service delivery capabilities across BPO, ITES, Creative, and Professional Services.

5. Non-Functional, Security & Performance Requirements

Key Performance Indicator	Target Threshold	Operational Condition
Public Showcase Latency	Under 800 Milliseconds	p95 page render time globally via CDN distribution.
Investor Application Routing	Under 5 Seconds	Synchronous fast-track packet generation across agency APIs.
System Availability	99.95 Percent Monthly Uptime	Applied to public showcase and investor landing portals.

6. Stage-Gate Execution & Verification Traceability Matrix

Requirement Code	Functional Feature / Subsystem	Target Stage Gate	Mandatory Verification Proof
BR-INV-01	Investor Concierge & Landing Pad	Gate G5 (Week 20)	Multi-agency workflow routing test logs, SEZ API integration sign-off.
BR-INV-02	Trade Mission & MOU Engine	Gate G5 (Week 20)	Matchmaking algorithm validation, MOU lifecycle audit logs.

BR-INV-03	Global Service Showcase	Gate G4 (Week 16)	CDN load testing benchmarks, public API rate-limiting validation.
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FUNCTIONAL REQUIREMENTS DOCUMENT (FRD) / SYSTEM REQUIREMENTS SPECIFICATION (SRS)

NIGERIA SERVICE EXPORT GATEWAY (NATEP / NCMSE)
MODULE 6: PORTAL 4 — GLOBAL PROMOTION, INVESTMENT FACILITATION &
ECOSYSTEM ENABLEMENT DOMAIN

System-of-Record (SoR) Architectural Matrix

Data Domain / Subsystem	Primary SoR Owner	Downstream Consuming Modules	Permitted Operations	Invariant Constraints
Investor Pipeline Store	Module 6 Landing Subsystem	NIPC, NEPZA, FMITI, Module 5	Create investor records, track landing status, log investment commitments.	Investor data must be protected under NDPA encryption standards; no unauthenticated access.
Trade Mission Ledger	Module 6 Delegation Subsystem	Foreign Ministry, NATEP Secretariat	Register delegations, schedule B2B sessions, log conversion outcomes.	B2B sessions can only pair foreign buyers with verified Module 4 exporters.
Public Service Index	Module 6 Showcase Subsystem	External Web, Global Buyers	Index public profiles, serve search	Displays non-confidential data only;

			requests, filter by CPC category.	sensitive commercial details masked.
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Detailed Subsystem Functional Specifications

1. Investor Concierge & Landing Pad Subsystem

Requirement Reference: FR-M6-01 (BRD Reference: BR-INV-01)

1. System Processing Logic & Business Rules:

- i. Foreign enterprises apply for fast-track establishment by submitting corporate documentation, proposed sector focus, and estimated job creation figures.
- ii. The subsystem generates an immutable Investor Application Token (IAT-YYYYMMDD-UUIDv4) and dispatches regulatory payload packages to NIPC, FIRS, and NEPZA via Module 2 API Gateway connectors.
- iii. Investors tracking setup can view real-time clearance status across each agency workspace.

2. User Story:

- i. **As an** International Enterprise Investor,
- ii. **I want** to apply for local business landing and SEZ space through a single digital platform,
- iii. **So that** I can establish operational presence in Nigeria without navigating disparate regulatory offices.

3. Acceptance Criteria (Gherkin Format):

○ i. Scenario 1: Fast-Track Landing Application Submission

- 1. **Given** an authenticated foreign investor submits an application for SEZ business setup,
- 2. **When** the Landing Pad processes the submission,
- 3. **Then** the system generates an Investor Application Token (IAT), logs the transaction in Module 2 WORM storage, and routes clearance payloads to partner agency endpoints within 5 seconds.

2. International Trade Mission & Delegation Matchmaking Subsystem

Requirement Reference: FR-M6-02 (BRD Reference: BR-INV-02)

1. System Processing Logic & Business Rules:

- i. Trade mission coordinators create mission events specifying sector focus, target foreign markets, and delegate quotas.
- ii. An automated matchmaking engine evaluates delegate purchasing requirements against Module 4 verified exporter profiles, generating prioritized B2B interaction schedules.

- iii. MOUs signed during missions are recorded under the mission entity and tracked against subsequent contract awards in Module 3.
2. **User Story:**
- i. **As a** NATEP Trade Mission Coordinator,
 - ii. **I want** the platform to automatically match foreign delegates with verified local service exporters,
 - iii. **So that** bilateral trade meetings yield high-probability commercial contracts.
3. **Acceptance Criteria (Gherkin Format):**
- i. **Scenario 1: Automated B2B Matchmaking Schedule Generation**
-
1. **Given** a trade mission featuring 20 foreign buyers in IT/ITES,
-
2. **When** the coordinator triggers the matchmaking routine,
-
3. **Then** the system pairs each delegate with top-rated, verified IT exporters from Module 4 and generates structured meeting agendas within 3 seconds.

Domain Event JSON Envelopes

1. Investor Landing Application Submitted Event Envelope (event.investor.landing.submitted.v1)

```
{
  "eventId": "evt_1a2b3c4d-5e6f-7a8b-9c0d-1e2f3a4b5c6d",
  "eventType": "event.investor.landing.submitted.v1",
  "eventTimestamp": "2026-09-09T17:15:00.000Z",
  "producer": "natep.portal4.landing-engine",
  "actor": {
    "uid": "UAID-20260901-88990011-2233-4455-6677-8899aabbccdd",
    "natepld": "NT-INV-55102",
    "entityType": "FOREIGN_INVESTOR",
    "verificationStatus": "VERIFIED"
  },
  "data": {
    "investorApplicationId": "IAT-20260909-4412-UUIDv4",
    "companyName": "Global Digital Solutions LLC",
    "originCountry": "USA",
    "targetSectorCpc": "83131",
    "proposedInvestmentUsd": 2500000.00,
    "projectedLocalJobs": 150,
  }
}
```

```
"requestedLandingZone": "ITANA_SEZ_HUB"
},
"context": {
  "correlationId": "corr_33445566-7788-9900-aabb-ccddeeff1122",
  "originatingIp": "192.168.10.45",
  "userAgent": "NATEP-Portal4-InvestorApp/1.0"
}
}
```

Non-Functional & System Level Performance Targets

1. **Global Content Delivery:** Public showcase profiles and investment collateral must be served through a multi-region CDN ensuring sub-800ms load times for users in Europe, North America, and Asia.
2. **Security & Privacy:** Compliance with NDPA 2023 regulations; sensitive financial and strategic setup details encrypted at rest using AES-256-GCM.
3. **Audit Readiness:** Every investor state transition, MOU logging, and trade mission pairing appends a hash-chained audit record to Module 2 WORM storage.

Stage-Gate Delivery & Verification Traceability

Requirement ID	Subsystem Component	Target Delivery Gate	Required Verification Artifact
FR-M6-01	Investor Concierge Subsystem	Stage Gate G5 (Week 20)	End-to-end investor routing test logs, agency integration verification.
FR-M6-02	Trade Mission Matchmaking Engine	Stage Gate G5 (Week 20)	Matchmaking algorithm performance logs, MOU outcome conversion report.

Module 7: BRD/FRD

BUSINESS REQUIREMENTS DOCUMENT (BRD)

NIGERIA SERVICE EXPORT GATEWAY (NATEP / NCMSE)

MODULE 7: PORTAL 5 — FINANCING, OUTCOMES-BASED CAPITAL & ESCROW
CAPITALIZATION DOMAIN

1. Document Control & Baseline Reference

Control Attribute	Specification
Document Title	Business Requirements Document (BRD) — Module 7: Portal 5 (Financing, Outcomes-Based Capital & Escrow Capitalization Domain)
Parent Programme	Nigeria Service Export Gateway (NATEP / NCMSE)
Executive Sponsor	Federal Ministry of Industry, Trade and Investment (FMITI)
Business Owner	NATEP / NCMSE Secretariat
Technical Authority	Architecture, Data, Security & Privacy Authority (ADSPA)
Baseline Source	Integrated Four-Part Terms of Reference (TOR) v1.1, BRD Modules 1–6 Baselines
Precedence	Applicable Law > Executive Decisions > Integrated TOR > BRD Modules 1–6 > BRD Module 7 > FRD/SRS Specifications

2. Executive Summary & Strategic Context

2.1 Strategic Rationale & Mandate

While Modules 3, 4, 5, and 6 establish market matchmaking, talent readiness, national analytics, and global promotion, **Module 7 (Portal 5: Financing & Outcomes-Based Capital)** operates as the capital allocation engine of the Nigeria Service Export Gateway. It bridges the

financing gap for high-growth service exporters, outsourcing institutions, and talent accelerators by blending public investment with private, outcomes-based capital.

Module 7 addresses three primary capital and liquidity bottlenecks:

- **Contract Pre-Financing Constraints:** High collateral requirements prevent verified IT/BPO exporters from securing upfront liquidity against active Module 3 international service contracts.
- **Skilling Capital Arbitrage:** Talent development programs lack scalable, risk-mitigated financing structures tied directly to verified export job placement metrics.
- **Escrow Liquidity Settlement Lag:** Capital delays in international multi-currency escrow releases reduce working capital velocity for scaling export enterprises.

2.2 Alignment with the Shared Foundation (Module 2)

Module 7 operates under **The Single-Platform Mandate** (Module 2). It consumes contract, job placement, and escrow events emitted by Modules 3, 4, and 5 via Module 2's Kafka Event Bus. Disbursement approvals write directly to Module 2's Cryptographic Hash-Chained Audit Trail (WORM storage), while all monetary transactions interface with Module 2's Multi-Currency Settlement Engine.

3. Institutional Context & System-of-Record (SoR) Scope

3.1 Business Capabilities Matrix

Capability Category	Strategic Goal	Core Functional Responsibility
Export Contract Factoring	Pre-Financing & Liquidity	Provide automated invoice and contract pre-financing against verified Module 3 escrow balances.
Outcomes-Based Financing	Capital Alignment	Deploy Social Impact Bonds (SIBs) and Income Share Agreement (ISA) pools for upskilling pipelines tied to verified job placements.
Capital Provider Marketplace	Private Capital Mobilization	Provide DFIs, commercial banks, and private equity firms direct visibility into verified trade yield data.
Automated Portfolio Risk Engine	Risk & Default Mitigation	Continuously evaluate default probabilities across financed contracts using

		historical Module 5 trade velocity data.
--	--	--

3.2 Authoritative Data Ownership Matrix

Data Domain / Entity	Authoritative Subsystem Owner	Permitted Operations	Forbidden Operations
Financing Facility Ledger	Module 7 Capital Subsystem	Originate financing facilities, disburse working capital, record repayments.	Overriding direct escrow balances or mutating underlying Module 3 contracts.
Outcomes & Placement Verification	Module 7 Outcomes Engine	Verify job placement triggers, calculate fund payout ratios, issue yield certificates.	Falsifying placement metrics without verified payroll or escrow remittance proofs.
Financial Risk Telemetry	Module 7 Risk Engine	Read-only aggregation of trade performance, assign credit scores, compute risk provisions.	Modifying live platform fee schedules or changing seller verification tiers directly.

4. Detailed Subsystem Business Requirements

4.1 Export Contract & Invoice Factoring Engine (BR-FIN-01)

- **BR-FIN-01.1 (Contract Liquidity Advances):** The subsystem shall enable verified Module 4 service exporters with active, locked Module 3 escrow contracts to draw up to 80 percent of contract value in pre-financing.
- **BR-FIN-01.2 (Automated Repayment Sweep):** Upon completion of project milestones and escrow release in Module 3, the platform shall automatically sweep principal plus interest into the capital provider's pool before disbursing the net yield to the exporter.

4.2 Outcomes-Based Capital & Upskilling Bonds (BR-FIN-02)

- **BR-FIN-02.1 (Outcomes-Tied Capital Tranches):** Training providers and talent accelerators shall receive capital disbursements contingent upon verified export job placement metrics logged across Modules 3 and 4.
- **BR-FIN-02.2 (Verification Triggers):** Capital release triggers require cryptographic proof

of contract execution, employer verification, and foreign currency salary remittance.

4.3 Capital Provider & Investor Gateway (BR-FIN-03)

- **BR-FIN-03.1 (Institutional Investor Dashboards):** Participating DFIs, local banks, and impact investors shall access real-time dashboards displaying portfolio yield, default rates, and active capital utilization.

5. Non-Functional, Security & Performance Requirements

Key Performance Indicator	Target Threshold	Operational Condition
Factoring Eligibility Evaluation	Under 3 Seconds	Automated credit scoring and contract valuation latency.
Disbursement Execution	Under 10 Seconds	Synchronous transaction processing via Module 2 Settlement Engine.
System Uptime	99.95 Percent Monthly Uptime	Applied to financial portal endpoints and underwriting models.

6. Stage-Gate Execution & Verification Traceability Matrix

Requirement Code	Functional Feature / Subsystem	Target Stage Gate	Mandatory Verification Proof
BR-FIN-01	Export Contract Factoring Engine	Gate G5 (Week 20)	Automated repayment sweep test logs, escrow lockup integration tests.
BR-FIN-02	Outcomes-Based Capital Engine	Gate G5 (Week 20)	Placement verification trigger logs, outcomes-based tranche payout reports.

BR-FIN-03	Investor Gateway & Risk Engine	Gate G5 (Week 20)	Portfolio risk calculation performance logs, investor RBAC validation.
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FUNCTIONAL REQUIREMENTS DOCUMENT (FRD) / SYSTEM REQUIREMENTS SPECIFICATION (SRS)

NIGERIA SERVICE EXPORT GATEWAY (NATEP / NCMSE)
MODULE 7: PORTAL 5 — FINANCING, OUTCOMES-BASED CAPITAL & ESCROW
CAPITALIZATION DOMAIN

System-of-Record (SoR) Architectural Matrix

Data Domain / Subsystem	Primary SoR Owner	Downstream Consuming Modules	Permitted Operations	Invariant Constraints
Capital Facility Store	Module 7 Financing Subsystem	FMITI, CBN, Module 5, Capital Providers	Originate loan facilities, compute yield, track repayment state.	Underwriting models must strictly utilize verified trade history from Module 5.
Outcomes Metric Store	Module 7 Outcomes Engine	NATEP Secretariat, Investors	Record placement proofs, trigger capital payouts, log bond performance.	Payout triggers require two-party cryptographic verification from Module 2.
Risk & Yield Ledger	Module 7 Risk Subsystem	Institutional Investors	Index default rates, update credit scores,	Strictly read-only data consumption;

			project capital yield.	cannot alter underlying operational debt states.
--	--	--	------------------------	--

Detailed Subsystem Functional Specifications

1. Export Contract & Invoice Factoring Subsystem

Requirement Reference: FR-M7-01 (BRD Reference: BR-FIN-01)

1. System Processing Logic & Business Rules:

- i. A verified exporter submits an active Module 3 escrow contract (minimum value 10,000 USD equivalent) for pre-financing.
- ii. The Risk Engine verifies contract validity, buyer reputation score in Module 3, and seller delivery history in Module 4.
- iii. Upon passing credit scoring, the subsystem generates an advance offer of up to 80 percent of contract value.
- iv. Acceptance triggers a dynamic escrow lien in Module 3, routing milestone payouts directly to the financing settlement pool until principal and interest are satisfied.

2. User Story:

- i. **As a** Service Exporter with a locked 50,000 USD contract,
- ii. **I want** to obtain immediate working capital pre-financing against my contract,
- iii. **So that** I can fund talent operations and project execution without operational delay.

3. Acceptance Criteria (Gherkin Format):

- i. **Scenario 1: Automated Pre-Financing Facility Approval**

■

1. **Given** an exporter with a locked 50,000 USD escrow contract applies for pre-financing,

■

2. **When** the Factoring Engine evaluates the contract and buyer trust score,

■

3. **Then** the system approves a 40,000 USD advance, places an automated lien on the Module 3 escrow contract, and logs the loan facility in Module 2 WORM storage within 3 seconds.

2. Outcomes-Based Capital & Upskilling Bond Subsystem

Requirement Reference: FR-M7-02 (BRD Reference: BR-FIN-02)

1. System Processing Logic & Business Rules:

- i. Capital pools are created by DFIs and private investors to fund talent upskilling initiatives.
- ii. Training providers receive baseline capital (e.g., 30 percent), with the remaining 70 percent released in tranches contingent upon verified export job placements.
- iii. The Outcomes Engine continuously queries Modules 3 and 4 for placement events matching candidate IDs, verifying 90-day retention and salary thresholds before

authorizing payout tranches.

2. User Story:

- i. **As an** Impact Capital Investor,
- ii. **I want** my investment capital disbursed to training providers only when candidates achieve verified export job placements,
- iii. **So that** capital allocation is directly tied to measurable job creation outcomes.

3. Acceptance Criteria (Gherkin Format):

- i. **Scenario 1: Verified Job Placement Outcome Payout**

■

1. **Given** a training provider achieves 100 verified talent placements in Module 3 contracts with active 90-day retention,

■

2. **When** the Outcomes Engine completes cross-verification against Module 4 job registers,

■

3. **Then** the system releases the corresponding 70 percent performance capital tranche and issues a cryptographic Verification Certificate within 5 seconds.

Domain Event JSON Envelopes

1. Contract Factoring Disbursed Event Envelope

(event.financing.factoring.disbursed.v1)

```
{
  "eventId": "evt_9f8e7d6c-5b4a-3f2e-1d0c-9b8a7f6e5d4c",
  "eventType": "event.financing.factoring.disbursed.v1",
  "eventTimestamp": "2026-09-09T17:20:00.000Z",
  "producer": "natep.portal5.factoring-engine",
  "actor": {
    "uid": "UAID-20260715-99887766-5544-3322-1100-aabbccddeeff",
    "natepld": "NT-EXP-44019",
    "entityType": "SERVICE_EXPORTER",
    "verificationStatus": "VERIFIED"
  },
  "data": {
    "facilityId": "FAC-20260909-8812-UUIDv4",
    "sourceContractId": "CTR-20260820-9901",
    "advanceAmountUsd": 40000.00,
    "totalContractValueUsd": 50000.00,
    "interestRatePercent": 2.50,
```

```

    "escrowLienStatus": "ACTIVE_LIEN_ATTACHED",
    "settlementPoolAccountId": "ACC-POOL-FIN-0091"
  },
  "context": {
    "correlationId": "corr_11223344-8899-0011-2233-445566778899",
    "originatingIp": "10.0.6.12",
    "userAgent": "NATEP-Portal5-Financing/1.0"
  }
}

```

Non-Functional & System Level Performance Targets

1. **Transaction & Liquidity Security:** Mandatory double-entry ledger bookkeeping for all financing facilities, backed by multi-signature transaction authorization via Module 2's Vault framework.
2. **Data Encryption:** Financial records, credit assessments, and investor profile details encrypted at rest using AES-256-GCM.
3. **Audit Trail Integrity:** All credit decisions, lien attachments, outcomes-based payout authorizations, and interest sweeps append immutable records to Module 2 WORM storage.

Stage-Gate Delivery & Verification Traceability

Requirement ID	Subsystem Component	Target Delivery Gate	Required Verification Artifact
FR-M7-01	Export Contract Factoring Engine	Stage Gate G5 (Week 20)	End-to-end factoring lifecycle test logs, escrow lien attachment logs.
FR-M7-02	Outcomes-Based Capital Engine	Stage Gate G5 (Week 20)	Placement trigger verification reports, outcomes payout audit traces.

Module 8: BRD/FRD

BUSINESS REQUIREMENTS DOCUMENT (BRD)

NIGERIA SERVICE EXPORT GATEWAY (NATEP / NCMSE)

MODULE 8: PORTAL 6 — DISPUTE RESOLUTION, Cross-Border CONTRACT
ARBITRATION & COMPLIANCE ENFORCEMENT DOMAIN

1. Document Control & Baseline Reference

Control Attribute	Specification
Document Title	Business Requirements Document (BRD) — Module 8: Portal 6 (Dispute Resolution, Cross-Border Contract Arbitration & Compliance Enforcement Domain)
Parent Programme	Nigeria Service Export Gateway (NATEP / NCMSE)
Executive Sponsor	Federal Ministry of Industry, Trade and Investment (FMITI)
Business Owner	NATEP / NCMSE Secretariat
Technical Authority	Architecture, Data, Security & Privacy Authority (ADSPA)
Baseline Source	Integrated Four-Part Terms of Reference (TOR) v1.1, BRD Modules 1–7 Baselines
Precedence	Applicable Law > Executive Decisions > Integrated TOR > BRD Modules 1–7 > BRD Module 8 > FRD/SRS Specifications

2. Executive Summary & Strategic Context

2.1 Strategic Rationale & Mandate

While Modules 3 through 7 handle trade matching, supplier readiness, analytics, global promotion, and contract financing, **Module 8 (Portal 6: Dispute Resolution & Arbitration)** serves as the legal protection and trust enforcement domain of the Nigeria Service Export

Gateway. It provides a legally binding, multi-jurisdictional digital dispute resolution framework to resolve commercial conflicts, escrow holdbacks, SLA breaches, and IP infringement claims between international buyers and Nigerian service exporters.

Module 8 addresses three critical trust barriers:

- **Cross-Border Enforcement Friction:** Litigating cross-border service trade disputes in traditional courts is slow and cost-prohibitive for SMEs and international enterprise off-takers.
- **Escrow Lockup Stalemates:** Unresolved buyer-seller disagreements leave Module 3 escrow capital frozen indefinitely without an automated, legally recognized mediation mechanism.
- **Non-Performance & Quality Arbitrage:** Absence of structured online dispute resolution (ODR) protocols damages Nigeria's national brand equity when delivery expectations fail.

2.2 Alignment with the Shared Foundation (Module 2)

Module 8 operates strictly under **The Single-Platform Mandate**. It ingests contract terms, milestone submissions, and escrow transaction states emitted by Module 3. All case evidence packages, mediator decisions, and final arbitral awards write directly to Module 2's Cryptographic Hash-Chained Audit Trail (WORM storage) and Evidence Vault.

3. Institutional Context & System-of-Record (SoR) Scope

3.1 Business Capabilities Matrix

Capability Category	Strategic Goal	Core Functional Responsibility
Online Dispute Resolution (ODR)	Fast-Track Conflict Resolution	Automated mediation workflows, structured buyer-seller negotiations, and automated settlement agreements.
Legally Binding Arbitration Engine	Cross-Border Legal Enforcement	Digital tribunal workspace for accredited arbitrators under UNCITRAL/NICArb rules with enforceable awards.
Escrow Freeze & Lien Management	Neutral Liquidity Control	Automated freezing, partial release, or full clawback of Module 3 escrow balances based on arbitral rulings.
Exporter Sanction &	Ecosystem Quality Control	Automated updating of

Integrity Index		exporter trust scores and suspension routines in Module 4 upon confirmed breach of contract.
------------------------	--	--

3.2 Authoritative Data Ownership Matrix

Data Domain / Entity	Authoritative Subsystem Owner	Permitted Operations	Forbidden Operations
Dispute Case Dossier	Module 8 Dispute Subsystem	Create case file, ingest contract telemetry, log evidence, track resolution status.	Mutating underlying Module 3 contracts or modifying historic escrow receipts.
Arbitral Awards & Decisions	Module 8 Tribunal Engine	Issue legally binding rulings, attach cryptographic signatures, authorize escrow release directives.	Falsifying tribunal decisions or altering signed arbitral awards after issuance.
Compliance & Sanction Register	Module 8 Sanctions Engine	Issue compliance warnings, apply trust score penalties, trigger Module 4 exporter profile suspensions.	Overriding direct statutory tax registration states maintained in Module 4.

4. Detailed Subsystem Business Requirements

4.1 Online Mediation & Early Settlement (BR-DSP-01)

- **BR-DSP-01.1 (Structured Direct Negotiation):** The system shall provide a 14-day automated negotiation window for buyers and sellers to reach mutual escrow settlement prior to formal arbitration.
- **BR-DSP-01.2 (AI-Assisted Mediation Recommendations):** The platform shall analyze dispute grounds against contract SLAs and historical Module 5 resolution data to propose non-binding settlement splits.

4.2 Cross-Border Digital Arbitration Tribunal (BR-DSP-02)

- **BR-DSP-02.1 (Accredited Arbitrator Portal):** Secure workspace for accredited international arbitrators (e.g., NICArb, ICC, LCIA) to review evidence packages, conduct virtual hearings, and issue binding awards.
- **BR-DSP-02.2 (Automated Escrow Directives):** Arbitral awards rendered in Portal 6 automatically execute corresponding fund distributions or clawbacks in Module 3's escrow engine without manual intervention.

4.3 Fraud Prevention & Vendor Sanctions (BR-DSP-03)

- **BR-DSP-03.1 (Sanctions Enforcement):** Confirmed fraudulent claims, intentional breach, or failure to abide by arbitral awards shall trigger immediate profile suspension across Module 4, Module 6 directories, and Module 7 financing eligibility.

5. Non-Functional, Security & Performance Requirements

Key Performance Indicator	Target Threshold	Operational Condition
Case Filing Processing Latency	Under 2 Seconds	Case creation, escrow hold placement, and notice dispatch.
Evidence Encryption	AES-256-GCM at Rest	End-to-end cryptographic protection of submitted commercial secrets.
System Uptime	99.95 Percent Monthly Uptime	Applied to tribunal workspaces, ODR portals, and arbitration endpoints.

6. Stage-Gate Execution & Verification Traceability Matrix

Requirement Code	Functional Feature / Subsystem	Target Stage Gate	Mandatory Verification Proof
BR-DSP-01	Online Dispute Resolution (ODR) Engine	Gate G5 (Week 20)	Automated negotiation workflows, AI recommendation precision logs.
BR-DSP-02	Digital Arbitration Tribunal Engine	Gate G5 (Week 20)	Arbitrator RBAC validation,

			automated escrow directive execution logs.
BR-DSP-03	Sanctions & Trust Enforcement Engine	Gate G5 (Week 20)	Automated suspension test suites, Module 4 trust score penalty verification.

FUNCTIONAL REQUIREMENTS DOCUMENT (FRD) / SYSTEM REQUIREMENTS SPECIFICATION (SRS)

NIGERIA SERVICE EXPORT GATEWAY (NATEP / NCMSE)

MODULE 8: PORTAL 6 — DISPUTE RESOLUTION, CROSS-BORDER CONTRACT

ARBITRATION & COMPLIANCE ENFORCEMENT DOMAIN

System-of-Record (SoR) Architectural Matrix

Data Domain / Subsystem	Primary SoR Owner	Downstream Consuming Modules	Permitted Operations	Invariant Constraints
Dispute Registry Store	Module 8 ODR Subsystem	Module 3, Module 4, Module 5	Log dispute tickets, record settlement terms, track resolution timers.	Disputes can only be filed against active or milestone-locked Module 3 contracts.
Tribunal Award Store	Module 8 Arbitration Engine	Module 3 Escrow, Module 2 Audit Vault	Ingest evidence, render awards, issue cryptographically signed	Arbitral decisions are final; cannot be overridden without formal court

			execution orders.	annulment orders.
Sanctions & Compliance Ledger	Module 8 Sanctions Engine	Module 4, Module 6, Module 7	Assign penalty points, suspend exporter credentials, issue public breach notices.	Sanction actions require multi-party compliance sign-off or validated tribunal awards.

Detailed Subsystem Functional Specifications

1. Online Dispute Resolution (ODR) & Mediation Subsystem

Requirement Reference: FR-M8-01 (BRD Reference: BR-DSP-01)

1. System Processing Logic & Business Rules:

- i. A buyer or seller opens a dispute ticket on an active Module 3 contract milestone.
- ii. The subsystem immediately emits an event.dispute.opened.v1 event, placing a temporary hold on disputed Module 3 escrow funds.
- iii. A 14-day ODR negotiation window is initialized. If parties agree to a settlement, the subsystem generates an immutable Settlement Agreement Token (SAT-YYYYMMDD-UUIDv4), releasing escrow funds per the agreed ratio.
- iv. If no agreement is reached within 14 days, the dispute escalates automatically to formal digital arbitration.

2. User Story:

- i. **As an** International Buyer experiencing non-delivery of a software milestone,
- ii. **I want** to file a structured dispute and freeze the associated escrow payout,
- iii. **So that** my capital is protected while we negotiate or arbitrate a solution.

3. Acceptance Criteria (Gherkin Format):

- i. **Scenario 1: Automated Dispute Filing and Escrow Hold**
 -
 - 1. **Given** an international buyer files a dispute against a 15,000 USD contract milestone,
 -
 - 2. **When** the ODR Subsystem processes the dispute ticket,
 -
 - 3. **Then** the system freezes the 15,000 USD escrow payout in Module 3, notifies the exporter, and opens the 14-day mediation workspace within 2 seconds.

2. Digital Cross-Border Arbitration Tribunal Subsystem

Requirement Reference: FR-M8-02 (BRD Reference: BR-DSP-02)

1. System Processing Logic & Business Rules:

- i. Escalated disputes are assigned to an accredited sole arbitrator selected via a random neutral rotation algorithm.
- ii. Both parties submit evidence packages encrypted with public keys in Module 2's Evidence Vault.
- iii. The arbitrator reviews evidence and issues a legally binding Arbitral Award Token (AAT-YYYYMMDD-UUIDv4).
- iv. The subsystem dispatches an execution command to Module 3, releasing escrow funds directly to the winning party's account.

2. User Story:

- i. **As an** Accredited Arbitrator,
- ii. **I want to** review cryptographically verified contract evidence and render a binding arbitral award,
- iii. **So that** cross-border commercial service disputes are legally resolved without traditional litigation delays.

3. Acceptance Criteria (Gherkin Format):

- i. **Scenario 1: Binding Award Execution and Escrow Settlement**
 -
 - 1. **Given** an arbitrator issues an award favoring the buyer for an 80 percent refund on a 20,000 USD dispute,
 -
 - 2. **When** the tribunal engine registers the signed Arbitral Award Token (AAT),
 -
 - 3. **Then** the system triggers Module 3 to return 16,000 USD to the buyer and disburse 4,000 USD to the exporter within 5 seconds.

Domain Event JSON Envelopes

1. Dispute Opened Event Envelope (event.dispute.opened.v1)

```
{
  "eventId": "evt_8a7b6c5d-4e3f-2a1b-0c9d-8e7f6a5b4c3d",
  "eventType": "event.dispute.opened.v1",
  "eventTimestamp": "2026-09-09T17:25:00.000Z",
  "producer": "natep.portal6.odr-engine",
  "actor": {
    "uuid": "UAID-20260510-11223344-5566-7788-9900-aabbccddeeff",
    "natepId": "NT-BUY-88120",
    "entityType": "FOREIGN_BUYER",
    "verificationStatus": "VERIFIED"
  },
}
```



```

"data": {
  "disputeId": "DSP-20260909-1102-UUIDv4",
  "contractId": "CTR-20260815-4421",
  "disputedMilestoneId": "MS-03",
  "disputedAmountUsd": 18500.00,
  "disputeReasonCode": "SLA_BREACH_QUALITY_NON_CONFORMANCE",
  "escrowHoldStatus": "FUNDS_FROZEN_IN_ESCROW"
},
"context": {
  "correlationId": "corr_99001122-3344-5566-7788-99aabbccddeeff",
  "originatingIp": "192.168.12.80",
  "userAgent": "NATEP-Portal6-ODR/1.0"
}
}

```

Non-Functional & System Level Performance Targets

1. **Legal Validity & Immutability:** All arbitration records, evidence hashes, and arbitral awards must be cryptographically hashed and anchored to Module 2's Audit Vault in compliance with the UNCITRAL Model Law on International Commercial Arbitration.
2. **Security & Confidentiality:** Strict role-based access control (RBAC); dispute evidence accessible only by assigned tribunal arbitrators, case parties, and legal clerks.
3. **Audit Trail Completeness:** Every mediation offer, counter-offer, arbitrator assignment, and award execution appends an immutable record to Module 2 WORM storage.

Stage-Gate Delivery & Verification Traceability

Requirement ID	Subsystem Component	Target Delivery Gate	Required Verification Artifact
FR-M8-01	ODR & Mediation Engine	Stage Gate G5 (Week 20)	Automated escrow freeze logs, mediation settlement execution test suites.
FR-M8-02	Digital Arbitration Tribunal	Stage Gate G5 (Week 20)	Arbitral award generation tests, automated Module 3 escrow release execution logs.

Module 4: Dump

Business Requirements Document (BRD) / Functional Requirements Document (FRD)
MODULE 4: PORTAL 2 — SUPPLY, READINESS, CERTIFICATION & INCENTIVES DOMAIN
Document Control & Baseline Alignment

1. **Document Title:** Business Requirements Document (BRD) / Functional Requirements Document (FRD) — Module 4: Portal 2 (Supply, Readiness, Certification & Incentives4Domain)
2. **Parent Programme:** Nigeria Service Export Gateway (NATEP / NCMSE)
3. **Baseline Alignment:** Integrated Four-Part TOR v1.1, BRD Module 1 Baseline, BRD Module 2 (Shared Platform Foundation Services) Baseline, and BRD Module 3 Baseline
4. **Formatting Policy:** Plain English specification standard. Numbers and Roman numerals are used strictly across all sections to maintain non-bulleted structure. Standard text is used exclusively for all terms, units, and metrics—avoiding mathematical symbols, formulas, or special characters.

System-of-Record (SoR) Architectural Matrix — Portal 2

Data Domain / Subsystem	Primary SoR Owner	Downstream Consuming Modules	Permitted Operations	Invariant Constraints
Exporter Profiles & Readiness	Portal 2 Exporter Registry	Portals 1, 3 & Observatory	Profile creation, readiness score computation, capability indexing.	Zero unverified corporate or identity claims permitted to bid on international RFPs.
Certification & Regulatory Validation	Portal 2 Regulatory Subsystem	Portals 1, 3 & Observatory	Credential validation, regulatory clearing, certificate issuance.	Direct linkage required with sector regulators (e.g., NITDA, COREN, NMDPRA) or verified evidence uploads.
Export Readiness & Assessment	Portal 2 Assessment Engine	Portal 1 & Observatory	Gap analysis execution, readiness	Automatic recalculation of readiness

Engine			tiering, diagnostic reporting.	tiers upon profile or certification updates.
Incentives & Trade Facilitation	Portal 2 Incentives Engine	Observatory & External Regulators	Grant application, tax incentive tracking, trade rebate processing.	Incentives restricted strictly to verified exporters meeting minimum readiness thresholds.
Exporter Trust & Badging Engine	Portal 2 Badging Subsystem	Portal 1, Observatory & Shared Foundation	Readiness badge issuance, entitlement unlocking, completion tracking.	Entitlements strictly bound to verified completion thresholds; manual overrides logged to Module 2 Evidence Vault.

Detailed Subsystem Functional Specifications

1. Exporter Onboarding & Profile Readiness Subsystem

Requirement Reference: FR-M4-01 (BRD Reference: BR-P2-01 through BR-P2-04)

1. System Processing Logic & Business Rules:

- i. Account registration is open and unblocked for Nigerian service exporters, talent, and professional agencies, requiring basic identity input and corporate details.
- ii. Exporters build comprehensive capability portfolios, including service categories (mapped to UN CPC codes), delivery capacity, past trade history, and team qualifications.
- iii. Profile completeness and verification depth automatically compute the Exporter's Profile Completion Percentage and assign symmetric trust badges:

■

1. **Registered Talent / Entity (1 percent to 39 percent completion):** Unlocks RFP browsing, capability portfolio creation, and access to market access guides.

■

2. **NATEP Verified Exporter (40 percent to 79 percent completion):** Unlocked via verified CAC registration, Tax Identification Number (TIN) validation, and corporate bank account verification. Permits bidding on open RFPs up to 50,000 USD and receiving automated match alerts.

■

3. **Top-Rated Export Partner (80 percent or higher completion):** Unlocked via sector regulatory clearance, verified trade history, and active escrow performance. Unlocks priority placement in AI matchmaking, direct enterprise invitations, and lower platform escrow fees.
 - iv. Every exporter dashboard displays a dynamic Next Unlock Prompt showing current permissions, completion percentage, and specific verification steps required to unlock the next trust tier.
2. **User Story:**
 - i. **As a** Nigerian Service Exporter or Talent Candidate,
 - ii. **I want** to build a verified trade profile and track my readiness progression through clear badges and completion metrics,
 - iii. **So that** I can establish credibility with international buyers and unlock access to high-value global export RFPs.

3. **Acceptance Criteria (Gherkin Format):**

- i. **Scenario 1: Progressive Exporter Onboarding and Badging**

■

1. **Given** a Nigerian service business registers and validates its Corporate Affairs Commission (CAC) registration number and Tax Identification Number,

■

2. **When** the verification engine confirms active standing via the Shared Regulatory Verification Engine,

■

3. **Then** the system assigns the NATEP Verified Exporter badge, updates profile completion above 40 percent, and enables bidding on RFPs up to 50,000 USD.
- ii. **Scenario 2: Display of Next Unlock Prompts**

■

1. **Given** an exporter profile achieves 60 percent completion,

■

2. **When** the user accesses their dashboard,

■

3. **Then** the system displays current active capabilities ("*Can bid on RFPs up to 50,000 USD*") and highlights the next milestone ("*Upload Sector Regulatory Certification to reach Top-Rated Exporter status and unlock Enterprise RFPs*").

2. Sectoral Regulatory Validation & Certification Subsystem

Requirement Reference: FR-M4-02 (BRD Reference: BR-P2-05 through BR-P2-08)

1. **System Processing Logic & Business Rules:**

- i. Exporters submit sector-specific professional certifications and regulatory clearances

(e.g., NITDA/NDPC for IT, COREN for engineering, CIBN for financial services, or international standards like ISO 27001).

- ii. Credential verification utilizes a hybrid model: automated verification against open regulatory databases where available, or fallback AI Document Verification for uploaded certificate documents.
- iii. Verified certifications are cryptographically hashed and indexed in Module 2's Evidence Vault to prevent credential reuse or fraudulent claims.
- iv. Expired or revoked certifications trigger an automated profile score degradation and notify the exporter to renew credentials within 30 days before trust badge downgrade occurs.

2. User Story:

- i. **As a** Professional Service Exporter,
- ii. **I want** my sector licenses and professional certifications verified and badged automatically,
- iii. **So that** foreign buyers have immediate, cryptographic proof of my regulatory compliance and technical competence.

3. Acceptance Criteria (Gherkin Format):

○ i. Scenario 1: Hybrid Certification Verification

■

1. **Given** an engineering firm uploads a COREN practice license,

■

2. **When** the system processes the document through the AI Verification Engine or regulatory API,

■

3. **Then** the credential status updates to VERIFIED, links to the exporter's public trust badge, and registers a cryptographic record in the Evidence Vault.

○ ii. Scenario 2: Expiration and Automated Badge Downgrade Warning

■

1. **Given** an exporter's professional certification is 30 days away from expiration,

■

2. **When** the daily cron review executes,

■

3. **Then** the system dispatches an automated renewal warning notification and highlights the pending risk on the exporter's dashboard.

3. Export Readiness Assessment & Diagnostic Subsystem

Requirement Reference: FR-M4-03 (BRD Reference: BR-P2-09 through BR-P2-12)

1. Explicit Operational Readiness Tier Hierarchy Definition:

- i. The platform enforces an explicit, ordinal three-tier hierarchy for operational export readiness.
- ii. Hierarchy levels are ranked strictly from lowest (Level 1) to highest (Level 3) capability:

■

1. Level 1 — Export Capacity Building (Lowest Readiness Level):

- *Target Entity:* Emerging entities and early-stage talent needing foundational

skill-building, basic certification support, and internal quality management setup.

- **System Rule:** Exporters at Level 1 are restricted from bidding on high-value or complex enterprise RFPs. They are automatically routed to training pathways and capacity-building resources.

■

2. **Level 2 — Export Emerging (Intermediate Readiness Level):**

- **Target Entity:** Growing businesses possessing basic operational capacity that require targeted market-access diagnostic actions and minor compliance support to scale.
- **System Rule:** Exporters at Level 2 can bid on standard international RFPs (up to baseline system thresholds) and receive automated gap-analysis action plans to reach Level 3.

■

3. **Level 3 — Export Ready (Highest Readiness Level):**

- **Target Entity:** Fully qualified, legally compliant, and operationally mature exporters capable of executing complex, high-value international contracts immediately.
- **System Rule:** Exporters at Level 3 receive maximum weight multipliers in Module 3's AI Matchmaking Engine, priority placement in enterprise search listings, and full eligibility for government export grants and incentives.

- **iii. Progression Direction:** Hierarchy advancement moves strictly upward from Level 1 to Level 2 to Level 3. Downgrades occur dynamically if verified credentials expire or compliance checks fail.

2. **Evaluation Parameters & Scoring Matrix:**

- i. The Assessment Engine evaluates readiness across five weighted parameters:

■

1. **Export Capacity (25 percent):** Operational scale, service volume, and resource bandwidth.

■

2. **Financial Stability (20 percent):** Cash flow metrics, corporate banking verification, and audit readiness.

■

3. **Quality Management Systems (20 percent):** Internal quality controls, ISO standards, and sector certifications.

■

4. **Cross-Border Delivery Experience (20 percent):** Verified international trade history and escrow contract fulfillments.

■

5. **Legal and IP Protection (15 percent):** Active regulatory clearances, contracts management, and IP safeguards.

- ii. Total computed diagnostic score (0 to 100 percent) maps directly to the operational hierarchy:

■

1. **Score under 50 percent:** Assigned to **Level 1 (Export Capacity Building)**.

■

2. **Score 50 percent to 79 percent:** Assigned to **Level 2 (Export Emerging)**.

■

3. **Score 80 percent and above:** Assigned to **Level 3 (Export Ready)**.

3. **User Story:**

- i. **As an** Emerging Nigerian Service Business,
- ii. **I want** to complete an automated export readiness diagnostic,
- iii. **So that** I can see my exact hierarchy placement and receive a clear action plan to ascend to Export Ready status.

4. **Acceptance Criteria (Gherkin Format):**

- i. **Scenario 1: Hierarchy Tier Assignment via Diagnostic Score**

■

1. **Given** an exporter completes the readiness assessment and achieves a calculated total score of 85 percent across all five evaluation parameters,

■

2. **When** the diagnostic engine processes the score,

■

3. **Then** the system assigns the exporter to **Level 3 (Export Ready)**, applies the maximum match multiplier in Module 3, and updates the public profile badge.

- ii. **Scenario 2: Automatic Action Plan for Level 1 & 2 Entities**

■

1. **Given** an exporter scores 45 percent and is assigned to **Level 1 (Export Capacity Building)**,

■

2. **When** the assessment report renders,

■

3. **Then** the system generates a tailored remediation dashboard highlighting missing parameters (e.g., missing COREN/NITDA certification or lack of quality management systems) needed to advance to Level 2.

4. **Export Incentives & Trade Facilitation Subsystem**

Requirement Reference: FR-M4-04 (BRD Reference: BR-P2-13 through BR-P2-16)

1. **System Processing Logic & Business Rules:**

- i. Verified exporters (NATEP Verified Exporter status or higher) gain access to apply for government-backed trade facilitation incentives, export expansion grants, tax credits, and subsidized trade missions.
- ii. Application processing cross-references platform trade history, verified export earnings logged via Module 3 escrow channels, and compliance status.
- iii. Incentive approvals follow a two-tier review workflow: automated pre-qualification by the platform rules engine followed by final review sign-off from designated NATEP / NEPC Administrators.
- iv. Granted incentives and tax credit certificates are digitally issued, logged in the Evidence Vault, and tracked against the exporter's lifetime trade performance.

2. **User Story:**

- i. **As a** Verified Service Exporter with active export earnings,
- ii. **I want** to apply for government export expansion incentives and tax rebates directly

- through the portal,
- iii. **So that** I can reduce operational costs and reinvest in expanding my international trade capacity.
- 3. **Acceptance Criteria (Gherkin Format):**
 - i. **Scenario 1: Pre-Qualification for Export Grants based on Verified Trade Volume**
 -
 - 1. **Given** an exporter achieves over 50,000 USD in verified export volume processed through platform escrow,
 -
 - 2. **When** the exporter submits an application for the Service Export Expansion Grant,
 -
 - 3. **Then** the system automatically pre-qualifies the application, attaches verified escrow receipts as evidence, and routes it to the Administrator review queue.

Non-Functional & Operational Performance Targets — Portal 2

1. **Performance & Scalability Metrics:**
 - i. **Readiness Score Computation Latency:** Real-time computation and display of readiness diagnostics within 2 seconds of assessment submission.
 - ii. **Certification AI Verification Processing Speed:** Document extraction, tamper check, and status update completed within 15 seconds of upload.
 - iii. **Profile Entitlement Calculation Speed:** Real-time computation and UI update of Profile Completion Percentage and feature permissions within 500 milliseconds of attribute submission.
2. **Security & Privacy Standards:**
 - i. Financial statements and internal corporate documentation submitted for readiness evaluation must be encrypted at rest using AES-256-GCM.
 - ii. Exporter compliance data and trade history disclosures are shared with third parties strictly under active NDPA 2023 consent grants.

Stage-Gate Delivery & Verification Traceability

Requirement ID	Subsystem Component	Target Delivery Gate	Required Verification Artifact
FR-M4-01	Exporter Onboarding & Profile Readiness	Stage Gate G3 (Week 10)	Unblocked registration logs, readiness badging evaluation scripts, and profile unlock UI test sign-off.
FR-M4-02	Sectoral Regulatory	Stage Gate G3	Regulatory verification API

	Validation	(Week 10)	integration logs, AI document scanner benchmarks, and Evidence Vault registration checks.
FR-M4-03	Export Readiness Diagnostic Engine	Stage Gate G4 (Week 16)	Diagnostic scoring test execution scripts, gap analysis generation checks, and matchmaking weight integration logs.
FR-M4-04	Export Incentives & Trade Facilitation	Stage Gate G4 (Week 16)	Incentive application workflow UAT sign-off, automated pre-qualification rule verification, and grant ledger audit logs.

Module 3: Dump

Business Requirements Document (BRD) / Functional Requirements Document (FRD)

MODULE 3: PORTAL 1 — DEMAND & MARKET ACCESS DOMAIN

Document Control & Baseline Alignment

1. **Document Title:** Business Requirements Document (BRD) / Functional Requirements Document (FRD) — Module 3: Portal 1 (Demand & Market Access Domain)
2. **Parent Programme:** Nigeria Service Export Gateway (NATEP / NCMSE)
3. **Baseline Alignment:** Integrated Four-Part TOR v1.1, BRD Module 1 Baseline, and BRD Module 2 (Shared Platform Foundation Services) Baseline
4. **Formatting Policy:** Plain English specification standard. Numbers and Roman numerals are used strictly across all sections to maintain non-bulleted structure.

System-of-Record (SoR) Architectural Matrix — Portal 1

Data Domain / Subsystem	Primary SoR Owner	Downstream Consuming Modules	Permitted Operations	Invariant Constraints
Buyer Profiles & Sourcing Scope	Portal 1 Buyer Registry	Portals 2, 3 & Observatory	Profile creation, sourcing criteria definition, JIT verification status updates.	Zero arbitrary buyer blocks on registration; action-based JIT gating enforced before opportunity publication.
Service Opportunities & RFPs	Portal 1 Opportunity Engine	Portals 2, 3 & Observatory	Demand posting, taxonomy tagging, document attaching, status transitions.	Mandatory mapping to WTO/GATS Modes of Supply and UN CPC taxonomy codes.
Matchmaking & Matching Engine	Portal 1 Matching Engine	Portals 2, 3 & Observatory	Profile indexing, score computation, notification dispatches, shortlist	Zero bias in match score computation; relies strictly on verified capabilities

			generation.	and criteria.
Opportunity Applications & Bids	Portal 1 Bidding Engine	Portal 2 & Observatory	Proposal submission, evidence linking, proposal status tracking.	Proposals blocked without active actor verification and valid consent grants.
Market Intelligence & Access Insights	Portal 1 Market Insights Subsystem	Observatory & External Buyers	Trade barrier publishing, regulatory insights, market trend analytics.	Public access restricted to standardized, aggregated, and non-confidential data.

Detailed Subsystem Functional Specifications

1. Global Buyer Onboarding & Hybrid JIT Verification Subsystem

Requirement Reference: FR-M3-01 (BRD Reference: BR-P1-01 through BR-P1-04)

1. System Processing Logic & Business Rules:

- i. Account registration is open and completely unblocked, requiring only corporate email domain validation (user@company.com) and basic profile setup.
- ii. High-value operational actions (e.g., publishing an RFP, sending direct sourcing invitations, or viewing unmasked exporter profiles) trigger a Just-In-Time (JIT) KYB verification step.
- iii. JIT verification executes through a two-stage hybrid evaluation pipeline:

■

1. **Primary Route (Open Public Registries):** The system queries integrated free/open-source global registries (such as GLEIF LEI API, UK Companies House API, EU VIES VAT API) using the buyer's provided business registration or tax number. Successful matches immediately assign the profile to an elevated Trust Tier.

■

2. **Secondary Fallback Route (AI Document Verification):** If no record is found in open APIs, the buyer is prompted to upload local incorporation certificates, tax filings, or trade licenses. The AI Verification Engine performs OCR extraction, visual tamper analysis, and domain cross-matching to grant tier approval.
- iv. The platform assigns neutral, non-indicting Trust Tiers:

-
- 1. **Standard Sourcing Partner (Tier 1):** Unlocked upon basic email and domain registration. Permits account browsing, RFP drafting, and platform navigation.
-
- 2. **Established Trade Partner (Tier 2):** Unlocked via successful AI Document Verification or Open API verification. Permits posting public RFPs, receiving exporter bids, and managing pipelines.
-
- 3. **Verified Enterprise Partner (Tier 3):** Unlocked via multi-attribute verification (e.g., matching Open API records + active payment/escrow method setup). Grants priority matching placement and direct talent pool engagement.
- v. Submissions that fail both Open API lookups and AI document confidence checks are queued for 1-click manual review by NATEP Desk Officers without locking the user out of their account.
- 2. **User Story:**
 - i. **As an** International Service Buyer,
 - ii. **I want** to register without upfront document blockers and complete Just-In-Time KYB verification via automated APIs or AI document scans only when I am ready to post a job,
 - iii. **So that** I experience a seamless onboarding process while establishing transparent business credibility on the platform.
- 3. **Acceptance Criteria (Gherkin Format):**
 - i. **Scenario 1: Zero-Friction Registration without Upfront Blockers**
 -
 - 1. **Given** an international buyer registers using a corporate email domain,
 -
 - 2. **When** the email verification link is confirmed,
 -
 - 3. **Then** the system creates the account under Standard Sourcing Partner status (Tier 1) and redirects the user to the portal dashboard without requesting corporate verification documents.
 - ii. **Scenario 2: Primary JIT Verification via Open APIs**
 -
 - 1. **Given** a Tier 1 buyer attempts to publish a new RFP and provides a UK Company Registration Number or global LEI,
 -
 - 2. **When** the JIT engine queries the Companies House or GLEIF API and validates active corporate standing,
 -
 - 3. **Then** the system automatically updates the buyer to Established Trade Partner status (Tier 2) and publishes the RFP immediately.
 - iii. **Scenario 3: Fallback AI Document Verification Flow**
 -
 - 1. **Given** a buyer's business number is not found in connected open public APIs,

-
- 2. **When** the buyer uploads a local certificate of incorporation or tax registration document,
-
- 3. **Then** the AI Verification Engine extracts entity attributes, validates document authenticity, and updates the account status to Tier 2 upon reaching a high confidence threshold.

2. Service Demand Opportunity & RFP Publishing Subsystem

Requirement Reference: FR-M3-02 (BRD Reference: BR-P1-05 through BR-P1-08)

1. System Processing Logic & Business Rules:

- i. Registered buyers construct Service Demand Opportunities by specifying title, scope of work, budget range, engagement model, delivery timelines, and required qualifications.
- ii. Every opportunity must be tagged with master classification codes corresponding to UN CPC Ver. 2.1, ISIC Rev. 4, and WTO/GATS Modes of Supply (Mode 1: Cross-Border, Mode 2: Consumption Abroad, Mode 3: Commercial Presence, Mode 4: Presence of Natural Persons).
- iii. Supporting attachments are processed via the Shared Evidence Vault (Module 2) for malware scanning and cryptographic SHA-256 deduplication.
- iv. Opportunity state transitions follow an extended lifecycle: DRAFT transitions to UNDER_REVIEW transitions to PUBLISHED transitions to CLOSED transitions to FILLED (or CANCELLED).
- v. Transition to FILLED occurs when a formal contract award event is registered between the buyer and a selected Nigerian service exporter.

2. User Story:

- i. **As a** Registered International Buyer,
- ii. **I want** to post structured Requests for Proposals (RFPs) and track their lifecycle through to candidate selection,
- iii. **So that** I can manage my demand pipeline transparently and signal when an opportunity has been successfully awarded and filled.

3. Acceptance Criteria (Gherkin Format):

- i. **Scenario 1: RFP Publishing with Mandatory Taxonomy Mapping**

-
- 1. **Given** a buyer completes an RFP form and selects valid WTO/GATS Modes of Supply and UN CPC service categories from the central taxonomy service,
-
- 2. **When** the buyer submits the opportunity for publication and passes JIT verification,
-
- 3. **Then** the system validates taxonomy completeness, persists the opportunity record, and broadcasts a standardized JSON domain event (event.opportunity.published.v1).

- ii. **Scenario 2: Opportunity Lifecycle Transition to FILLED**

-
- 1. **Given** a published opportunity is in CLOSED status following bid evaluations,

-
- 2. **When** the buyer executes a contract award selection for a qualified Nigerian service exporter,
-
- 3. **Then** the system transitions the opportunity status to FILLED, records the export conversion metric, and notifies all non-selected bidders.

3. Automated Matchmaking & Supplier Discovery Subsystem

Requirement Reference: FR-M3-03 (BRD Reference: BR-P1-09 through BR-P1-12)

1. System Processing Logic & Business Rules:

- i. Upon opportunity publication, the Matchmaking Engine executes an automated algorithm evaluating verified exporter capabilities, certifications, past performance, and capacity against opportunity requirements.
- ii. Match scoring operates on a scale of 0 to 100 points, calculated using weighted parameters: Category Alignment (30%), Verified Certifications (25%), Past Performance/Export Experience (25%), and Capacity/Availability (20%).
- iii. Exporters achieving a match score of 75 points or higher are automatically notified via system notifications and email dispatches.
- iv. The algorithm relies strictly on cryptographically verified actor profiles and active NDPA 2023 consent grants; unverified credentials or revoked consent fields are excluded from match evaluation.

2. User Story:

- i. **As a** Verified Nigerian Service Exporter or Professional Talent,
- ii. **I want** the system to automatically match my verified profile and capabilities against new trade opportunities,
- iii. **So that** I am immediately alerted to relevant global contracts without manually searching through job listings.

3. Acceptance Criteria (Gherkin Format):

- i. **Scenario 1: Automated Match Calculation and Notification**

-
- 1. **Given** a new opportunity is published in the Software Engineering export category,
-
- 2. **When** the Matchmaking Engine evaluates active exporter profiles against the requirement parameters,
-
- 3. **Then** the system identifies candidates scoring 75 points or above and dispatches automated match notifications within 5 minutes of publication.

- ii. **Scenario 2: Strict Exclusion of Unverified Profile Attributes**

-
- 1. **Given** an exporter profile contains self-declared certifications that have not been validated by the Regulatory Validation Subsystem,
-
- 2. **When** the Matchmaking Engine calculates the weighted match score,
-

3. **Then** it assigns 0 points to unverified certification fields to ensure scoring integrity.

4. Proposal Submission & Bid Management Subsystem

Requirement Reference: FR-M3-04 (BRD Reference: BR-P1-13 through BR-P1-16)

1. System Processing Logic & Business Rules:

- i. Eligible exporters and talent candidates submit formal proposals against published opportunities before specified deadline timestamps.
- ii. Proposal submission requires linking verified credentials, technical proposals, commercial schedules, and explicit consent grants (via Shared Consent Engine) allowing buyers to view disclosure packages.
- iii. Bidding channels support single-entity submissions as well as consortium/joint-venture bids where a primary lead exporter links co-bidders under formal delegation agreements.
- iv. Late submissions are automatically blocked by the system API upon reaching the exact UTC deadline timestamp.

2. User Story:

- i. **As a** Qualified Service Exporter,
- ii. **I want** to submit encrypted, consent-backed commercial and technical proposals against open global RFPs,
- iii. **So that** I can bid on international contracts securely while protecting my proprietary business data.

3. Acceptance Criteria (Gherkin Format):

- i. **Scenario 1: Timely Bid Submission with Active Consent**

■

1. **Given** a verified exporter submits a technical and financial proposal prior to the opportunity deadline timestamp,

■

2. **When** the exporter attaches required compliance evidence and executes an active consent grant for buyer review,

■

3. **Then** the system registers the bid, generates an immutable submission receipt, and updates the bid status to SUBMITTED.

- ii. **Scenario 2: Automated Deadline Enforcement**

■

1. **Given** an opportunity deadline has passed based on the official system UTC clock,

■

2. **When** an exporter attempts to execute POST /api/v1/opportunities/{id}/bids,

■

3. **Then** the system rejects the submission with HTTP 400 Bad Request and records a late attempt log entry.

5. Market Access Intelligence & Regulatory Insights Subsystem

Requirement Reference: FR-M3-05 (BRD Reference: BR-P1-17 through BR-P1-20)

1. System Processing Logic & Business Rules:

- i. Portal 1 serves as the central repository for international trade agreements, WTO/GATS

tariff and non-tariff commitments, bilateral trade arrangements, and destination-market regulatory guidelines.

- ii. Information dispatches are indexed by target country, service sector (CPC code), and Mode of Supply to allow filtered querying.
- iii. Updates to trade policies or foreign market access constraints are maintained by NATEP Administrators and undergo review workflow before public release.
- iv. Public REST API endpoints serve non-confidential trade intelligence using Redis-backed caching layers for optimized performance.

2. User Story:

- i. **As a** Nigerian Exporter planning international expansion,
- ii. **I want** to search and review destination market regulatory requirements, visa rules, and trade barriers by service category,
- iii. **So that** I can ensure full compliance with foreign trade laws prior to entering international contracts.

3. Acceptance Criteria (Gherkin Format):

○ i. Scenario 1: Destination Market Regulatory Search

■

1. **Given** an exporter queries market access rules for IT services targeting United Kingdom (Mode 1 and Mode 4),

■

2. **When** the query request reaches the Market Access REST endpoint,

■

3. **Then** the system returns structured regulatory guides, visa requirements, and relevant tax treaties filtered by the target sector and supply mode.

○ ii. Scenario 2: Dynamic Caching of Market Intelligence Data

■

1. **Given** a high volume of users query general market intelligence insights,

■

2. **When** the request is processed,

■

3. **Then** the API layer serves responses directly from Redis cache with response latency under 50 milliseconds.

Non-Functional & Operational Performance Targets — Portal 1

1. Performance & Scalability Metrics:

- i. **RFP Search & Query Latency:** Under 100 milliseconds for p95 searches across active opportunity datasets.
- ii. **Automated Match Processing Window:** Full match computation and notification dispatch within 5 minutes of opportunity publication.
- iii. **Concurrent Bid Handling:** Capability to process up to 1,000 simultaneous bid submissions during peak deadline windows without data degradation or response timeout.
- iv. **AI Verification Processing Speed:** Document OCR extraction, fraud evaluation, and

status update completed within 15 seconds of upload.

2. Security & Privacy Standards:

- i. Proposal financial details and technical disclosures must be encrypted at rest using AES-256-GCM.
- ii. Access to unmasked exporter proposals is restricted strictly to designated verifiers and foreign buyers possessing active, non-expired NDPA 2023 consent grants.

Stage-Gate Delivery & Verification Traceability

Requirement ID	Subsystem Component	Target Delivery Gate	Required Verification Artifact
FR-M3-01	Global Buyer Onboarding & Hybrid JIT Verification	Stage Gate G3 (Week 10)	Unblocked registration logs, Open API adapter test reports, and AI document verification engine benchmark results.
FR-M3-02	Opportunity & RFP Publishing Engine	Stage Gate G3 (Week 10)	Taxonomy mapping validation report, RFP lifecycle status transition tests (including FILLED), and UAT sign-off.
FR-M3-03	Automated Matchmaking Subsystem	Stage Gate G4 (Week 16)	Match scoring algorithm test execution scripts and notification dispatch logs.
FR-M3-04	Proposal Submission & Bid Management	Stage Gate G4 (Week 16)	Encrypted proposal persistence verification, consent check logs, and deadline test scripts.
FR-M3-05	Market Access &	Stage Gate G4	Market access

	Trade Intelligence	(Week 16)	repository query load test reports and API benchmark results.
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Functional Requirements Document (FRD) / System Requirements Specification (SRS)

MODULE 3: PORTAL 1 — DEMAND & MARKET ACCESS DOMAIN

Document Control & Baseline Alignment

1. **Document Title:** Functional Requirements Document (FRD) / System Requirements Specification (SRS) — Module 3: Portal 1 (Demand & Market Access Domain)
2. **Parent Programme:** Nigeria Service Export Gateway (NATEP / NCMSE)
3. **Baseline Alignment:** Integrated Four-Part TOR v1.1, BRD Module 1 Baseline, and BRD Module 2 (Shared Platform Foundation Services) Baseline
4. **Formatting Policy:** Plain English specification standard. Numbers and Roman numerals are used strictly across all sections to maintain non-bulleted structure. Standard text is used exclusively for all terms, units, and metrics—avoiding mathematical symbols or formulas.

Expanded System-of-Record (SoR) Architectural Matrix — Portal 1

Data Domain / Subsystem	Primary SoR Owner	Downstream Consuming Modules	Permitted Operations	Invariant Constraints
Buyer Profiles & Sourcing Scope	Portal 1 Buyer Registry	Portals 2, 3 & Observatory	Profile creation, sourcing criteria definition, JIT verification status updates.	Zero arbitrary buyer blocks on registration; action-based JIT gating enforced before opportunity publication.
Service Opportunities & RFPs	Portal 1 Opportunity Engine	Portals 2, 3 & Observatory	Demand posting, taxonomy tagging, document attaching,	Mandatory mapping to WTO/GATS Modes of Supply and UN CPC taxonomy

			status transitions.	codes.
Matchmaking & Matching Engine	Portal 1 Matching Engine	Portals 2, 3 & Observatory	Profile indexing, score computation, notification dispatches, shortlist generation.	Zero bias in match score computation; relies strictly on verified capabilities and criteria.
Opportunity Applications & Bids	Portal 1 Bidding Engine	Portal 2 & Observatory	Proposal submission, evidence linking, proposal status tracking.	Proposals blocked without active actor verification and valid consent grants.
Market Intelligence & Access Insights	Portal 1 Market Insights Subsystem	Observatory & External Buyers	Trade barrier publishing, regulatory insights, market trend analytics.	Public access restricted to standardized, aggregated, and non-confidential data.
Trust, Badging & Progress Engine	Portal 1 Trust & Verification Subsystem	Portals 2, 3, Observatory & Shared Foundation	Profile score computation, badge issuance, entitlement unlocking, escrow access control.	Capability entitlements are strictly bound to verified completion thresholds; manual override logged to Module 2 Evidence Vault.

Detailed Subsystem Functional Specifications

1. Global Buyer Onboarding & Hybrid JIT Verification Subsystem

Requirement Reference: FR-M3-01 (BRD Reference: BR-P1-01 through BR-P1-04)

1. System Processing Logic & Business Rules:

- i. Account registration is open and completely unblocked, requiring only corporate email domain validation (user@company.com) and basic profile setup.
- ii. High-value operational actions (e.g., publishing an RFP, sending direct sourcing invitations, or viewing unmasked exporter profiles) trigger a Just-In-Time (JIT) KYB verification step.
- iii. JIT verification executes through a two-stage hybrid evaluation pipeline:

■

1. **Primary Route (Open Public Registries):** The system queries integrated free/open-source global registries (such as GLEIF LEI API, UK Companies House API, EU VIES VAT API) using the buyer's provided business registration or tax number. Successful matches immediately assign the profile to an elevated Trust Tier.

■

2. **Secondary Fallback Route (AI Document Verification):** If no record is found in open APIs, the buyer is prompted to upload local incorporation certificates, tax filings, or trade licenses. The AI Verification Engine performs OCR extraction, visual tamper analysis, and domain cross-matching to grant tier approval.
- iv. The platform assigns neutral, non-indicting Trust Tiers:

■

1. **Standard Sourcing Partner (Tier 1):** Unlocked upon basic email and domain registration. Permits account browsing, RFP drafting, and platform navigation.

■

2. **Established Trade Partner (Tier 2):** Unlocked via successful AI Document Verification or Open API verification. Permits posting public RFPs, receiving exporter bids, and managing pipelines.

■

3. **Verified Enterprise Partner (Tier 3):** Unlocked via multi-attribute verification (e.g., matching Open API records plus active payment/escrow method setup). Grants priority matching placement and direct talent pool engagement.
- v. Submissions that fail both Open API lookups and AI document confidence checks are queued for 1-click manual review by NATEP Desk Officers without locking the user out of their account.

2. User Story:

- i. **As an** International Service Buyer,
- ii. **I want** to register without upfront document blockers and complete Just-In-Time KYB verification via automated APIs or AI document scans only when I am ready to post a job,
- iii. **So that** I experience a seamless onboarding process while establishing transparent business credibility on the platform.

3. Acceptance Criteria (Gherkin Format):

○ i. Scenario 1: Zero-Friction Registration without Upfront Blockers

-
- 1. **Given** an international buyer registers using a corporate email domain,
-
- 2. **When** the email verification link is confirmed,
-
- 3. **Then** the system creates the account under Standard Sourcing Partner status (Tier 1) and redirects the user to the portal dashboard without requesting corporate verification documents.

○ ii. Scenario 2: Primary JIT Verification via Open APIs

-
- 1. **Given** a Tier 1 buyer attempts to publish a new RFP and provides a UK Company Registration Number or global LEI,
-
- 2. **When** the JIT engine queries the Companies House or GLEIF API and validates active corporate standing,
-
- 3. **Then** the system automatically updates the buyer to Established Trade Partner status (Tier 2) and publishes the RFP immediately.

○ iii. Scenario 3: Fallback AI Document Verification Flow

-
- 1. **Given** a buyer's business number is not found in connected open public APIs,
-
- 2. **When** the buyer uploads a local certificate of incorporation or tax registration document,
-
- 3. **Then** the AI Verification Engine extracts entity attributes, validates document authenticity, and updates the account status to Tier 2 upon reaching a high confidence threshold.

2. Service Demand Opportunity & RFP Publishing Subsystem

Requirement Reference: FR-M3-02 (BRD Reference: BR-P1-05 through BR-P1-08)

1. System Processing Logic & Business Rules:

- i. Registered buyers construct Service Demand Opportunities by specifying title, scope of work, budget range, engagement model, delivery timelines, and required qualifications.
- ii. Every opportunity must be tagged with master classification codes corresponding to UN CPC Ver. 2.1, ISIC Rev. 4, and WTO/GATS Modes of Supply (Mode 1: Cross-Border, Mode 2: Consumption Abroad, Mode 3: Commercial Presence, Mode 4: Presence of Natural Persons).
- iii. Supporting attachments are processed via the Shared Evidence Vault (Module 2) for malware scanning and cryptographic SHA-256 deduplication.
- iv. Opportunity state transitions follow an extended lifecycle: DRAFT transitions to

UNDER_REVIEW transitions to PUBLISHED transitions to CLOSED transitions to FILLED (or CANCELLED).

- v. Transition to FILLED occurs when a formal contract award event is registered between the buyer and a selected Nigerian service exporter.

2. User Story:

- i. **As a** Registered International Buyer,
- ii. **I want** to post structured Requests for Proposals (RFPs) and track their lifecycle through to candidate selection,
- iii. **So that** I can manage my demand pipeline transparently and signal when an opportunity has been successfully awarded and filled.

3. Acceptance Criteria (Gherkin Format):

○ i. Scenario 1: RFP Publishing with Mandatory Taxonomy Mapping

■

1. **Given** a buyer completes an RFP form and selects valid WTO/GATS Modes of Supply and UN CPC service categories from the central taxonomy service,

■

2. **When** the buyer submits the opportunity for publication and passes JIT verification,

■

3. **Then** the system validates taxonomy completeness, persists the opportunity record, and broadcasts a standardized JSON domain event (event.opportunity.published.v1).

○ ii. Scenario 2: Opportunity Lifecycle Transition to FILLED

■

1. **Given** a published opportunity is in CLOSED status following bid evaluations,

■

2. **When** the buyer executes a contract award selection for a qualified Nigerian service exporter,

■

3. **Then** the system transitions the opportunity status to FILLED, records the export conversion metric, and notifies all non-selected bidders.

3. Automated Matchmaking & Supplier Discovery Subsystem

Requirement Reference: FR-M3-03 (BRD Reference: BR-P1-09 through BR-P1-12)

1. System Processing Logic & Business Rules:

- i. Upon opportunity publication, the Matchmaking Engine executes an automated algorithm evaluating verified exporter capabilities, certifications, past performance, and capacity against opportunity requirements.
- ii. Match scoring operates on a scale of 0 to 100 points, calculated using weighted parameters: Category Alignment (30 percent), Verified Certifications (25 percent), Past Performance / Export Experience (25 percent), and Capacity / Availability (20 percent).
- iii. Exporters achieving a match score of 75 points or higher are automatically notified via system notifications and email dispatches.
- iv. The algorithm relies strictly on cryptographically verified actor profiles and active NDPA 2023 consent grants; unverified credentials or revoked consent fields are excluded from match evaluation.

2. User Story:

- i. **As a** Verified Nigerian Service Exporter or Professional Talent,
- ii. **I want** the system to automatically match my verified profile and capabilities against new trade opportunities,
- iii. **So that** I am immediately alerted to relevant global contracts without manually searching through job listings.

3. Acceptance Criteria (Gherkin Format):

○ i. Scenario 1: Automated Match Calculation and Notification

■

1. **Given** a new opportunity is published in the Software Engineering export category,

■

2. **When** the Matchmaking Engine evaluates active exporter profiles against the requirement parameters,

■

3. **Then** the system identifies candidates scoring 75 points or above and dispatches automated match notifications within 5 minutes of publication.

○ ii. Scenario 2: Strict Exclusion of Unverified Profile Attributes

■

1. **Given** an exporter profile contains self-declared certifications that have not been validated by the Regulatory Validation Subsystem,

■

2. **When** the Matchmaking Engine calculates the weighted match score,

■

3. **Then** it assigns 0 points to unverified certification fields to ensure scoring integrity.

4. Proposal Submission & Bid Management Subsystem

Requirement Reference: FR-M3-04 (BRD Reference: BR-P1-13 through BR-P1-16)

1. System Processing Logic & Business Rules:

- i. Eligible exporters and talent candidates submit formal proposals against published opportunities before specified deadline timestamps.
- ii. Proposal submission requires linking verified credentials, technical proposals, commercial schedules, and explicit consent grants (via Shared Consent Engine) allowing buyers to view disclosure packages.
- iii. Bidding channels support single-entity submissions as well as consortium/joint-venture bids where a primary lead exporter links co-bidders under formal delegation agreements.
- iv. Late submissions are automatically blocked by the system API upon reaching the exact UTC deadline timestamp.

2. User Story:

- i. **As a** Qualified Service Exporter,
- ii. **I want** to submit encrypted, consent-backed commercial and technical proposals against open global RFPs,
- iii. **So that** I can bid on international contracts securely while protecting my proprietary business data.

3. Acceptance Criteria (Gherkin Format):

○ i. Scenario 1: Timely Bid Submission with Active Consent

■

1. **Given** a verified exporter submits a technical and financial proposal prior to the opportunity deadline timestamp,

■

2. **When** the exporter attaches required compliance evidence and executes an active consent grant for buyer review,

■

3. **Then** the system registers the bid, generates an immutable submission receipt, and updates the bid status to SUBMITTED.

○ ii. Scenario 2: Automated Deadline Enforcement

■

1. **Given** an opportunity deadline has passed based on the official system UTC clock,

■

2. **When** an exporter attempts to execute POST /api/v1/opportunities/{id}/bids,

■

3. **Then** the system rejects the submission with HTTP 400 Bad Request and records a late attempt log entry.

5. Market Access Intelligence & Regulatory Insights Subsystem

Requirement Reference: FR-M3-05 (BRD Reference: BR-P1-17 through BR-P1-20)

1. System Processing Logic & Business Rules:

- i. Portal 1 serves as the central repository for international trade agreements, WTO/GATS tariff and non-tariff commitments, bilateral trade arrangements, and destination-market regulatory guidelines.
- ii. Information dispatches are indexed by target country, service sector (CPC code), and Mode of Supply to allow filtered querying.
- iii. Updates to trade policies or foreign market access constraints are maintained by NATEP Administrators and undergo review workflow before public release.
- iv. Public REST API endpoints serve non-confidential trade intelligence using Redis-backed caching layers for optimized performance.

2. User Story:

- i. **As a** Nigerian Exporter planning international expansion,
- ii. **I want** to search and review destination market regulatory requirements, visa rules, and trade barriers by service category,
- iii. **So that** I can ensure full compliance with foreign trade laws prior to entering international contracts.

3. Acceptance Criteria (Gherkin Format):

○ i. Scenario 1: Destination Market Regulatory Search

■

1. **Given** an exporter queries market access rules for IT services targeting United Kingdom (Mode 1 and Mode 4),

■

2. **When** the query request reaches the Market Access REST endpoint,
-
3. **Then** the system returns structured regulatory guides, visa requirements, and relevant tax treaties filtered by the target sector and supply mode.
- ii. **Scenario 2: Dynamic Caching of Market Intelligence Data**
-
1. **Given** a high volume of users query general market intelligence insights,
-
2. **When** the request is processed,
-
3. **Then** the API layer serves responses directly from Redis cache with response latency under 50 milliseconds.

6. Symmetric Trust Badging & Progressive Feature Unlocking Subsystem

Requirement Reference: FR-M3-06 (BRD Reference: BR-P1-21 through BR-P1-25)

1. **System Processing Logic & Business Rules:**
 - i. The platform maintains dual trust badging pipelines for both **Buyers (Demand Side)** and **Exporters/Talent (Supply Side)** to ensure mutual credibility before contract execution.
 - ii. The Profile Completion Percentage is computed dynamically using weighted attribute completion across two dimensions: *Data Completeness (40 percent)* and *Verification Depth (60 percent)*.
 - iii. System permissions and feature entitlements are dynamically governed by the user's computed Profile Completion Percentage and active Badge Status:
 -
 - 1. **Unverified Tier (less than 40 percent completion):** Platform browsing, public directory search, drafting RFPs or proposals, and accessing free market intelligence.
 -
 - 2. **Verified Trade Partner (40 percent to 79 percent completion):** Publishing active RFPs, receiving or submitting bids, direct messaging, and unmasking basic counterparty profiles.
 -
 - 3. **Enterprise / Top-Rated Trade Partner (80 percent or higher completion plus Verified Payment Setup):** Unlocked priority placement in AI matching, high-value contract execution (greater than 50,000 USD), automated milestone escrow processing, and direct candidate or buyer invitations.
 - iv. Every profile dashboard must display a **Next Unlock Prompt** that explicitly states:
 -
 - 1. The user's current Profile Completion Percentage.
 -
 - 2. Current authorized actions ("What you can do now").
 -
 - 3. Missing verification attributes and the specific higher-tier features they will unlock upon completion ("Next Step to Unlock X").

2. User Story:

- i. **As a** Registered User (Buyer or Exporter),
- ii. **I want** a dynamic score tracker that clearly displays my profile completion rate, current entitlements, and pending verification milestones,
- iii. **So that** I understand exactly how to upgrade my trust status and unlock high-value trading features.

3. Acceptance Criteria (Gherkin Format):

- i. **Scenario 1: Dynamic Calculation of Completion Percentage & Entitlement Unlock**

■

1. **Given** an exporter reaches a 40 percent completion rate by verifying CAC registration and tax ID,

■

2. **When** the profile engine updates the score,

■

3. **Then** the system assigns the **NATEP Verified Exporter** badge and enables bid submission on open RFPs up to 50,000 USD.

- ii. **Scenario 2: Progressive Display of Next Unlock Prompts**

■

1. **Given** a buyer account is at 65 percent completion (Established Trade Partner),

■

2. **When** the buyer views their account dashboard,

■

3. **Then** the UI displays the missing verification step ("*Add Verified Multi-Currency Escrow / Payment Method*") and highlights the upcoming entitlement ("*Unlocks Direct Talent Sourcing & Enterprise Match Priority*").

Symmetric Badging & Entitlement Framework Matrix

User Role	Trust Badge Name	Verification Requirements	Profile Completion Threshold	Unlocked Platform Capabilities
Buyer	Registered Sourcing Partner	Corporate email validation (user@company.com).	1 percent to 39 percent	Browse exporter directories, view public market access insights, and draft RFPs in local sandbox.
Buyer	Established Trade Partner	Primary JIT API match (UK	40 percent to 79 percent	Publish public RFPs, receive

		Companies House / GLEIF LEI) OR successful AI Document Verification.		unmasked exporter proposals, and send direct sourcing invitations.
Buyer	Verified Enterprise Buyer	Verified cross-border escrow connection + active corporate payment method setup.	80 percent to 100 percent	Execute high-value contracts (greater than 50,000 USD), access priority AI candidate matching, and issue direct hire agreements.
Exporter	Registered Talent / Entity	Basic email validation + preliminary skill/corporate intake.	1 percent to 39 percent	Search published RFPs, create capability portfolio, and access market access regulatory guides.
Exporter	NATEP Verified Exporter	Verified CAC registration, Tax Identification Number (TIN), and bank account validation.	40 percent to 79 percent	Submit formal bids on RFPs up to 50,000 USD, receive automated match alerts, and access shortlisting.
Exporter	Top-Rated Export	Sector regulatory clearance (e.g.,	80 percent to 100 percent	Priority placement in AI

	Partner	NITDA, COREN) + minimum 3 verified escrow transactions (greater than 10,000 USD).		Matchmaking, direct invitation to Enterprise RFPs, and lower platform escrow fees.
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7. Cross-Border Settlement & Verified Escrow Subsystem

Requirement Reference: FR-M3-07 (BRD Reference: BR-P1-26 through BR-P1-30)

1. System Processing Logic & Business Rules:

- i. The platform integrates with Central Bank of Nigeria (CBN) licensed banking institutions and cross-border escrow gateways to manage contract funding and milestone releases.
- ii. Payment channel verification operates on a JIT basis when a buyer initiates a contract award.
- iii. Contract execution workflow operates via milestone-based escrow locks:

■

1. **Funding Phase:** Buyer deposits contract value into a segregated multi-currency escrow account (USD, EUR, GBP, NGN).

■

2. **Execution & Evidence Phase:** Exporter completes work milestones and uploads deliverable artifacts to Module 2's Evidence Vault.

■

3. **Inspection & Approval Phase:** Buyer has a fixed window (default: 7 business days) to accept deliverables.

■

4. **Disbursement Phase:** Upon buyer approval (or automatic expiration of the inspection window), escrow funds disburse directly to the exporter's verified account.

- iv. Disputes lock escrow funds and trigger a NATEP Dispute Officer review, evaluating submitted deliverables against original RFP specifications logged in Module 2.

2. User Story:

- i. **As a** Verified Buyer or Exporter,
- ii. **I want** payments secured in multi-currency escrow accounts and disbursed automatically upon milestone completion,
- iii. **So that** both parties are protected against non-performance and payment default.

3. Acceptance Criteria (Gherkin Format):

- i. **Scenario 1: Automated Milestone Escrow Disbursement**

■

1. **Given** an exporter uploads a completed milestone deliverable to the Evidence Vault,

■

2. **When** the buyer approves the deliverable within the 7-day review window,
-
3. **Then** the payment gateway automatically releases the milestone escrow funds to the exporter's verified bank account.

Non-Functional & Operational Performance Targets — Extended Portal 1

1. Performance & Scalability Metrics:

- i. **RFP Search & Query Latency:** Under 100 milliseconds for p95 searches across active opportunity datasets.
- ii. **Automated Match Processing Window:** Full match computation and notification dispatch within 5 minutes of opportunity publication.
- iii. **Concurrent Bid Handling:** Capability to process up to 1,000 simultaneous bid submissions during peak deadline windows without data degradation or response timeout.
- iv. **AI Verification Processing Speed:** Document OCR extraction, fraud evaluation, and status update completed within 15 seconds of upload.
- v. **Profile Entitlement Calculation Speed:** Real-time computation and UI update of Profile Completion Percentage and feature permissions within 500 milliseconds of attribute submission.

2. Security & Privacy Standards:

- i. Proposal financial details and technical disclosures must be encrypted at rest using AES-256-GCM.
- ii. Access to unmasked exporter proposals is restricted strictly to designated verifiers and foreign buyers possessing active, non-expired NDPA 2023 consent grants.

Complete Stage-Gate Delivery & Verification Traceability

Requirement ID	Subsystem Component	Target Delivery Gate	Required Verification Artifact
FR-M3-01	Global Buyer Onboarding & Hybrid JIT Verification	Stage Gate G3 (Week 10)	Unblocked registration logs, Open API adapter test reports, and AI document verification engine benchmark results.
FR-M3-02	Opportunity & RFP Publishing Engine	Stage Gate G3 (Week 10)	Taxonomy mapping validation report, RFP lifecycle status

			transition tests (including FILLED), and UAT sign-off.
FR-M3-03	Trust-Weighted AI Matchmaking Subsystem	Stage Gate G4 (Week 16)	Vector similarity scoring test scripts, weight distribution verification logs, and match notification benchmarks.
FR-M3-04	Proposal Submission & Bid Management	Stage Gate G4 (Week 16)	Encrypted proposal persistence verification, consent check logs, and deadline test scripts.
FR-M3-05	Market Access & Trade Intelligence	Stage Gate G4 (Week 16)	Market access repository query load test reports and API benchmark results.
FR-M3-06	Symmetric Badging & Entitlement Engine	Stage Gate G3 (Week 10)	Profile completion percentage calculation test scripts, dynamic badge evaluation logs, and permission gating integration tests.
FR-M3-07	Cross-Border Escrow & Payment Gateway	Stage Gate G4 (Week 16)	Payment gateway API integration logs, milestone escrow lock/release unit tests, and dispute resolution workflow

			sign-off.
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